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# Linear Algebra

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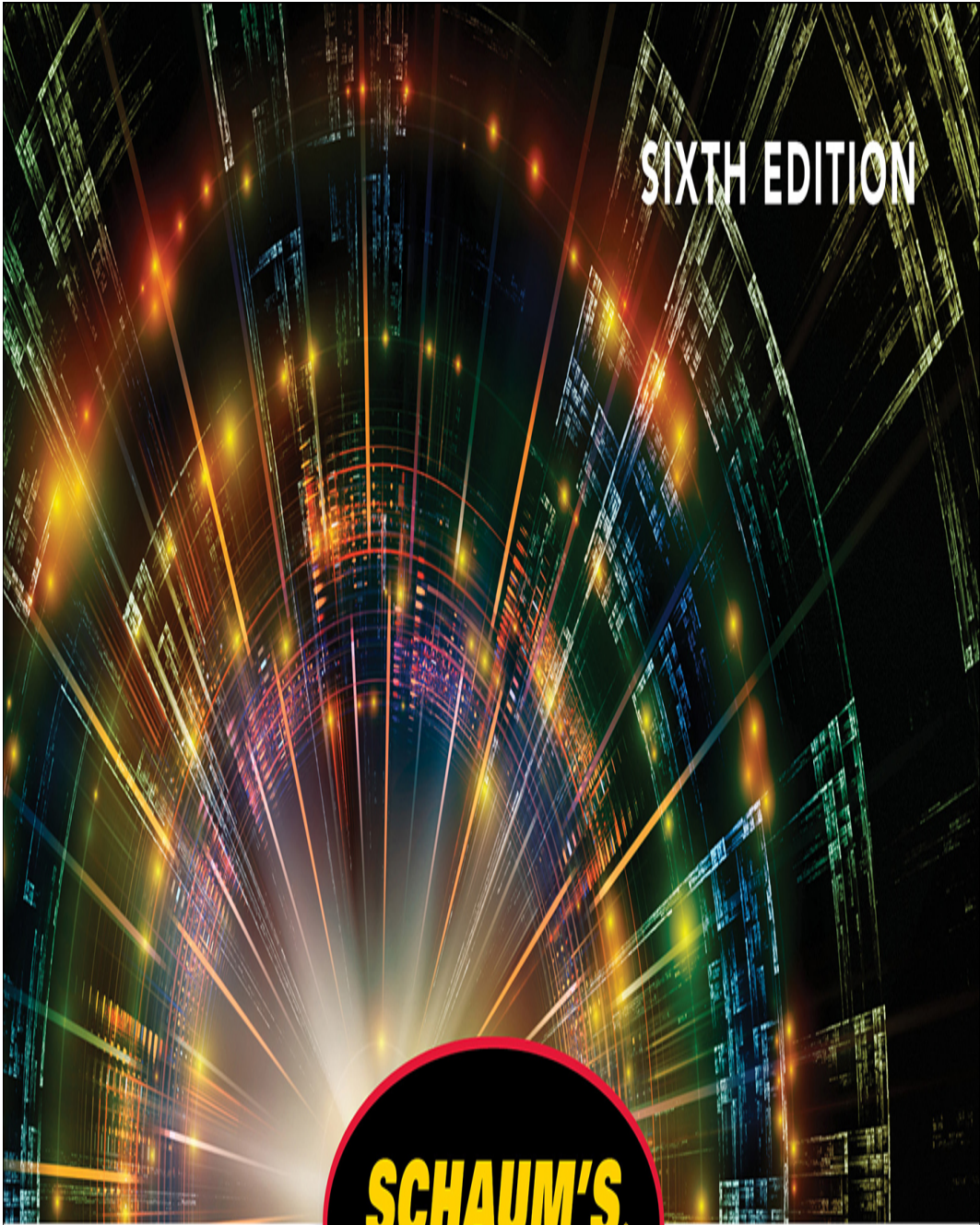
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# ***Linear Algebra***

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*Sixth Edition*

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*Temple University*

***Marc Lars Lipson, PhD***

*University of Virginia*

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**Schaum's Outline Series**



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ISBN: 978-1-26-001145-6

MHID: 1-26-001145-3

The material in this eBook also appears in the print version of this title: ISBN: 978-1-26-001144-9, MHID: 1-26-001144-5.

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# ***Preface***

Linear algebra has in recent years become an essential part of the mathematical background required by mathematicians and mathematics teachers, engineers, computer scientists, physicists, economists, and statisticians, among others. This requirement reflects the importance and wide applications of the subject matter.

This book is designed for use as a textbook for a formal course in linear algebra or as a supplement to all current standard texts. It aims to present an introduction to linear algebra which will be found helpful to all readers regardless of their fields of specification. More material has been included than can be covered in most first courses. This has been done to make the book more flexible, to provide a useful book of reference, and to stimulate further interest in the subject.

Each chapter begins with clear statements of pertinent definitions, principles, and theorems together with illustrative and other descriptive material. This is followed by graded sets of solved and supplementary problems. The solved problems serve to illustrate and amplify the theory, and to provide the repetition of basic principles so vital to effective learning. Numerous proofs, especially those of all essential theorems, are included among the solved problems. The supplementary problems serve as a complete review of the material of each chapter.

The first three chapters treat vectors in Euclidean space, matrix algebra, and systems of linear equations. These chapters provide the motivation and basic computational

tools for the abstract investigations of vector spaces and linear mappings which follow. After chapters on inner product spaces and orthogonality and on determinants, there is a detailed discussion of eigenvalues and eigenvectors giving conditions for representing a linear operator by a diagonal matrix. This naturally leads to the study of various canonical forms, specifically, the triangular, Jordan, and rational canonical forms. Later chapters cover linear functions and the dual space  $V^*$ , and bilinear, quadratic, and Hermitian forms. The last chapter treats linear operators on inner product spaces.

The main changes in the sixth edition are that some parts in Appendix D have been added to the main part of the text, that is, Chapter Four and Chapter Eight. There are also many additional solved and supplementary problems.

Finally, we wish to thank the staff of the McGraw-Hill Schaum's Outline Series, especially Diane Grayson, for their unfailing cooperation.

SEYMOUR LIPSCHUTZ  
MARC LARS LIPSON

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## ***List of Symbols***

$A = [a_{ij}]$ , matrix, 27  
 $\bar{A} = [\bar{a}_{ij}]$ , conjugate matrix, 38  
 $|A|$ , determinant, 266, 270  
 $A^*$ , adjoint, 379  
 $A^H$ , conjugate transpose, 38  
 $A^T$ , transpose, 33  
 $A^+$ , Moore-Penrose inverse, 420  
 $A_{ij}$ , minor, 271  
 $A(I, J)$ , minor, 275  
 $A(V)$ , linear operators, 176  
 $\text{adj } A$ , adjoint (classical), 273  
 $A \sim B$ , row equivalence, 72  
 $A \simeq B$ , congruence, 362  
 $\mathbf{C}$ , complex numbers, 11  
 $\mathbf{C}^n$ , complex  $n$ -space, 13  
 $C[a, b]$ , continuous functions, 230  
 $C(f)$ , companion matrix, 306  
 $\text{colsp } (A)$ , column space, 120  
 $d(u, v)$ , distance, 5, 243  
 $\text{diag}(a_{11}, \dots, a_{nn})$ , diagonal matrix, 35  
 $\text{diag}(A_{11}, \dots, A_{nn})$ , block diagonal, 40  
 $\det(A)$ , determinant, 270  
 $\dim V$ , dimension, 124  
 $\{e_1, \dots, e_n\}$ , usual basis, 125  
 $E_k$ , projections, 386

$f: A \rightarrow B$ , mapping, 166  
 $F(X)$ , function space, 114  
 $G \circ F$ , composition, 175  
 $\text{Hom}(V, U)$ , homomorphisms, 176  
**i, j, k**, 9  
 $I_n$ , identity matrix, 33  
 $\text{Im } F$ , image, 171  
 $J(\lambda)$ , Jordan block, 331  
 $K$ , field of scalars, 112  
 $\text{Ker } F$ , kernel, 171  
 $m(t)$ , minimal polynomial, 305  
 $\mathbf{M}_{m,n}$ ,  $m \times n$  matrices, 114  
 $n$ -space, 5, 13, 229, 242  
 $\mathbf{P}(t)$ , polynomials, 114  
 $\mathbf{P}_n(t)$ , polynomials, 114  
 $\text{proj}(u, v)$ , projection, 6, 236  
 $\text{proj}(u, V)$ , projection, 237  
 $\mathbf{Q}$ , rational numbers, 11  
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 $\text{sgn } \sigma$ , sign, parity, 269  
 $\text{span}(S)$ , linear span, 119  
 $\text{tr}(A)$ , trace, 33  
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 $T$ -invariant, 329  
 $T^t$ , transpose, 353  
 $\|u\|$ , norm, 5, 13, 229, 243  
 $[u]_S$ , coordinate vector, 130  
 $u \cdot v$ , dot product, 4, 13

$\langle u, v \rangle$ , inner product, 228, 240  
 $u \times v$ , cross product, 10  
 $u \otimes v$ , tensor product, 398  
 $u \wedge v$ , exterior product, 403  
 $u \oplus v$ , direct sum, 129, 329  
 $V \cong U$ , isomorphism, 132, 171  
 $V \otimes W$ , tensor product, 398  
 $V^*$ , dual space, 351  
 $V^{**}$ , second dual space, 352  
 $\bigwedge^r V$ , exterior product, 403  
 $W^0$ , annihilator, 353  
 $\bar{z}$ , complex conjugate, 12  
 $Z(v, T)$ ,  $T$ -cyclic subspace, 332  
 $\delta_{ij}$ , Kronecker delta, 37  
 $\Delta(t)$ , characteristic polynomial, 296  
 $\lambda$ , eigenvalue, 298  
 $\Sigma$ , summation symbol, 29

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# CHAPTER 1

## *Vectors in $R^n$ and $C^n$ , Spatial Vectors*

### 1.1 Introduction

---

There are two ways to motivate the notion of a vector: one is by means of lists of numbers and subscripts, and the other is by means of certain objects in physics. We discuss these two ways below.

Here we assume the reader is familiar with the elementary properties of the field of real numbers, denoted by **R**. On the other hand, we will review properties of the field of complex numbers, denoted by **C**. In the context of vectors, the elements of our number fields are called scalars.

Although we will restrict ourselves in this chapter to vectors whose elements come from **R** and then from **C**, many of our operations also apply to vectors whose entries come from some arbitrary field  $K$ .

### Lists of Numbers

---

Suppose the weights (in pounds) of eight students are listed as follows:

156, 125, 145, 134, 178, 145, 162, 193

One can denote all the values in the list using only one symbol, say  $w$ , but with different subscripts; that is,

$w_1, w_2, w_3, w_4, w_5, w_6, w_7, w_8$

Observe that each subscript denotes the position of the value in the list. For example,

$w_1 = 156$ , the first number,  $w_2 = 125$ , the second number, ...

Such a list of values,

$w = (w_1, w_2, w_3, \dots, w_8)$

is called a *linear array* or *vector*.

## Vectors in Physics

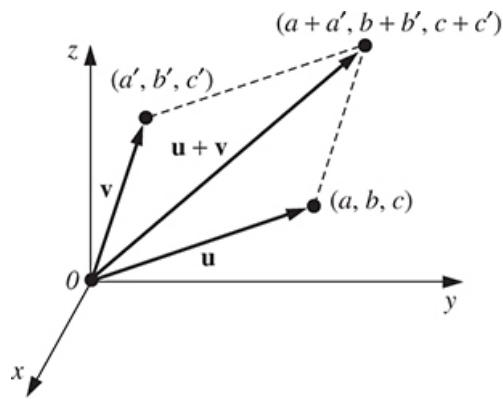
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Many physical quantities, such as temperature and speed, possess only “magnitude.” These quantities can be represented by real numbers and are called *scalars*. On the other hand, there are also quantities, such as force and velocity, that possess both “magnitude” and “direction.” These quantities, which can be represented by arrows having appropriate lengths and directions and emanating from some given reference point  $O$ , are called *vectors*.

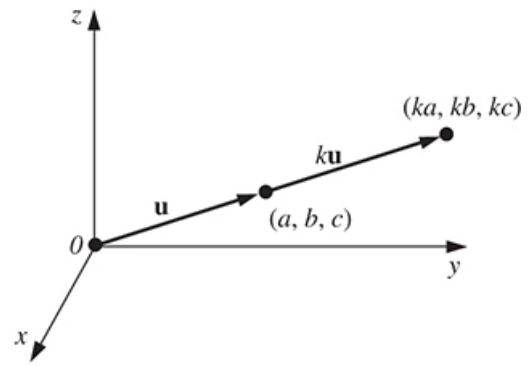
Now we assume the reader is familiar with the space  $\mathbf{R}^3$  where all the points in space are represented by ordered triples of real numbers. Suppose the origin of the axes in  $\mathbf{R}^3$  is chosen as the reference point  $O$  for the vectors discussed above. Then every vector is uniquely determined by the coordinates of its endpoint, and vice versa.

There are two important operations, vector addition and scalar multiplication, associated with vectors in physics. The definition of these operations and the relationship between these operations and the endpoints of the vectors are as follows.

- (i) **Vector Addition:** The resultant  $\mathbf{u} + \mathbf{v}$  of two vectors  $\mathbf{u}$  and  $\mathbf{v}$  is obtained by the *parallelogram law*; that is,  $\mathbf{u} + \mathbf{v}$  is the diagonal of the parallelogram formed by  $\mathbf{u}$  and  $\mathbf{v}$ . Furthermore, if  $(a, b, c)$  and  $(a', b', c')$  are the endpoints of the vectors  $\mathbf{u}$  and  $\mathbf{v}$ , then  $(a + a', b + b', c + c')$  is the endpoint of the vector  $\mathbf{u} + \mathbf{v}$ . These properties are pictured in Fig. 1-1(a).



(a) Vector Addition



(b) Scalar Multiplication

**Figure 1-1**

- (ii) **Scalar Multiplication:** The product  $k\mathbf{u}$  of a vector  $\mathbf{u}$  by a real number  $k$  is obtained by multiplying the magnitude of  $\mathbf{u}$  by  $k$  and retaining the same direction if  $k > 0$  or the opposite direction if  $k < 0$ . Also, if  $(a, b, c)$  is the endpoint of the vector  $\mathbf{u}$ , then  $(ka, kb, kc)$  is the endpoint of the vector  $k\mathbf{u}$ . These properties are pictured in Fig. 1-1(b).

Mathematically, we identify the vector  $\mathbf{u}$  with its  $(a, b, c)$  and write  $\mathbf{u} = (a, b, c)$ . Moreover, we call the ordered triple

$(a, b, c)$  of real numbers a point or vector depending upon its interpretation. We generalize this notion and call an  $n$ -tuple  $(a_1, a_2, \dots, a_n)$  of real numbers a vector. However, special notation may be used for the vectors in  $\mathbf{R}^3$  called *spatial* vectors (Section 1.6).

## 1.2 Vectors in $\mathbf{R}^n$

---

The set of all  $n$ -tuples of real numbers, denoted by  $\mathbf{R}^n$ , is called *n-space*. A particular  $n$ -tuple in  $\mathbf{R}^n$ , say

$$u = (a_1, a_2, \dots, a_n)$$

is called a *point* or *vector*. The numbers  $a_i$  are called the *coordinates*, *components*, *entries*, or *elements* of  $u$ .

Moreover, when discussing the space  $\mathbf{R}^n$ , we use the term *scalar* for the elements of  $\mathbf{R}$ .

Two vectors,  $u$  and  $v$ , are *equal*, written  $u = v$ , if they have the same number of components and if the corresponding components are equal. Although the vectors  $(1, 2, 3)$  and  $(2, 3, 1)$  contain the same three numbers, these vectors are not equal because corresponding entries are not equal.

The vector  $(0, 0, \dots, 0)$  whose entries are all 0 is called the *zero vector* and is usually denoted by  $0$ .

### EXAMPLE 1.1

(a) The following are vectors:

$$(2, -5), \quad (7, 9), \quad (0, 0, 0), \quad (3, 4, 5)$$

The first two vectors belong to  $\mathbf{R}^2$ , whereas the last two belong to  $\mathbf{R}^3$ . The third is the zero vector in  $\mathbf{R}^3$ .

(b) Find  $x, y, z$  such that  $(x - y, x + y, z - 1) = (4, 2, 3)$ .

By definition of equality of vectors, corresponding entries must be equal. Thus,

$$x - y = 4, \quad x + y = 2, \quad z - 1 = 3$$

Solving the above system of equations yields  $x = 3$ ,  $y = -1$ ,  $z = 4$ .

## Column Vectors

---

Sometimes a vector in  $n$ -space  $\mathbf{R}^n$  is written vertically rather than horizontally. Such a vector is called a *column vector*, and, in this context, the horizontally written vectors in [Example 1.1](#) are called *row vectors*. For example, the following are column vectors with 2, 2, 3, and 3 components, respectively:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ -4 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}, \quad \begin{bmatrix} 1.5 \\ \frac{2}{3} \\ -15 \end{bmatrix}$$

We also note that any operation defined for row vectors is defined analogously for column vectors.

## 1.3 Vector Addition and Scalar Multiplication

---

Consider two vectors  $u$  and  $v$  in  $\mathbf{R}^n$ , say

$$u = (a_1, a_2, \dots, a_n) \quad \text{and} \quad v = (b_1, b_2, \dots, b_n)$$

Their *sum*, written  $u + v$ , is the vector obtained by adding corresponding components from  $u$  and  $v$ . That is,

$$u + v = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

The *product*, of the vector  $u$  by a real number  $k$ , written  $ku$ , is the vector obtained by multiplying each component of  $u$  by  $k$ . That is,

$$ku = k(a_1, a_2, \dots, a_n) = (ka_1, ka_2, \dots, ka_n)$$

Observe that  $u + v$  and  $ku$  are also vectors in  $\mathbf{R}^n$ . The sum of vectors with different numbers of components is not defined.

Negatives and subtraction are defined in  $\mathbf{R}^n$  as follows:

$$-u = (-1)u \quad \text{and} \quad u - v = u + (-v)$$

The vector  $-u$  is called the *negative* of  $u$ , and  $u - v$  is called the *difference* of  $u$  and  $v$ .

Now suppose we are given vectors  $u_1, u_2, \dots, u_m$  in  $\mathbf{R}^n$  and scalars  $k_1, k_2, \dots, k_m$  in  $\mathbf{R}$ . We can multiply the vectors by the corresponding scalars and then add the resultant scalar products to form the vector

$$v = k_1u_1 + k_2u_2 + k_3u_3 + \dots + k_mu_m$$

Such a vector  $v$  is called a *linear combination* of the vectors  $u_1, u_2, \dots, u_m$ .

## EXAMPLE 1.2

(a) Let  $u = (2, 4, -5)$  and  $v = (1, -6, 9)$ . Then

$$u + v = (2 + 1, 4 + (-6), -5 + 9) = (3, -2, 4)$$

$$7u = (7(2), 7(4), 7(-5)) = (14, 28, -35)$$

$$-v = (-1)(1, -6, 9) = (-1, 6, -9)$$

$$3u - 5v = (6, 12, -15) + (-5, 30, -45) = (1, 42, -60)$$

(b) The zero vector  $0 = (0, 0, \dots, 0)$  in  $\mathbf{R}^n$  is similar to the scalar 0 in that, for any vector  $u = (a_1, a_2, \dots, a_n)$ .

$$u + 0 = (a_1 + 0, a_2 + 0, \dots, a_n + 0) = (a_1, a_2, \dots, a_n) = u$$

(c) Let  $u = \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}$  and  $v = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$ . Then  $2u - 3v = \begin{bmatrix} 4 \\ 6 \\ -8 \end{bmatrix} + \begin{bmatrix} -9 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} -5 \\ 9 \\ -2 \end{bmatrix}$ .

Basic properties of vectors under the operations of vector addition and scalar multiplication are described in the following theorem.

**THEOREM 1.1:** For any vectors  $u, v, w$  in  $\mathbf{R}^n$  and any scalars  $k, K$  in  $\mathbf{R}$ ,

(i)  $(u + v) + w = u + (v + w)$ ,

(ii)  $u + 0 = u$ ,

(iii)  $u + (-u) = 0$ ,

(iv)  $u + v = v + u$ ,

(v)  $k(u + v) = ku + kv$ ,

(vi)  $(k + K)u = ku + Ku$ ,

(vii)  $(kK)u = k(Ku)$ ,

(viii)  $1u = u$ .

We postpone the proof of [Theorem 1.1](#) until [Chapter 2](#), where it appears in the context of matrices ([Problem 2.3](#)).

Suppose  $u$  and  $v$  are vectors in  $\mathbf{R}^n$  for which  $u = kv$  for some nonzero scalar  $k$  in  $\mathbf{R}$ . Then  $u$  is called a *multiple* of  $v$ . Also,  $u$  is said to be in the *same* or *opposite direction* as  $v$  according to whether  $k > 0$  or  $k < 0$ .

## 1.4 Dot (Inner) Product

---

Consider arbitrary vectors  $u$  and  $v$  in  $\mathbf{R}^n$ ; say,

$$u = (a_1, a_2, \dots, a_n) \quad \text{and} \quad v = (b_1, b_2, \dots, b_n)$$

The *dot product* or *inner product* of  $u$  and  $v$  is denoted and defined by

$$u \cdot v = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

That is,  $u \cdot v$  is obtained by multiplying corresponding components and adding the resulting products. The vectors  $u$  and  $v$  are said to be *orthogonal* (or *perpendicular*) if their dot product is zero—that is, if  $u \cdot v = 0$ .

### EXAMPLE 1.3

(a) Let  $u = (1, -2, 3)$ ,  $v = (4, 5, -1)$ ,  $w = (2, 7, 4)$ . Then,

$$u \cdot v = 1(4) - 2(5) + 3(-1) = 4 - 10 - 3 = -9$$

$$u \cdot w = 2 - 14 + 12 = 0, \quad v \cdot w = 8 + 35 - 4 = 39$$

Thus,  $u$  and  $w$  are orthogonal.

(b) Let  $u = \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}$  and  $v = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$ . Then  $u \cdot v = 6 - 3 + 8 = 11$ .

(c) Suppose  $u = (1; 2; 3; 4)$  and  $v = (6; k; -8; 2)$ . Find  $k$  so that  $u$  and  $v$  are orthogonal.

First obtain  $u \cdot v = 6 + 2k - 24 + 8 = -10 + 2k$ . Then set  $u \cdot v = 0$  and solve for  $k$ :

$$-10 + 2k = 0 \quad \text{or} \quad 2k = 10 \quad \text{or} \quad k = 5$$

Basic properties of the dot product in  $\mathbf{R}^n$  (proved in [Problem 1.13](#)) follow.

**THEOREM 1.2:** For any vectors  $u, v, w$  in  $\mathbf{R}^n$  and any scalar  $k$  in  $\mathbf{R}$  :

$$(i) (u + v) \cdot w = u \cdot w + v \cdot w,$$

$$(ii) (ku) \cdot v = k(u \cdot v),$$

$$(iii) u \cdot v = v \cdot u,$$

$$(iv) u \cdot u \geq 0, \text{ and } u \cdot u = 0 \text{ iff } u = 0.$$

Note that (ii) says that we can “take  $k$  out” from the first position in an inner product. By (iii) and (ii),

$$u \cdot (kv) = (kv) \cdot u = k(v \cdot u) = k(u \cdot v)$$

That is, we can also “take  $k$  out” from the second position in an inner product.

The space  $\mathbf{R}^n$  with the above operations of vector addition, scalar multiplication, and dot product is usually called *Euclidean  $n$ -space*.

## **Norm (Length) of a Vector**

---

The *norm* or *length* of a vector  $u$  in  $\mathbf{R}^n$ , denoted by  $\|u\|$ , is defined to be the nonnegative square root of  $u \cdot u$ . In particular, if  $u = (a_1, a_2, \dots, a_n)$ , then

$$\|u\| = \sqrt{u \cdot u} = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$$

That is,  $\|u\|$  is the square root of the sum of the squares of the components of  $u$ . Thus,  $\|u\| \geq 0$ , and  $\|u\| = 0$  if and only if  $u = 0$ .

A vector  $u$  is called a *unit vector* if  $\|u\| = 1$  or, equivalently, if  $u \cdot u = 1$ . For any nonzero vector  $v$  in  $\mathbf{R}^n$ , the vector

$$\hat{v} = \frac{1}{\|v\|} v = \frac{v}{\|v\|}$$

is the unique unit vector in the same direction as  $v$ . The process of finding  $\hat{v}$  from  $v$  is called *normalizing*  $v$ .

#### EXAMPLE 1.4

(a) Suppose  $u = (1, -2, -4, 5, 3)$ . To find  $\|u\|$ , we can first find  $\|u\|^2 = u \cdot u$  by squaring each component of  $u$  and adding, as follows:

$$\|u\|^2 = 1^2 + (-2)^2 + (-4)^2 + 5^2 + 3^2 = 1 + 4 + 16 + 25 + 9 = 55$$

Then  $\|u\| = \sqrt{55}$ .

(b) Let  $v = (1, -3, 4, 2)$  and  $w = (\frac{1}{2}, -\frac{1}{6}, \frac{5}{6}, \frac{1}{6})$ . Then

$$\|v\| = \sqrt{1 + 9 + 16 + 4} = \sqrt{30} \quad \text{and} \quad \|w\| = \sqrt{\frac{9}{36} + \frac{1}{36} + \frac{25}{36} + \frac{1}{36}} = \sqrt{\frac{36}{36}} = \sqrt{1} = 1$$

Thus  $w$  is a unit vector, but  $v$  is not a unit vector. However, we can normalize  $v$  as follows:

$$\hat{v} = \frac{v}{\|v\|} = \left( \frac{1}{\sqrt{30}}, \frac{-3}{\sqrt{30}}, \frac{4}{\sqrt{30}}, \frac{2}{\sqrt{30}} \right)$$

This is the unique unit vector in the same direction as  $v$ .

The following formula (proved in [Problem 1.14](#)) is known as the Schwarz inequality or Cauchy-Schwarz inequality. It is used in many branches of mathematics.

**THEOREM 1.3** (Schwarz): For any vectors  $u, v$  in  $\mathbf{R}^n$ ,  $|u \cdot v| \leq \|u\| \|v\|$ .

Using the above inequality, we also prove ([Problem 1.15](#)) the following result known as the “triangle inequality” or Minkowski’s inequality.

**THEOREM 1.4** (Minkowski): For any vectors  $u, v$  in  $\mathbf{R}^n$ ,  $\|u + v\| \leq \|u\| + \|v\|$ .

## **Distance, Angles, Projections**

---

The *distance* between vectors  $u = (a_1, a_2, \dots, a_n)$  and  $v = (b_1, b_2, \dots, b_n)$  in  $\mathbf{R}^n$  is denoted and defined by

$$d(u, v) = \|u - v\| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \cdots + (a_n - b_n)^2}$$

One can show that this definition agrees with the usual notion of distance in the Euclidean plane  $\mathbf{R}^2$  or space  $\mathbf{R}^3$ .

The *angle*  $\theta$  between nonzero vectors  $u, v$  in  $\mathbf{R}^n$  is defined by

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}$$

This definition is well defined, because, by the Schwarz inequality ([Theorem 1.3](#)),

$$-1 \leq \frac{u \cdot v}{\|u\| \|v\|} \leq 1$$

Note that if  $u \cdot v = 0$ , then  $\theta = 90^\circ$  (or  $\theta = \pi/2$ ). This then agrees with our previous definition of orthogonality.

The *projection* of a vector  $u$  onto a nonzero vector  $v$  is the vector denoted and defined by

$$\text{proj}(u, v) = \frac{u \cdot v}{\|v\|^2} v = \frac{u \cdot v}{v \cdot v} v$$

We show below that this agrees with the usual notion of vector projection in physics.

**EXAMPLE 1.5**

(a) Suppose  $u = (1, 2, 3)$  and  $v = (2, 4, 5)$ . Then

$$d(u, v) = \sqrt{(1-2)^2 + (-2-4)^2 + (3-5)^2} = \sqrt{1+36+4} = \sqrt{41}$$

To find  $\cos \theta$ , where  $\theta$  is the angle between  $u$  and  $v$ , we first find

$$u \cdot v = 2 - 8 + 15 = 9, \quad \|u\|^2 = 1 + 4 + 9 = 14, \quad \|v\|^2 = 4 + 16 + 25 = 45$$

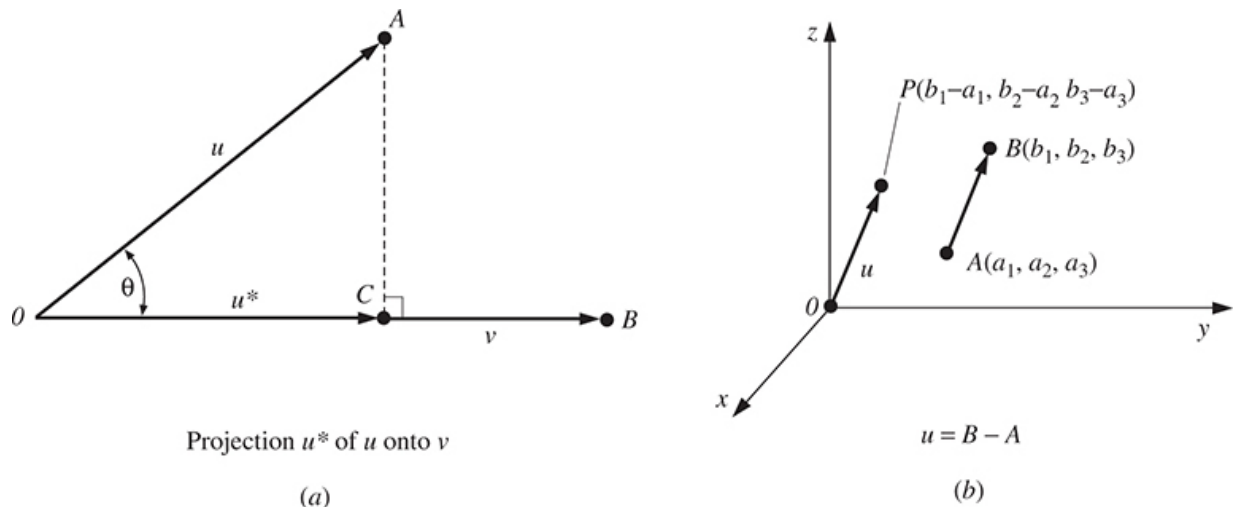
Then

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|} = \frac{9}{\sqrt{14}\sqrt{45}}$$

Also,

$$\text{proj}(u, v) = \frac{u \cdot v}{\|v\|^2} v = \frac{9}{45} (2, 4, 5) = \frac{1}{5} (2, 4, 5) = \left( \frac{2}{5}, \frac{4}{5}, 1 \right)$$

(b) Consider the vectors  $u$  and  $v$  in [Fig. 1-2](#) (a) (with respective endpoints  $A$  and  $B$ ). The (perpendicular) projection of  $u$  onto  $v$  is the vector  $u^*$  with magnitude



**Figure 1-2**

$$\|u^*\| = \|u\| \cos \theta = \|u\| \frac{u \cdot v}{\|u\| \|v\|} = \frac{u \cdot v}{\|v\|}$$

To obtain  $u^*$ , we multiply its magnitude by the unit vector in the direction of  $v$ , obtaining

$$u^* = \|u^*\| \frac{v}{\|v\|} = \frac{u \cdot v}{\|v\|} \frac{v}{\|v\|} = \frac{u \cdot v}{\|v\|^2} v$$

This is the same as the above definition of  $\text{proj}(u, v)$ .

## 1.5 Located Vectors, Hyperplanes, Lines, Curves in $\mathbf{R}^n$

---

This section distinguishes between an  $n$ -tuple  $P(a_i) \equiv P(a_1, a_2, \dots, a_n)$  viewed as a point in  $\mathbf{R}^n$  and an  $n$ -tuple  $u = [c_1, c_2, \dots, c_n]$  viewed as a vector (arrow) from the origin  $O$  to the point  $C(c_1, c_2, \dots, c_n)$ .

### Located Vectors

---

Any pair of points  $A(a_i)$  and  $B(b_i)$  in  $\mathbf{R}^n$  defines the *located vector* or *directed line segment* from  $A$  to  $B$ , written  $\overrightarrow{AB}$ . We identify  $\overrightarrow{PQ}$  with the vector

$$u = B - A = [b_1 - a_1, b_2 - a_2, \dots, b_n - a_n]$$

because  $\overrightarrow{PQ}$  and  $u$  have the same magnitude and direction. This is pictured in Fig. 1-2 (b) for the points  $A(a_1, a_2, a_3)$  and  $B(b_1, b_2, b_3)$  in  $\mathbf{R}^3$  and the vector  $u = B - A$  which has the endpoint  $P(b_1 - a_1, b_2 - a_2, b_3 - a_3)$ .

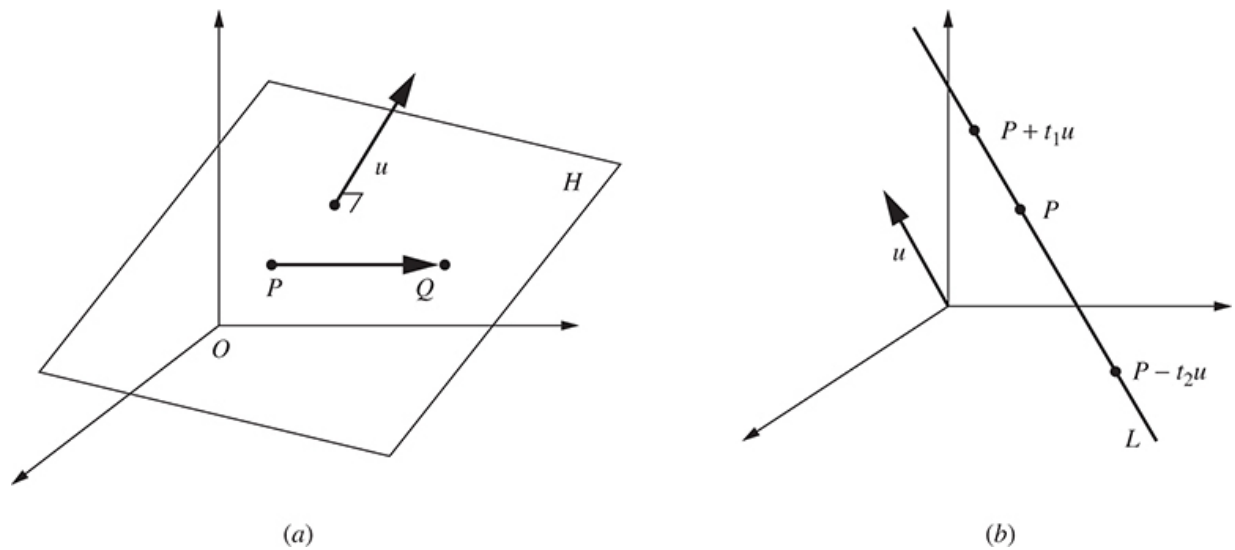
## Hyperplanes

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A *hyperplane*  $H$  in  $\mathbf{R}^n$  is the set of points  $(x_1, x_2, \dots, x_n)$  that satisfy a linear equation

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where the vector  $u = [a_1, a_2, \dots, a_n]$  of coefficients is not zero. Thus a hyperplane  $H$  in  $\mathbf{R}^2$  is a line, and a hyperplane  $H$  in  $\mathbf{R}^3$  is a plane. We show below, as pictured in Fig. 1-3 (a) for  $\mathbf{R}^3$ , that  $u$  is orthogonal to any directed line segment  $v = \overrightarrow{PQ}$ , where  $P(p_i)$  and  $Q(q_i)$  are points in  $H$ : [For this reason, we say that  $u$  is *normal* to  $H$  and that  $H$  is *normal* to  $u$ .]



**Figure 1-3**

Because  $P(p_i)$  and  $Q(q_i)$  belong to  $H$ , they satisfy the above hyperplane equation—that is,

$$a_1 p_1 + a_2 p_2 + \cdots + a_n p_n = b \quad \text{and} \quad a_1 q_1 + a_2 q_2 + \cdots + a_n q_n = b$$

Let  $v = \overrightarrow{PQ} = Q - P = [q_1 - p_1, q_2 - p_2, \dots, q_n - p_n]$

Then

$$\begin{aligned} u \cdot v &= a_1(q_1 - p_1) + a_2(q_2 - p_2) + \cdots + a_n(q_n - p_n) \\ &= (a_1 q_1 + a_2 q_2 + \cdots + a_n q_n) - (a_1 p_1 + a_2 p_2 + \cdots + a_n p_n) = b - b = 0 \end{aligned}$$

Thus  $v = \overrightarrow{PQ}$  is orthogonal to  $u$ , as claimed.

## Lines in $\mathbb{R}^n$

The line  $L$  in  $\mathbb{R}^n$  passing through the point  $P(b_1, b_2, \dots, b_n)$  and in the direction of a nonzero vector  $u = [a_1, a_2, \dots, a_n]$  consists of the points  $X(x_1, x_2, \dots, x_n)$  that satisfy

$$X = P + tu \quad \text{or} \quad \begin{cases} x_1 = a_1t + b_1 \\ x_2 = a_2t + b_2 \\ \dots\dots\dots \\ x_n = a_nt + b_n \end{cases} \quad \text{or} \quad L(t) = (a_it + b_i)$$

where the *parameter*  $t$  takes on all real values. Such a line  $L$  in  $\mathbf{R}^3$  is pictured in Fig. 1-3 (b).

### EXAMPLE 1.6

- (a) Let  $H$  be the plane in  $\mathbf{R}^3$  corresponding to the linear equation  $2x - 5y + 7z = 4$ . Observe that  $P(1, 1, 1)$  and  $Q(5, 4, 2)$  are solutions of the equation. Thus  $P$  and  $Q$  and the directed line segment

$$v = \overrightarrow{PQ} = Q - P = [5 - 1, 4 - 1, 2 - 1] = [4, 3, 1]$$

lie on the plane  $H$ . The vector  $u = [2, 5, 7]$  is normal to  $H$ , and, as expected,

$$u \cdot v = [2, -5, 7] \cdot [4, 3, 1] = 8 - 15 + 7 = 0$$

That is,  $u$  is orthogonal to  $v$ .

- (b) Find an equation of the hyperplane  $H$  in  $\mathbf{R}^4$  that passes through the point  $P(1, 3, 4, 2)$  and is normal to the vector  $u = [4, 2, 5, 6]$ .

The coefficients of the unknowns of an equation of  $H$  are the components of the normal vector  $u$ ; hence, the equation of  $H$  must be of the form

$$4x_1 - 2x_2 + 5x_3 + 6x_4 = k$$

Substituting  $P$  into this equation, we obtain

$$4(1) - 2(3) + 5(-4) + 6(2) = k \quad \text{or} \quad 4 - 6 - 20 + 12 = k \quad \text{or} \quad k = -10$$

Thus,  $4x_1^2 + 2x_2 + 5x_3 + 6x_4 = 10$  is the equation of  $H$ .

- (c) Find the parametric representation of the line  $L$  in  $\mathbf{R}^4$  passing through the point  $P(1, 2, 3, -4)$  and in the direction of  $u = [5, 6, 7, 8]$ . Also, find the point  $Q$  on  $L$  when  $t = 1$ .

Substitution in the above equation for  $L$  yields the following parametric representation:

$$x_1 = 5t + 1, \quad x_2 = 6t + 2, \quad x_3 = -7t + 3, \quad x_4 = 8t - 4$$

or, equivalently,

$$L(t) = (5t + 1, 6t + 2, -7t + 3, 8t - 4)$$

Note that  $t = 0$  yields the point  $P$  on  $L$ . Substitution of  $t = 1$  yields the point  $Q(6, 8, 4, 4)$  on  $L$ .

## **Curves in $\mathbf{R}^n$**

---

Let  $D$  be an interval (finite or infinite) on the real line  $\mathbf{R}$ . A continuous function  $F: D \rightarrow \mathbf{R}^n$  is a *curve* in  $\mathbf{R}^n$ . Thus, to each point  $t \in D$  there is assigned the following point in  $\mathbf{R}^n$ :

$$F(t) = [F_1(t), F_2(t), \dots, F_n(t)]$$

Moreover, the derivative (if it exists) of  $F(t)$  yields the vector

$$V(t) = \frac{dF(t)}{dt} = \left[ \frac{dF_1(t)}{dt}, \frac{dF_2(t)}{dt}, \dots, \frac{dF_n(t)}{dt} \right]$$

which is tangent to the curve. Normalizing  $V(t)$  yields

$$\mathbf{T}(t) = \frac{V(t)}{\|V(t)\|}$$

Thus,  $\mathbf{T}(t)$  is the unit tangent vector to the curve. (Unit vectors with geometrical significance are often presented in bold type.)

**EXAMPLE 1.7** Consider the curve  $F(t) = [\sin t, \cos t, t]$  in  $\mathbf{R}^3$ . Taking the derivative of  $F(t)$  [or each component of  $F(t)$ ] yields

$$V(t) = [\cos t, -\sin t, 1]$$

which is a vector tangent to the curve. We normalize  $V(t)$ . First we obtain

$$\|V(t)\|^2 = \cos^2 t + \sin^2 t + 1 = 1 + 1 = 2$$

Then the unit tangent vector  $\mathbf{T}(t)$  to the curve follows:

$$\mathbf{T}(t) = \frac{V(t)}{\|V(t)\|} = \left[ \frac{\cos t}{\sqrt{2}}, \frac{-\sin t}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

## 1.6 Vectors in $\mathbf{R}^3$ (Spatial Vectors), $\mathbf{i}, \mathbf{j}, \mathbf{k}$ Notation

---

Vectors in  $\mathbf{R}^3$ , called *spatial vectors*, appear in many applications, especially in physics. In fact, a special notation is frequently used for such vectors as follows:

$\mathbf{i} = [1, 0, 0]$  denotes the unit vector in the  $x$  direction

$\mathbf{j} = [0, 1, 0]$  denotes the unit vector in the  $y$  direction

$\mathbf{k} = [0, 0, 1]$  denotes the unit vector in the  $z$  direction

Then any vector  $u = [a, b, c]$  in  $\mathbf{R}^3$  can be expressed uniquely in the form

$$u = [a, b, c] = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$$

Because the vectors  $\mathbf{i}$ ,  $\mathbf{j}$ ,  $\mathbf{k}$  are unit vectors and are mutually orthogonal, we obtain the following dot products:

$$\mathbf{i} \cdot \mathbf{i} = 1, \quad \mathbf{j} \cdot \mathbf{j} = 1, \quad \mathbf{k} \cdot \mathbf{k} = 1 \quad \text{and} \quad \mathbf{i} \cdot \mathbf{j} = 0, \quad \mathbf{i} \cdot \mathbf{k} = 0, \quad \mathbf{j} \cdot \mathbf{k} = 0$$

Furthermore, the vector operations discussed above may be expressed in the  $\mathbf{ijk}$  notation as follows. Suppose

$$u = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \quad \text{and} \quad v = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$$

Then

$$u + v = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j} + (a_3 + b_3)\mathbf{k} \quad \text{and} \quad cu = ca_1\mathbf{i} + ca_2\mathbf{j} + ca_3\mathbf{k}$$

where  $c$  is a scalar. Also,

$$u \cdot v = a_1b_1 + a_2b_2 + a_3b_3 \quad \text{and} \quad \|u\| = \sqrt{u \cdot u} = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

**EXAMPLE 1.8** Suppose  $u = 3\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$  and  $v = 4\mathbf{i} + 8\mathbf{j} + 7\mathbf{k}$ .

(a) To find  $u + v$ , add corresponding components, obtaining

$$u + v = 7\mathbf{i} + 13\mathbf{j} + 5\mathbf{k}$$

(b) To find  $3u - 2v$ , first multiply by the scalars and then add:

$$3u - 2v = (9\mathbf{i} + 15\mathbf{j} + 6\mathbf{k}) + (-8\mathbf{i} + 16\mathbf{j} - 14\mathbf{k}) = \mathbf{i} + 31\mathbf{j} - 20\mathbf{k}$$

(c) To find  $u \cdot v$ , multiply corresponding components and then add:

$$u \cdot v = 12 + 40 - 14 = 38$$

(d) To find  $\|u\|$ , take the square root of the sum of the squares of the components:

$$\|u\| = \sqrt{9 + 25 + 4} = \sqrt{38}$$

## Cross Product

---

There is a special operation for vectors  $u$  and  $v$  in  $\mathbf{R}^3$  that is not defined in  $\mathbf{R}^n$  for  $n \neq 3$ . This operation is called the *cross product* and is denoted by  $u \times v$ . One way to easily remember the formula for  $u \times v$  is to use the determinant (of order two) and its negative, which are denoted and defined as follows:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad \text{and} \quad -\begin{vmatrix} a & b \\ c & d \end{vmatrix} = bc - ad$$

Here  $a$  and  $d$  are called the *diagonal* elements and  $b$  and  $c$  are the *nondiagonal* elements. Thus, the determinant is the product  $ad$  of the diagonal elements minus the product  $bc$  of the nondiagonal elements, but vice versa for the negative of the determinant.

Now suppose  $u = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$  and  $v = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ . Then

$$\begin{aligned} u \times v &= (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \\ &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \mathbf{k} \end{aligned}$$

That is, the three components of  $u \times v$  are obtained from the array

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$$

(which contain the components of  $u$  above the component of  $v$ ) as follows:

- (1) Cover the first column and take the determinant.
- (2) Cover the second column and take the negative of the determinant.
- (3) Cover the third column and take the determinant.

Note that  $u \times v$  is a vector; hence,  $u \times v$  is also called the *vector product* or *outer product* of  $u$  and  $v$ .

**EXAMPLE 1.9** Find  $u \times v$  where:

$$(a) u = 4\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}, v = 2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}, (b) u = [2, -1, 5], v = [3, 7, 6].$$

$$(a) \text{ Use } \begin{bmatrix} 4 & 3 & 6 \\ 2 & 5 & -3 \end{bmatrix} \text{ to get } u \times v = (-9 - 30)\mathbf{i} + (12 + 12)\mathbf{j} + (20 - 6)\mathbf{k} = -39\mathbf{i} + 24\mathbf{j} + 14\mathbf{k}$$

$$(b) \text{ Use } \begin{bmatrix} 2 & -1 & 5 \\ 3 & 7 & 6 \end{bmatrix} \text{ to get } u \times v = [-6 - 35, 15 - 12, 14 + 3] = [-41, 3, 17]$$

**Remark:** The cross products of the vectors  $\mathbf{i}, \mathbf{j}, \mathbf{k}$  are as follows:

$$\begin{array}{lll} \mathbf{i} \times \mathbf{j} = \mathbf{k}, & \mathbf{j} \times \mathbf{k} = \mathbf{i}, & \mathbf{k} \times \mathbf{i} = \mathbf{j} \\ \mathbf{j} \times \mathbf{i} = -\mathbf{k}, & \mathbf{k} \times \mathbf{j} = -\mathbf{i}, & \mathbf{i} \times \mathbf{k} = -\mathbf{j} \end{array}$$

Thus, if we view the triple  $(\mathbf{i}, \mathbf{j}, \mathbf{k})$  as a cyclic permutation, where  $\mathbf{i}$  follows  $\mathbf{k}$  and hence  $\mathbf{k}$  precedes  $\mathbf{i}$ , then the product of two of them in the given direction is the third one, but the product of two of them in the opposite direction is the negative of the third one.

Two important properties of the cross product are contained in the following theorem.

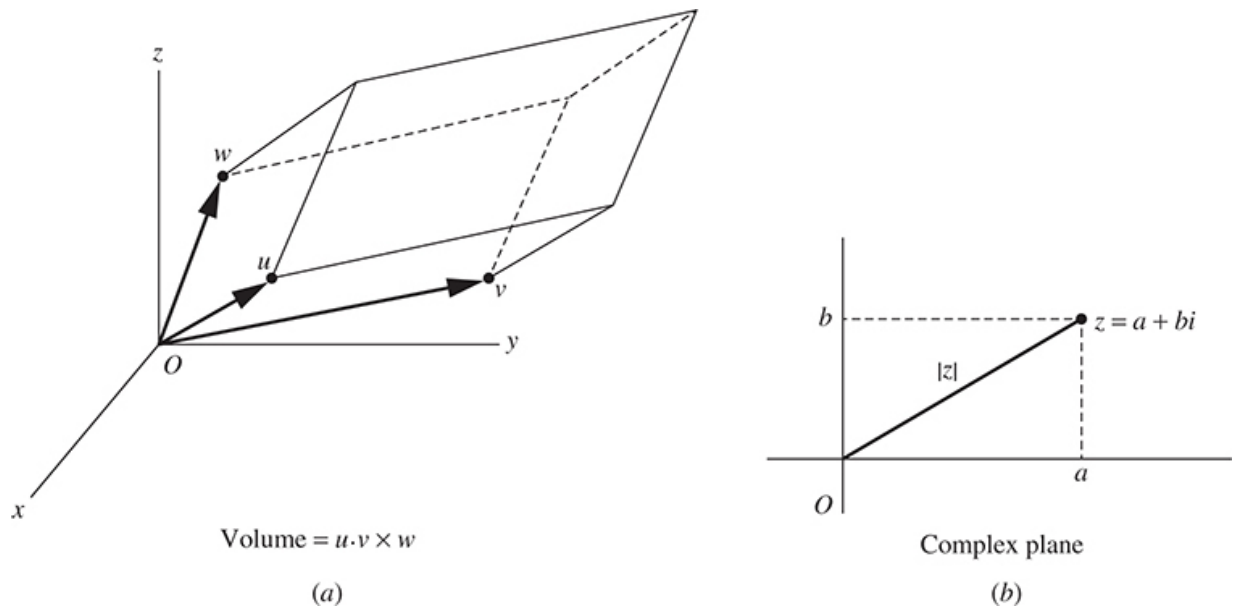
**THEOREM 1.5:** Let  $u, v, w$  be vectors in  $\mathbf{R}^3$ .

- (a) The vector  $u \times v$  is orthogonal to both  $u$  and  $v$ .

(b) The absolute value of the “triple product”

$$u \cdot v \times w$$

represents the volume of the parallelepiped formed by the vectors  $u$ ,  $v$ ,  $w$ . [See Fig. 1-4 (a).]



**Figure 1-4**

We note that the vectors  $u$ ,  $v$ ,  $u \times v$  form a right-handed system, and that the following formula gives the magnitude of  $u \times v$ :

$$\|u \times v\| = \|u\| \|v\| \sin \theta$$

where  $\theta$  is the angle between  $u$  and  $v$ .

## 1.7 Complex Numbers

The set of complex numbers is denoted by  $\mathbf{C}$ . Formally, a complex number is an ordered pair  $(a, b)$  of real numbers

where equality, addition, and multiplication are defined as follows:

$$\begin{aligned}(a, b) &= (c, d) \quad \text{if and only if } a = c \text{ and } b = d \\(a, b) + (c, d) &= (a + c, b + d) \\(a, b) \cdot (c, d) &= (ac - bd, ad + bc)\end{aligned}$$

We identify the real number  $a$  with the complex number  $(a, 0)$ ; that is,

$$a \leftrightarrow (a, 0)$$

This is possible because the operations of addition and multiplication of real numbers are preserved under the correspondence; that is,

$$(a, 0) + (b, 0) = (a + b, 0) \quad \text{and} \quad (a, 0) \cdot (b, 0) = (ab, 0)$$

Thus we view  $\mathbf{R}$  as a subset of  $\mathbf{C}$ , and replace  $(a, 0)$  by  $a$  whenever convenient and possible.

We note that the set  $\mathbf{C}$  of complex numbers with the above operations of addition and multiplication is a *field* of numbers, like the set  $\mathbf{R}$  of real numbers and the set  $\mathbf{Q}$  of *rational numbers*.

The complex number  $(0, 1)$  is denoted by  $i$ . It has the important property that

$$i^2 = ii = (0, 1)(0, 1) = (-1, 0) = -1 \quad \text{or} \quad i = \sqrt{-1}$$

Accordingly, any complex number  $z = (a, b)$  can be written in the form

$$z = (a, b) = (a, 0) + (0, b) = (a, 0) + (b, 0) \cdot (0, 1) = a + bi$$

The above notation  $z = a + bi$ , where  $a \equiv \operatorname{Re} z$  and  $b \equiv \operatorname{Im} z$  are called, respectively, the *real* and *imaginary parts* of  $z$ , is more convenient than  $(a, b)$ . In fact, the sum and product of complex numbers  $z = a + bi$  and  $w = c + di$  can be derived by simply using the commutative and distributive laws and  $i^2 = -1$ :

$$\begin{aligned} z + w &= (a + bi) + (c + di) = a + c + bi + di = (a + c) + (b + d)i \\ zw &= (a + bi)(c + di) = ac + bci + adi + bdi^2 = (ac - bd) + (bc + ad)i \end{aligned}$$

We also define the *negative* of  $z$  and subtraction in  $\mathbf{C}$  by

$$-z = -1z \quad \text{and} \quad w - z = w + (-z)$$

**Warning:** The letter  $i$  representing  $\sqrt{-1}$  has no relationship whatsoever to the vector  $\mathbf{i} = [1, 0, 0]$  in Section 1.6.

## Complex Conjugate, Absolute Value

Consider a complex number  $z = a + bi$ . The *conjugate* of  $z$  is denoted and defined by

$$\bar{z} = \overline{a + bi} = a - bi$$

Then  $z\bar{z} = (a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$ . Note that  $z$  is real if and only if  $\bar{z} = z$ .

The *absolute value* of  $z$ , denoted by  $|z|$ , is defined to be the nonnegative square root of  $z\bar{z}$ . Namely,

$$|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$$

Note that  $|z|$  is equal to the norm of the vector  $(a, b)$  in  $\mathbf{R}^2$ .

Suppose  $z \neq 0$ . Then the inverse  $z^{-1}$  of  $z$  and division in  $\mathbf{C}$  of  $w$  by  $z$  are given, respectively, by

$$z^{-1} = \frac{\bar{z}}{z\bar{z}} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i \quad \text{and} \quad \frac{w}{z} = \frac{w\bar{z}}{z\bar{z}} = w z^{-1}$$

**EXAMPLE 1.10** Suppose  $z = 2 + 3i$  and  $w = 5 - 2i$ . Then

$$z + w = (2 + 3i) + (5 - 2i) = 2 + 5 + 3i - 2i = 7 + i$$

$$zw = (2 + 3i)(5 - 2i) = 10 + 15i - 4i - 6i^2 = 16 + 11i$$

$$\bar{z} = \overline{2 + 3i} = 2 - 3i \quad \text{and} \quad \bar{w} = \overline{5 - 2i} = 5 + 2i$$

$$\frac{w}{z} = \frac{5 - 2i}{2 + 3i} = \frac{(5 - 2i)(2 - 3i)}{(2 + 3i)(2 - 3i)} = \frac{4 - 19i}{13} = \frac{4}{13} - \frac{19}{13}i$$

$$|z| = \sqrt{4 + 9} = \sqrt{13} \quad \text{and} \quad |w| = \sqrt{25 + 4} = \sqrt{29}$$

## Complex Plane

---

Recall that the real numbers  $\mathbf{R}$  can be represented by points on a line. Analogously, the complex numbers  $\mathbf{C}$  can be represented by points in the plane. Specifically, we let the point  $(a, b)$  in the plane represent the complex number  $a + bi$  as shown in Fig. 1-4 (b). In such a case,  $|z|$  is the distance from the origin  $O$  to the point  $z$ . The plane with this representation is called the *complex plane*, just like the line representing  $\mathbf{R}$  is called the *real line*.

## 1.8 Vectors in $\mathbf{C}^n$

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The set of all  $n$ -tuples of complex numbers, denoted by  $\mathbf{C}^n$ , is called *complex  $n$ -space*. Just as in the real case, the elements of  $\mathbf{C}^n$  are called *points* or *vectors*, the elements of  $\mathbf{C}$  are called *scalars*, and vector addition in  $\mathbf{C}^n$  and scalar multiplication on  $\mathbf{C}^n$  are given by

$$\begin{aligned} [z_1, z_2, \dots, z_n] + [w_1, w_2, \dots, w_n] &= [z_1 + w_1, z_2 + w_2, \dots, z_n + w_n] \\ z[z_1, z_2, \dots, z_n] &= [zz_1, zz_2, \dots, zz_n] \end{aligned}$$

where the  $z_i$ ,  $w_i$ , and  $z$  belong to  $\mathbf{C}$ .

**EXAMPLE 1.11** Consider vectors  $u = [2 + 3i, 4 - i, 3]$  and  $v = [3 - 2i, 5i, 4 - 6i]$  in  $\mathbf{C}^3$ . Then

$$\begin{aligned} u + v &= [2 + 3i, 4 - i, 3] + [3 - 2i, 5i, 4 - 6i] = [5 + i, 4 + 4i, 7 - 6i] \\ (5 - 2i)u &= [(5 - 2i)(2 + 3i), (5 - 2i)(4 - i), (5 - 2i)(3)] = [16 + 11i, 18 - 13i, 15 - 6i] \end{aligned}$$

## **Dot (Inner) Product in $\mathbf{C}^n$**

---

Consider vectors  $u = [z_1, z_2, \dots, z_n]$  and  $v = [w_1, w_2, \dots, w_n]$  in  $\mathbf{C}^n$ . The *dot* or *inner product* of  $u$  and  $v$  is denoted and defined by

$$u \cdot v = z_1 \bar{w}_1 + z_2 \bar{w}_2 + \cdots + z_n \bar{w}_n$$

This definition reduces to the real case because  $\bar{w}_i = w_i$  when  $w_i$  is real. The norm of  $u$  is defined by

$$\|u\| = \sqrt{u \cdot u} = \sqrt{z_1 \bar{z}_1 + z_2 \bar{z}_2 + \cdots + z_n \bar{z}_n} = \sqrt{|z_1|^2 + |z_2|^2 + \cdots + |z_n|^2}$$

We emphasize that  $u \cdot u$  and so  $\|u\|$  are real and positive when  $u \neq 0$  and 0 when  $u = 0$ .

**EXAMPLE 1.12** Consider vectors  $u = [2 + 3i, 4 - i, 3 + 5i]$  and  $v = [3 - 4i, 5i, 4 - 2i]$  in  $\mathbf{C}^3$ . Then

$$\begin{aligned} u \cdot v &= (2 + 3i)(\overline{3 - 4i}) + (4 - i)(\overline{5i}) + (3 + 5i)(\overline{4 - 2i}) \\ &= (2 + 3i)(3 + 4i) + (4 - i)(-5i) + (3 + 5i)(4 + 2i) \\ &= (-6 + 13i) + (-5 - 20i) + (2 + 26i) = -9 + 19i \\ u \cdot u &= |2 + 3i|^2 + |4 - i|^2 + |3 + 5i|^2 = 4 + 9 + 16 + 1 + 9 + 25 = 64 \\ \|u\| &= \sqrt{64} = 8 \end{aligned}$$

The space  $\mathbf{C}^n$  with the above operations of vector addition, scalar multiplication, and dot product, is called *complex Euclidean n- space*. [Theorem 1.2](#) for  $\mathbf{R}^n$  also holds for  $\mathbf{C}^n$  if we replace  $u \cdot v = v \cdot u$  by

$$u \cdot v = \overline{u} \cdot v$$

On the other hand, the Schwarz inequality ([Theorem 1.3](#)) and Minkowski's inequality ([Theorem 1.4](#)) are true for  $\mathbf{C}^n$  with no changes.

## SOLVED PROBLEMS

### Vectors in $\mathbf{R}^n$

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**1.1.** Determine which of the following vectors are equal:

$$u_1 = (1, 2, 3), \quad u_2 = (2, 3, 1), \quad u_3 = (1, 3, 2), \quad u_4 = (2, 3, 1)$$

Vectors are equal only when corresponding entries are equal; hence, only  $u_2 = u_4$ .

**1.2.** Let  $u = (2, -7, 1)$ ,  $v = (-3, 0, 4)$ ,  $w = (0, 5, -8)$ . Find:

(a)  $3u - 4v$ ,

(b)  $2u + 3v - 5w$ .

First perform the scalar multiplication and then the vector addition.

(a)

$$3u - 4v = 3(2, -7, 1) - 4(-3, 0, 4) = (6, -21, 3) + (12, 0, -16) = (18, -21, -13)$$

(b)

$$2u + 3v - 5w = (4, -14, 2) + (-9, 0, 12) + (0, -25, 40) = (-5, -39, 54)$$

**1.3.** Let  $u = \begin{bmatrix} 5 \\ 3 \\ -4 \end{bmatrix}$ ,  $v = \begin{bmatrix} -1 \\ 5 \\ 2 \end{bmatrix}$ ,  $w = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$

(a)  $5u - 2v$ ,

(b)  $-2u + 4v - 3w$ .

First perform the scalar multiplication and then the vector addition:

(a)  $5u - 2v = 5 \begin{bmatrix} 5 \\ 3 \\ -4 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 25 \\ 15 \\ -20 \end{bmatrix} + \begin{bmatrix} 2 \\ -10 \\ -4 \end{bmatrix} = \begin{bmatrix} 27 \\ 5 \\ -24 \end{bmatrix}$

(b)  $-2u + 4v - 3w = \begin{bmatrix} -10 \\ -6 \\ 8 \end{bmatrix} + \begin{bmatrix} -4 \\ 20 \\ 8 \end{bmatrix} + \begin{bmatrix} -9 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} -23 \\ 17 \\ 22 \end{bmatrix}$

**1.4.** Find  $x$  and  $y$ , where: (a)  $(x, 3) = (2, x + y)$ , (b)  $(4, y) = x(2, 3)$ .

(a) Because the vectors are equal, set the corresponding entries equal to each other, yielding

$$x = 2, \quad 3 = x + y$$

Solve the linear equations, obtaining  $x = 2$ ,  $y = 1$ .

(b) First multiply by the scalar  $x$  to obtain  $(4, y) = (2x, 3x)$ . Then set corresponding entries equal to each other to obtain

$$4 = 2x, \quad y = 3x$$

Solve the equations to yield  $x = 2$ ,  $y = 6$ .

**1.5.** Write the vector  $v = (1, 2, 5)$  as a linear combination of the vectors  $u_1 = (1, 1, 1)$ ,  $u_2 = (1, 2, 3)$ ,  $u_3 = (2, 1, 1)$ .

We want to express  $v$  in the form  $v = xu_1 + yu_2 + zu_3$  with  $x, y, z$  as yet unknown. First we have

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} x + y + 2z \\ x + 2y - z \\ x + 3y + z \end{bmatrix}$$

(It is more convenient to write vectors as columns than as rows when forming linear combinations.) Set corresponding entries equal to each other to obtain

$$\begin{array}{lll} x + y + 2z = 1 & & x + y + 2z = 1 \\ x + 2y - z = -2 & \text{or} & y - 3z = -3 \\ x + 3y + z = 5 & & 2y - z = 4 \end{array} \quad \text{or} \quad \begin{array}{lll} x + y + 2z = 1 & & x + y + 2z = 1 \\ & & y - 3z = -3 \\ & & 5z = 10 \end{array}$$

This unique solution of the triangular system is  $x = -6$ ,  $y = 3$ ,  $z = 2$ . Thus,  $v = -6u_1 + 3u_2 + 2u_3$ .

**1.6.** Write  $v = (2, 5, 3)$  as a linear combination of

$$u_1 = (1, -3, 2), u_2 = (2, -4, -1), u_3 = (1, -5, 7).$$

Find the equivalent system of linear equations and then solve. First,

$$\begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = x \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ -4 \\ -1 \end{bmatrix} + z \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix} = \begin{bmatrix} x + 2y + z \\ -3x - 4y - 5z \\ 2x - y + 7z \end{bmatrix}$$

Set the corresponding entries equal to each other to obtain

$$\begin{array}{lll} x + 2y + z = 2 & & x + 2y + z = 2 \\ -3x - 4y - 5z = -5 & \text{or} & 2y - 2z = 1 \\ 2x - y + 7z = 3 & & -5y + 5z = -1 \end{array} \quad \text{or} \quad \begin{array}{lll} x + 2y + z = 2 & & x + 2y + z = 2 \\ & & 2y - 2z = 1 \\ & & 0 = 3 \end{array}$$

The third equation,  $0x + 0y + 0z = 3$ , indicates that the system has no solution. Thus,  $v$  cannot be written as a linear combination of the vectors  $u_1, u_2, u_3$ .

## **Dot (Inner) Product, Orthogonality, Norm in $\mathbb{R}^n$**

**1.7.** Find  $u \times v$  where:

(a)  $u = (2, 5, 6)$  and  $v = (8, 2, 3)$ ,

(b)  $u = (4, 2, 3, 5, -1)$  and  $v = (2, 6, 1, 4, 8)$ .

Multiply the corresponding components and add:

(a)  $u \cdot v = 2(8) + 5(2) + 6(-3) = 16 + 10 - 18 = 12$

(b)  $u \cdot v = 8 + 12 + 3 - 20 - 8 = -5$

**1.8.** Let  $u = (5, 4, 1)$ ,  $v = (3, -4, 1)$ ,  $w = (1, -2, 3)$ . Which pair of vectors, if any, are perpendicular (orthogonal)?

Find the dot product of each pair of vectors:

$$u \cdot v = 15 - 16 + 1 = 0, \quad v \cdot w = 3 + 8 + 3 = 14, \quad u \cdot w = 5 - 8 + 3 = 0$$

Thus,  $u$  and  $v$  are orthogonal,  $u$  and  $w$  are orthogonal, but  $v$  and  $w$  are not.

**1.9.** Find  $k$  so that  $u$  and  $v$  are orthogonal, where:

(a)  $u = (1, k, -3)$  and  $v = (2, -5, 4)$ ,

(b)  $u = (2, 3, k, -4, 1, 5)$  and  $v = (6, -1, 3, 7, 2k)$ .

Compute  $u \cdot v$ , set  $u \cdot v$  equal to 0, and then solve for  $k$ :

(a)  $u \cdot v = 1(2) + k(-5) - 3(4) = -5k - 10$ . Then  $-5k - 10 = 0$ , or  $k = -2$ .

(b)  $u \cdot v = 12 - 3k - 12 + 7 + 10k = 7k + 7$ . Then  $7k + 7 = 0$ , or  $k = -1$ .

- 1.10.** Find  $\|u\|$ , where: (a)  $u = (3, -12, -4)$ , (b)  $u = (2, -3, 8, -7)$ .

First find  $\|u\|^2 = u \cdot u$  by squaring the entries and adding. Then  $\|u\| = \sqrt{\|u\|^2}$

(a)

$$\|u\|^2 = (3)^2 + (-12)^2 + (-4)^2 = 9 + 144 + 16 = 169. \text{ Then } \|u\| = \sqrt{169} = 13.$$

(b)  $\|u\|^2 = 4 + 9 + 64 + 49 = 126$ . Then  $\|u\| = \sqrt{126}$ .

- 1.11.** Recall that *normalizing* a nonzero vector  $v$  means finding the unique unit vector  $\hat{v}$  in the same direction as  $v$ , where

$$\hat{v} = \frac{1}{\|v\|} v$$

Normalize: (a)  $u = (3, -4)$ , (b)  $v = (4, -2, -3, 8)$ , (c)  $w = (\frac{1}{2}, \frac{2}{3}, -\frac{1}{4})$ .

(a) First find  $\|u\| = \sqrt{9 + 16} = \sqrt{25} = 5$ . Then divide each entry of  $u$  by 5, obtaining  $\hat{u} = (\frac{3}{5}, -\frac{4}{5})$ .

(b) Here  $\|v\| = \sqrt{16 + 4 + 9 + 64} = \sqrt{93}$ . Then

$$\hat{v} = \left( \frac{4}{\sqrt{93}}, \frac{-2}{\sqrt{93}}, \frac{-3}{\sqrt{93}}, \frac{8}{\sqrt{93}} \right)$$

(c) Note that  $w$  and any positive multiple of  $w$  will have the same normalized form. Hence, first multiply  $w$  by 12 to “clear fractions”—that is, first find  $w' = 12w = (6, 8, -3)$ . Then

$$\|w'\| = \sqrt{36 + 64 + 9} = \sqrt{109} \text{ and } \hat{w} = \hat{w'} = \left( \frac{6}{\sqrt{109}}, \frac{8}{\sqrt{109}}, \frac{-3}{\sqrt{109}} \right)$$

- 1.12.** Let  $u = (1, -3, 4)$  and  $v = (3, 4, 7)$ . Find:

(a)  $\cos \theta$ , where  $\theta$  is the angle between  $u$  and  $v$ ;

(b)  $\text{proj}(u, v)$ , the projection of  $u$  onto  $v$ ;

(c)  $d(u, v)$ , the distance between  $u$  and  $v$ .

First find  $u \cdot v = 3 \cdot 12 + 28 = 19$ ,  $\|u\|^2 = 1 + 9 + 16 = 26$ ,  $\|v\|^2 = 9 + 16 + 49 = 74$ . Then

$$(a) \cos \theta = \frac{u \cdot v}{\|u\| \|v\|} = \frac{19}{\sqrt{26} \sqrt{74}},$$

(b)

$$\text{proj}(u, v) = \frac{u \cdot v}{\|v\|^2} v = \frac{19}{74} (3, 4, 7) = \left( \frac{57}{74}, \frac{76}{74}, \frac{133}{74} \right) = \left( \frac{57}{74}, \frac{38}{37}, \frac{133}{74} \right),$$

$$(c) d(u, v) = \|u - v\| = \|(-2, -7, -3)\| = \sqrt{4 + 49 + 9} = \sqrt{62}.$$

**1.13.** Prove [Theorem 1.2](#): For any  $u, v, w$  in  $\mathbf{R}^n$  and  $k$  in  $\mathbf{R}$  :

(i)  $(u + v) \cdot w = u \cdot w + v \cdot w$ , (ii)  $(ku) \cdot v = k(u \cdot v)$ , (iii)  $u \cdot v = v \cdot u$ , (iv)  $u \cdot u \geq 0$ , and  $u \cdot u = 0$  iff  $u = 0$ .

Let  $u = (u_1, u_2, \dots, u_n)$ ,  $v = (v_1, v_2, \dots, v_n)$ ,  $w = (w_1, w_2, \dots, w_n)$ .

(i) Because  $u + v = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$ ,

$$\begin{aligned} (u + v) \cdot w &= (u_1 + v_1)w_1 + (u_2 + v_2)w_2 + \dots + (u_n + v_n)w_n \\ &= u_1w_1 + v_1w_1 + u_2w_2 + \dots + u_nw_n + v_nw_n \\ &= (u_1w_1 + u_2w_2 + \dots + u_nw_n) + (v_1w_1 + v_2w_2 + \dots + v_nw_n) \\ &= u \cdot w + v \cdot w \end{aligned}$$

(ii) Because  $ku = (ku_1, ku_2, \dots, ku_n)$ ,

$$(ku) \cdot v = ku_1v_1 + ku_2v_2 + \dots + ku_nv_n = k(u_1v_1 + u_2v_2 + \dots + u_nv_n) = k(u \cdot v)$$

$$\begin{aligned} (iii) \quad u \cdot v &= u_1v_1 + u_2v_2 + \dots + u_nv_n = v_1u_1 + v_2u_2 + \dots \\ &\quad + v_nu_n = v \cdot u \end{aligned}$$

(iv) Because  $u_i^2$  is nonnegative for each  $i$ , and because the sum of nonnegative real numbers is nonnegative,

$$u \cdot u = u_1^2 + u_2^2 + \cdots + u_n^2 \geq 0$$

Furthermore,  $u \cdot u = 0$  iff  $u_i = 0$  for each  $i$ , that is, iff  $u = 0$ .

**1.14.** Prove [Theorem 1.3](#) (Schwarz):  $|u \cdot v| \leq \|u\| \|v\|$ .

For any real number  $t$ , and using [Theorem 1.2](#), we have

$$0 \leq (tu + v) \cdot (tu + v) = t^2(u \cdot u) + 2t(u \cdot v) + (v \cdot v) = \|u\|^2 t^2 + 2(u \cdot v)t + \|v\|^2$$

Let  $a = \|u\|^2$ ,  $b = 2(u \cdot v)$ ,  $c = \|v\|^2$ . Then, for every value of  $t$ ,  $at^2 + bt + c \geq 0$ . This means that the quadratic polynomial cannot have two real roots. This implies that the discriminant  $D = b^2 - 4ac \leq 0$  or, equivalently,  $b^2 \leq 4ac$ . Thus,

$$4(u \cdot v)^2 \leq 4\|u\|^2 \|v\|^2$$

Dividing by 4 gives us our result.

**1.15.** Prove [Theorem 1.4](#) (Minkowski):  $\|u + v\| \leq \|u\| + \|v\|$ .

By the Schwarz inequality and other properties of the dot product,

$$\|u + v\|^2 = (u + v) \cdot (u + v) = (u \cdot u) + 2(u \cdot v) + (v \cdot v) \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$$

Taking the square root of both sides yields the desired inequality.

## **Points, Lines, Hyperplanes in $\mathbb{R}^n$**

---

Here we distinguish between an  $n$ -tuple  $P(a_1, a_2, \dots, a_n)$  viewed as a point in  $\mathbf{R}^n$  and an  $n$ -tuple  $u = [c_1, c_2, \dots, c_n]$  viewed as a vector (arrow) from the origin  $O$  to the point  $C(c_1, c_2, \dots, c_n)$ .

**1.16.** Find the vector  $u$  identified with the directed line segment  $\overrightarrow{PQ}$  for the points:

- (a)  $P(1, -2, 4)$  and  $Q(6, 1, -5)$  in  $\mathbf{R}^3$ ,      (b)  $P(2, 3, -6, 5)$  and  $Q(7, 1, 4, -8)$  in  $\mathbf{R}^4$ .

$$(a) \ u = \overrightarrow{PQ} = Q - P = [6 - 1, 1 - (-2), -5 - 4] = [5, 3, -9]$$

(b)

$$u = \overrightarrow{PQ} = Q - P = [7 - 2, 1 - 3, 4 + 6, -8 - 5] = [5, -2, 10, -13]$$

**1.17.** Find an equation of the hyperplane  $H$  in  $\mathbf{R}^4$  that passes through  $P(3, -4, 1, -2)$  and is normal to  $u = [2, 5, -6, -3]$ .

The coefficients of the unknowns of an equation of  $H$  are the components of the normal vector  $u$ . Thus, an equation of  $H$  is of the form  $2x_1 + 5x_2 - 6x_3 - 3x_4 = k$ . Substitute  $P$  into this equation to obtain  $k = -26$ . Thus, an equation of  $H$  is  $2x_1 + 5x_2 - 6x_3 - 3x_4 = -26$ .

**1.18.** Find an equation of the plane  $H$  in  $\mathbf{R}^3$  that contains  $P(1, -3, -4)$  and is parallel to the plane  $H$  determined by the equation  $3x - 6y + 5z = 2$ .

The planes  $H$  and  $H$  are parallel if and only if their normal directions are parallel or antiparallel (opposite direction). Hence, an equation of  $H$  is of the form  $3x - 6y + 5z = k$ . Substitute  $P$  into this equation to obtain  $k = 1$ . Then an equation of  $H$  is  $3x - 6y + 5z = 1$ .

**1.19.** Find a parametric representation of the line  $L$  in  $\mathbf{R}^4$  passing through  $P(4, -2, 3, 1)$  in the direction of  $u = [2,$

5, -7, 8.

Here  $L$  consists of the points  $X(x_i)$  that satisfy

$$X = P + tu \quad \text{or} \quad x_i = a_i t + b_i \quad \text{or} \quad L(t) = (a_i t + b_i)$$

where the parameter  $t$  takes on all real values. Thus we obtain

$$x_1 = 4 + 2t, \quad x_2 = -2 + 2t, \quad x_3 = 3 - 7t, \quad x_4 = 1 + 8t \quad \text{or} \quad L(t) = (4 + 2t, -2 + 2t, 3 - 7t, 1 + 8t)$$

**1.20.** Let  $C$  be the curve  $F(t) = (t^2, 3t - 2, t^3, t^2 + 5)$  in  $\mathbf{R}^4$ , where  $0 \leq t \leq 4$ .

- (a) Find the point  $P$  on  $C$  corresponding to  $t = 2$ .
- (b) Find the initial point  $Q$  and terminal point  $Q$  of  $C$ .
- (c) Find the unit tangent vector  $\mathbf{T}$  to the curve  $C$  when  $t = 2$ .

(a) Substitute  $t = 2$  into  $F(t)$  to get  $P = f(2) = (4, 4, 8, 9)$ .

(b) The parameter  $t$  ranges from  $t = 0$  to  $t = 4$ . Hence,  $Q = f(0) = (0, 2, 0, 5)$  and  $Q = F(4) = (16, 10, 64, 21)$ .

(c) Take the derivative of  $F(t)$ —that is, of each component of  $F(t)$ —to obtain a vector  $V$  that is tangent to the curve:

$$V(t) = \frac{dF(t)}{dt} = [2t, 3, 3t^2, 2t]$$

Now find  $V$  when  $t = 2$ ; that is, substitute  $t = 2$  in the equation for  $V(t)$  to obtain  $V = V(2) = [4, 3, 12, 4]$ . Then normalize  $V$  to obtain the desired unit tangent vector  $\mathbf{T}$ . We have

$$\|V\| = \sqrt{16 + 9 + 144 + 16} = \sqrt{185} \quad \text{and} \quad \mathbf{T} = \left[ \frac{4}{\sqrt{185}}, \frac{3}{\sqrt{185}}, \frac{12}{\sqrt{185}}, \frac{4}{\sqrt{185}} \right]$$

## Spatial Vectors (Vectors in $\mathbf{R}^3$ ), ijk Notation, Cross Product

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**1.21.** Let  $u = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$ ,  $v = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ ,  $w = \mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$ .  
Find:

- (a)  $u + v$ ,      (b)  $2u - 3v + 4w$ ,      (c)  $u \cdot v$  and  $u \cdot w$ ,      (d)  $\|u\|$  and  $\|v\|$ .

Treat the coefficients of  $\mathbf{i}$ ,  $\mathbf{j}$ ,  $\mathbf{k}$  just like the components of a vector in  $\mathbf{R}^3$ .

(a) Add corresponding coefficients to get  $u + v = 5\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ .

(b) First perform the scalar multiplication and then the vector addition:

$$\begin{aligned} 2u - 3v + 4w &= (4\mathbf{i} - 6\mathbf{j} + 8\mathbf{k}) + (-9\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}) + (4\mathbf{i} + 20\mathbf{j} + 12\mathbf{k}) \\ &= -\mathbf{i} + 11\mathbf{j} + 26\mathbf{k} \end{aligned}$$

(c) Multiply corresponding coefficients and then add:

$$u \cdot v = 6 - 3 - 8 = -5 \quad \text{and} \quad u \cdot w = 2 - 15 + 12 = -1$$

(d) The norm is the square root of the sum of the squares of the coefficients:

$$\|u\| = \sqrt{4 + 9 + 16} = \sqrt{29} \quad \text{and} \quad \|v\| = \sqrt{9 + 1 + 4} = \sqrt{14}$$

**1.22.** Find the (parametric) equation of the line  $L$ :

- (a) through the points  $P(1, 3, 2)$  and  $Q(2, 5, 6)$ ;  
(b) containing the point  $P(1, -2, 4)$  and perpendicular to the plane  $H$  given by the equation  $3x + 5y + 7z =$

15.

- (a) First find  $v = \overrightarrow{PQ} = Q - P = [1, 2, -8] = \mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$ . Then
- (b) Because  $L$  is perpendicular to  $H$ , the line  $L$  is in the same direction as the normal vector  $\mathbf{N} = 3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$  to  $H$ . Thus,

$$L(t) = (t + 1, 2t + 3, -8t + 2) = (t + 1)\mathbf{i} + (2t + 3)\mathbf{j} + (-8t + 2)\mathbf{k}$$

**1.23.** Let  $S$  be the surface  $xy^2 + 2yz = 16$  in  $\mathbf{R}^3$ .

- (a) Find the normal vector  $\mathbf{N}(x, y, z)$  to the surface  $S$ .
- (b) Find the tangent plane  $H$  to  $S$  at the point  $P(1, 2, 3)$ .
- (a) The formula for the normal vector to a surface  $F(x, y, z) = 0$  is

$$\mathbf{N}(x, y, z) = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}$$

where  $F_x, F_y, F_z$  are the partial derivatives. Using  $F(x, y, z) = xy^2 + 2yz - 16$ , we obtain

$$F_x = y^2, \quad F_y = 2xy + 2z, \quad F_z = 2y$$

Thus,  $\mathbf{N}(x, y, z) = y^2\mathbf{i} + (2xy + 2z)\mathbf{j} + 2y\mathbf{k}$ .

- (b) The normal to the surface  $S$  at the point  $P$  is

$$\mathbf{N}(P) = \mathbf{N}(1, 2, 3) = 4\mathbf{i} + 10\mathbf{j} + 4\mathbf{k}$$

Hence,  $\mathbf{N} = 2\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$  is also normal to  $S$  at  $P$ .

Thus an equation of  $H$  has the form  $2x + 5y + 2z = c$ . Substitute  $P$  in this equation to obtain  $c = 18$ .

Thus the tangent plane  $H$  to  $S$  at  $P$  is  $2x + 5y + 2z = 18$ .

**1.24.** Evaluate the following determinants and negative of determinants of order two:

$$(a) \quad (i) \begin{vmatrix} 3 & 4 \\ 5 & 9 \end{vmatrix}, \quad (ii) \begin{vmatrix} 2 & -1 \\ 4 & 3 \end{vmatrix}, \quad (iii) \begin{vmatrix} 4 & -5 \\ 3 & -2 \end{vmatrix}$$

$$(b) \quad (i) -\begin{vmatrix} 3 & 6 \\ 4 & 2 \end{vmatrix}, \quad (ii) -\begin{vmatrix} 7 & -5 \\ 3 & 2 \end{vmatrix}, \quad (iii) -\begin{vmatrix} 4 & -1 \\ 8 & -3 \end{vmatrix}$$

$$\text{Use } \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \text{ and } -\begin{vmatrix} a & b \\ c & d \end{vmatrix} = bc - ad. \text{ Thus,}$$

$$(a) \quad (i) 27 - 20 = 7, \quad (ii) 6 + 4 = 10, \quad (iii) 8 + 15 = 7.$$

$$(b) \quad (i) 24 - 6 = 18, \quad (ii) 15 - 14 = 1, \quad (iii) 8 + 12 = 20.$$

**1.25.** Let  $u = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ ,  $v = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ ,  $w = \mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$ .

Find: (a)  $u \times v$ , (b)  $u \times w$

$$(a) \quad \text{Use } \begin{bmatrix} 2 & -3 & 4 \\ 3 & 1 & -2 \end{bmatrix} \text{ to get } u \times v = (6 - 4)\mathbf{i} + (12 + 4)\mathbf{j} + (2 + 9)\mathbf{k} = 2\mathbf{i} + 16\mathbf{j} + 11\mathbf{k}.$$

$$(b) \quad \text{Use } \begin{bmatrix} 2 & -3 & 4 \\ 1 & 5 & 3 \end{bmatrix} \text{ to get } u \times w = (-9 - 20)\mathbf{i} + (4 - 6)\mathbf{j} + (10 + 3)\mathbf{k} = -29\mathbf{i} - 2\mathbf{j} + 13\mathbf{k}.$$

**1.26.** Find  $u \times v$ , where: (a)  $u = (1, 2, 3)$ ,  $v = (4, 5, 6)$ ; (b)  $u = (-4, 7, 3)$ ,  $v = (6, -5, 2)$ .

$$(a) \quad \text{Use } \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \text{ to get } u \times v = [12 - 15, 12 - 6, 5 - 8] = [-3, 6, -3].$$

$$(b) \quad \text{Use } \begin{bmatrix} -4 & 7 & 3 \\ 6 & -5 & 2 \end{bmatrix} \text{ to get } u \times v = [14 + 15, 18 + 8, 20 - 42] = [29, 26, -22].$$

**1.27.** Find a unit vector  $u$  orthogonal to  $v = [1, 3, 4]$  and  $w = [2, -6, -5]$ .

First find  $v \times w$ , which is orthogonal to  $v$  and  $w$ .

The array  $\begin{bmatrix} 1 & 3 & 4 \\ 2 & -6 & -5 \end{bmatrix}$  gives  $v \times w = [-15 + 24, 8 + 5, -6 - 61] = [9, 13, -12]$ .  
 Normalize  $v \times w$  to get  $u = [9/\sqrt{394}, 13/\sqrt{394}, -12/\sqrt{394}]$ .

### 1.28. Let

$u = (a_1, a_2, a_3)$  and  $v = (b_1, b_2, b_3)$  so  $u \times v = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1)$

. Prove:

(a)  $u \times v$  is orthogonal to  $u$  and  $v$  [Theorem 1.5 (a)].

(b)  $\|u \times v\|^2 = (u \cdot u)(v \cdot v)(u \cdot v)^2$  (Lagrange's identity).

(a) We have

$$\begin{aligned} u \cdot (u \times v) &= a_1(a_2b_3 - a_3b_2) + a_2(a_3b_1 - a_1b_3) + a_3(a_1b_2 - a_2b_1) \\ &= a_1a_2b_3 - a_1a_3b_2 + a_2a_3b_1 - a_1a_2b_3 + a_1a_3b_2 - a_2a_3b_1 = 0 \end{aligned}$$

Thus,  $u \times v$  is orthogonal to  $u$ . Similarly,  $u \times v$  is orthogonal to  $v$ .

(b) We have

$$\|u \times v\|^2 = (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \quad (1)$$

$$(u \cdot u)(v \cdot v) - (u \cdot v)^2 = (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \quad (2)$$

Expansion of the right-hand sides of (1) and (2) establishes the identity.

## Complex Numbers, Vectors in $\mathbb{C}^1$

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**1.29.** Suppose  $z = 5 + 3i$  and  $w = 2 - 4i$ . Find: (a)  $z + w$ ,  
 (b)  $z - w$ , (c)  $zw$ .

Use the ordinary rules of algebra together with  $i^2 = -1$  to obtain a result in the standard form  $a + bi$ .

$$(a) \quad z + w = (5 + 3i) + (2 - 4i) = 7 - i$$

$$(b) z - w = (5 + 3i) - (2 - 4i) = 5 + 3i - 2 + 4i = 3 + 7i$$

$$(c) zw = (5 + 3i)(2 - 4i) = 10 - 14i - 12i^2 = 10 - 14i + 12 = 22 - 14i$$

**1.30.** Simplify:  $(5 + 3i)(2 - 7i)$ , (b)  $(4 - 3i)^2$ , (c)  $(1 + 2i)^3$ .

$$(a) (5 + 3i)(2 - 7i) = 10 + 6i - 35i - 21i^2 = 31 - 29i$$

$$(b) (4 - 3i)^2 = 16 - 24i + 9i^2 = 7 - 24i$$

$$(c) (1 + 2i)^3 = 1 + 6i + 12i^2 + 8i^3 = 1 + 6i - 12 - 8i = -11 - 2i$$

**1.31.** Simplify:

$$(a) i^0, i^3, i^4, (b) i^5, i^6, i^7, i^8, (c) i^{39}, i^{174}, i^{252}, i^{317}.$$

(a)

$$i^0 = 1, \quad i^3 = i^2(i) = (-1)(i) = -i, \quad i^4 = (i^2)(i^2) = (-1)(-1) = 1$$

(b)

$$i^5 = (i^4)(i) = (1)(i) = i, \quad i^6 = (i^4)(i^2) = (1)(i^2) = i^2 = -1, \quad i^7 = i^3 = -i, \quad i^8 = i^4 = 1$$

(c) Using  $i^4 = 1$  and  $i^4 = 1$  and  $i^n = i^{4q+r} = (i^4)^q i^r = 1^q i^r = i^r$ ,  
divide the exponent  $n$  by 4 to obtain the remainder  $r$ .

$$i^{39} = i^{4(9)+3} = (i^4)^9 i^3 = 1^9 i^3 = i^3 = -i, \quad i^{174} = i^2 = -1, \quad i^{252} = i^0 = 1, \quad i^{317} = i^1 = i$$

**1.32.** Find the complex conjugate of each of the following:

(a)

$$(a) 6 + 4i, 7 - 5i, 4 + i, -3 - i, \quad (b) 6, -3, 4i, -9i.$$

(a)

$$\overline{6 + 4i} = 6 - 4i, \quad \overline{7 - 5i} = 7 + 5i, \quad \overline{4 + i} = 4 - i, \quad \overline{-3 - i} = -3 + i$$

$$(b) \overline{6} = 6, \quad \overline{-3} = -3, \quad \overline{4i} = -4i, \quad \overline{-9i} = 9i$$

(Note that the conjugate of a real number is the original number, but the conjugate of a pure imaginary number is the negative of the original number.)

**1.33.** Find  $z\bar{z}$  and  $|z|$  when  $z = 3 + 4i$ .

For  $z = a + bi$ , use

$z = a + bi$ , use  $z\bar{z} = a^2 + b^2$  and  $z = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}$ .

$$z\bar{z} = 9 + 16 = 25, \quad |z| = \sqrt{25} = 5$$

**1.34.** Simplify  $\frac{2 - 7i}{5 + 3i}$ .

To simplify a fraction  $z / w$  of complex numbers, multiply both numerator and denominator by  $\bar{w}$ , the conjugate of the denominator:

$$\frac{2 - 7i}{5 + 3i} = \frac{(2 - 7i)(5 - 3i)}{(5 + 3i)(5 - 3i)} = \frac{-11 - 41i}{34} = -\frac{11}{34} - \frac{41}{34}i$$

**1.35.** Prove: For any complex numbers

$z, w \in \mathbf{C}$ , (i)  $\overline{z + w} = \bar{z} + \bar{w}$ , (ii)  $\overline{z\bar{w}} = \bar{z}\bar{\bar{w}}$ , (iii)  $\bar{\bar{z}} = z$ .

Suppose  $z = a + bi$  and  $w = c + di$  where  $a, b, c, d \in \mathbf{R}$ .

$$\begin{aligned} \text{(i)} \quad \overline{z + w} &= \overline{(a + bi) + (c + di)} = \overline{(a + c) + (b + d)i} \\ &= (a + c) - (b + d)i = a + c - bi - di \\ &= (a - bi) + (c - di) = \bar{z} + \bar{w} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \overline{z\bar{w}} &= \overline{(a + bi)(c + di)} = \overline{(ac - bd) + (ad + bc)i} \\ &= (ac - bd) - (ad + bc)i = (a - bi)(c - di) = \bar{z}\bar{\bar{w}} \end{aligned}$$

$$\text{(iii)} \quad \bar{\bar{z}} = a + bi = \overline{a - bi} = a - (-b)i = a + bi = z$$

**1.36.** Prove: For any complex numbers  $z, w \in \mathbf{C}$ ,  $|zw| = |z||w|$ .

By (ii) of [Problem 1.35](#),

$$|zw|^2 = (zw)(\overline{zw}) = (zw)(\bar{z}\bar{\bar{w}}) = (z\bar{z})(w\bar{w}) = |z|^2|w|^2$$

The square root of both sides gives us the desired result.

**1.37.** Prove: For any complex numbers

$$z, w \in \mathbf{C}, |z + w| \leq |z| + |w|.$$

Suppose  $z = a + bi$  and  $w = c + di$  where  $a, b, c, d \in \mathbf{R}$ . Consider the vectors  $u = (a, b)$  and  $v = (c, d)$  in  $\mathbf{R}^2$ . Note that

$$|z| = \sqrt{a^2 + b^2} = \|u\|, \quad |w| = \sqrt{c^2 + d^2} = \|v\|$$

and

$$|z + w| = |(a + c) + (b + d)i| = \sqrt{(a + c)^2 + (b + d)^2} = \|(a + c, b + d)\| = \|u + v\|$$

By Minkowski's inequality ([Problem 1.15](#)),  $\|u + v\| \leq \|u\| + \|v\|$ , and so

$$|z + w| = |(a + c) + (b + d)i| = \sqrt{(a + c)^2 + (b + d)^2} = \|(a + c, b + d)\| = \|u + v\|$$

**1.38.** Find the dot products  $u \cdot v$  and  $v \cdot u$  where: (a)  $u = (1 - 2i, 3 + i)$ ,  $v = (4 + 2i, 5 - 6i)$ , (b)  $u = (3 - 2i, 4i, 1 + 6i)$ ,  $v = (5 + i, 2 - 3i, 7 + 2i)$ .

Recall that conjugates of the second vector appear in the dot product

$$(z_1, \dots, z_n) \cdot (w_1, \dots, w_n) = z_1 \bar{w}_1 + \dots + z_n \bar{w}_n$$

$$\begin{aligned} \text{(a)} \quad u \cdot v &= (1 - 2i)(\overline{4 + 2i}) + (3 + i)(\overline{5 - 6i}) \\ &= (1 - 2i)(4 - 2i) + (3 + i)(5 + 6i) = -10i + 9 + 23i = 9 + 13i \\ v \cdot u &= (4 + 2i)(\overline{1 - 2i}) + (5 - 6i)(\overline{3 + i}) \\ &= (4 + 2i)(1 + 2i) + (5 - 6i)(3 - i) = 10i + 9 - 23i = 9 - 13i \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad u \cdot v &= (3 - 2i)(\overline{5 + i}) + (4i)(\overline{2 - 3i}) + (1 + 6i)(\overline{7 + 2i}) \\
 &= (3 - 2i)(5 - i) + (4i)(2 + 3i) + (1 + 6i)(7 - 2i) = 20 + 35i \\
 v \cdot u &= (5 + i)(\overline{3 - 2i}) + (2 - 3i)(\overline{4i}) + (7 + 2i)(\overline{1 + 6i}) \\
 &= (5 + i)(3 + 2i) + (2 - 3i)(-4i) + (7 + 2i)(1 - 6i) = 20 - 35i
 \end{aligned}$$

In both cases,  $u \cdot v = \overline{v \cdot u}$ . This holds true in general, as seen in [Problem 1.40](#).

**1.39.** Let  $u = (7 - 2i, 2 + 5i)$  and  $v = (1 + i, -3 - 6i)$ . Find:

$$\text{(a)} \quad u + v, \quad \text{(b)} \quad 2iu, \quad \text{(c)} \quad (3 - i)v, \quad \text{(d)} \quad u \cdot v, \quad \text{(e)} \quad \|u\| \text{ and } \|v\|.$$

$$\text{(a)} \quad u + v = (7 - 2i + 1 + i, 2 + 5i - 3 - 6i) = (8 - i, -1 - i)$$

$$\text{(b)} \quad 2iu = (14i - 4i^2, 4i + 10i^2) = (4 + 14i, -10 + 4i)$$

$$\text{(c)} \quad (3 - i)v = (3 + 3i - i - i^2, -9 - 18i + 3i + 6i^2) = (4 + 2i, -15 - 15i)$$

$$\begin{aligned}
 \text{(d)} \quad u \cdot v &= (7 - 2i)(\overline{1 + i}) + (2 + 5i)(\overline{-3 - 6i}) \\
 &= (7 - 2i)(1 - i) + (2 + 5i)(-3 + 6i) = 5 - 9i - 36 - 3i = -31 - 12i
 \end{aligned}$$

$$\text{(e)} \quad \|u\| = \sqrt{7^2 + (-2)^2 + 2^2 + 5^2} = \sqrt{82} \text{ and } \|v\| = \sqrt{1^2 + 1^2 + (-3)^2 + (-6)^2} = \sqrt{47}$$

**1.40.** Prove: For any vectors  $u, v \in \mathbf{C}^n$  and any scalar

$z \in \mathbf{C}$ , (i)  $u \cdot v = \overline{v \cdot u}$ , (ii)  $(zu) \cdot v = z(u \cdot v)$ ,

(iii)  $u \cdot (zv) = \bar{z}(u \cdot v)$ .

Suppose  $u = (z_1, z_2, \dots, z_n)$  and  $v = (w_1, w_2, \dots, w_n)$ .

(i) Using the properties of the conjugate,

$$\begin{aligned}
 \overline{v \cdot u} &= \overline{w_1 \bar{z}_1 + w_2 \bar{z}_2 + \dots + w_n \bar{z}_n} = \overline{w_1 \bar{z}_1} + \overline{w_2 \bar{z}_2} + \dots + \overline{w_n \bar{z}_n} \\
 &= \bar{w}_1 z_1 + \bar{w}_2 z_2 + \dots + \bar{w}_n z_n = z_1 \bar{w}_1 + z_2 \bar{w}_2 + \dots + z_n \bar{w}_n = u \cdot v
 \end{aligned}$$

(ii) Because  $zu = (zz_1, zz_2, \dots, zz_n)$ ,

$$(zu) \cdot v = zz_1 \bar{w}_1 + zz_2 \bar{w}_2 + \dots + zz_n \bar{w}_n = z(z_1 \bar{w}_1 + z_2 \bar{w}_2 + \dots + z_n \bar{w}_n) = z(u \cdot v)$$

(Compare with [Theorem 1.2](#) on vectors in  $\mathbf{R}^n$ .)

(iii) Using (i) and (ii),

$$u \cdot (zv) = \overline{(zv) \cdot u} = z(\overline{v \cdot u}) = \bar{z}(\overline{v \cdot u}) = \bar{z}(u \cdot v)$$

## SUPPLEMENTARY PROBLEMS

### Vectors in $\mathbf{R}^n$

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**1.41.** Let  $u = (1, -2, 4)$ ,  $v = (3, 5, 1)$ ,  $w = (2, 1, -3)$ . Find:

(a)  $3u - 2v$ ; (b)  $5u + 3v - 4w$ ; (c)  $u \cdot v$ ,  $u \cdot w$ ,  $v \cdot w$ ; (d)  $\|u\|$ ,  $\|v\|$ ,  $\|w\|$ ; (e)  $\cos \theta$ , where  $\theta$  is the angle between  $u$  and  $v$ ; (f)  $d(u, v)$ ; (g)  $\text{proj}(u, v)$ .

**1.42.** Repeat [Problem 1.41](#) for vectors

$$u = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}, v = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, w = \begin{bmatrix} 3 \\ -2 \\ 6 \end{bmatrix}.$$

**1.43.** Let  $u = (2, -5, 4, 6, -3)$  and  $v = (5, -2, 1, -7, -4)$ .

Find:

(a)  $4u - 3v$ ; (b)  $5u + 2v$ ; (c)  $u \cdot v$ ; (d)  $\|u\|$  and  $\|v\|$ ; (e)  $\text{proj}(u, v)$ ; (f)  $d(u, v)$ .

**1.44.** Normalize each vector:

(a)  $u = (5, -7)$ ; (b)  $v = (1, 2, 2, 4)$ ; (c)  $w = \left(\frac{1}{2}, -\frac{1}{3}, \frac{3}{4}\right)$ .

**1.45.** Let  $u = (1, 2, -2)$ ,  $v = (3, -12, 4)$ , and  $k = -3$ .

(a) Find  $\|u\|$ ,  $\|v\|$ ,  $\|u + v\|$ ,  $\|ku\|$ .

(b) Verify that  $\|ku\| = |k| \|u\|$  and  $\|u + v\| \leq \|u\| + \|v\|$ .

**1.46.** Find  $x$  and  $y$  where:

(a)  $(x, y + 1) = (y, 2, 6)$ ; (b)  $x(2, y) = y(1, 2)$ .

**1.47.** Find  $x, y, z$  where  $(x, y + 1, y + z) = (2x + y, 4, 3z)$ .

**1.48.** Write  $v = (2, 5)$  as a linear combination of  $u_1$  and  $u_2$ , where:

(a)  $u_1 = (1, 2)$  and  $u_2 = (3, 5)$ ;

(b)  $u_1 = (3, -4)$  and  $u_2 = (2, -3)$ .

**1.49.** Write  $v = \begin{bmatrix} 9 \\ -3 \\ 16 \end{bmatrix}$  as a linear combination of

$$u_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, u_2 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}, u_3 = \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}.$$

**1.50.** Find  $k$  so that  $u$  and  $v$  are orthogonal, where:

(a)  $u = (3, k, -2)$ ,  $v = (6, -4, -3)$ ;

(b)  $u = (5, k, -4, 2)$ ,  $v = (1, 3, 2, 2k)$ ;

(c)  $u = (1, 7, k + 2, 2)$ ,  $v = (3, k, -3, k)$ .

## **Located Vectors, Hyperplanes, Lines in $\mathbf{R}^n$**

**1.51.** Find the vector  $v$  identified with the directed line segment  $\vec{PQ}$  for the points:

(a)  $P(2, 3, -7)$  and  $Q(1, 6, 5)$  in  $\mathbf{R}^3$ ;

(b)  $P(1, -8, -4, 6)$  and  $Q(3, 5, 2, 4)$  in  $\mathbf{R}^4$ .

**1.52.** Find an equation of the hyperplane  $H$  in  $\mathbf{R}^4$  that:

(a) contains  $P(1, 2, -3, 2)$  and is normal to  $u = [2, 3, -5, 6]$ ;

(b) contains  $P(3, -1, 2, 5)$  and is parallel to  $2x_1 - 3x_2 + 5x_3 - 7x_4 = 4$ .

**1.53.** Find a parametric representation of the line in  $\mathbf{R}^4$  that:

(a) passes through the points  $P(1, 2, 1, 2)$  and  $Q(3, 5, 7, -9)$ ;

(b) passes through  $P(1, 1, 3, 3)$  and is perpendicular to the hyperplane  $2x_1 + 4x_2 + 6x_3 - 8x_4 = 5$ .

### Spatial Vectors (Vectors in $\mathbf{R}^3$ ), $\mathbf{i}, \mathbf{j}, \mathbf{k}$ Notation

**1.54.** Given  $u = 3\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$ ,  $v = 2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ ,  $w = 4\mathbf{i} + 7\mathbf{j} + 2\mathbf{k}$ . Find:

(a)  $2u - 3v$ ; (b)  $3u + 4v - 2w$ ; (c)  $u \cdot v$ ,  $u \cdot w$ ,  $v \cdot w$ ; (d)  $\|u\|$ ,  $\|v\|$ ,  $\|w\|$ .

**1.55.** Find the equation of the plane  $H$ :

(a) with normal  $\mathbf{N} = 3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$  and containing the point  $P(1, 2, 3)$ ;

(b) parallel to  $4x + 3y - 2z = 11$  and containing the point  $Q(2, -1, 3)$ .

**1.56.** Find the (parametric) equation of the line  $L$ :

(a) through the point  $P(2, 5, -3)$  and in the direction of  $v = 4\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$ ;

(b) perpendicular to the plane  $2x - 3y + 7z = 4$  and containing  $P(1, -5, 7)$ .

**1.57.** Consider the following curve  $C$  in  $\mathbf{R}^3$  where  $0 \leq t \leq 5$ :

$$F(t) = t^3\mathbf{i} - t^2\mathbf{j} + (2t - 3)\mathbf{k}$$

(a) Find the point  $P$  on  $C$  corresponding to  $t = 2$ .

- (b) Find the initial point  $Q$  and the terminal point  $Q$ .  
 (c) Find the unit tangent vector  $\mathbf{T}$  to the curve  $C$  when  $t = 2$ .

**1.58.** Consider a moving body  $B$  whose position at time  $t$  is given by  $R(t) = t^2\mathbf{i} + t^3\mathbf{j} + 2t\mathbf{k}$ . [Then  $V(t) = dR(t)/dt$  and  $A(t) = dV(t)/dt$  denote, respectively, the velocity and acceleration of  $B$ .] When  $t = 1$ , find for the body  $B$ :  
 (a) position; (b) velocity  $\mathbf{v}$ ; (c) speed  $s$ ; (d) acceleration  $\mathbf{a}$ .

**1.59.** Find a normal vector  $\mathbf{N}$  and the tangent plane  $H$  to each surface at the given point:

- (a) surface  $x^2y + 3yz = 20$  and point  $P(1, 3, 2)$ ;  
 (b) surface  $x^2 + 3y^2 - 5z^2 = 160$  and point  $P(3, 2, 1)$ .

## Cross Product

---

**1.60.** Evaluate the following determinants and negative of determinants of order two:

$$(a) \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix}, \quad \begin{vmatrix} 3 & -6 \\ 1 & -4 \end{vmatrix}, \quad \begin{vmatrix} -4 & -2 \\ 7 & -3 \end{vmatrix}$$

$$(b) -\begin{vmatrix} 6 & 4 \\ 7 & 5 \end{vmatrix}, \quad -\begin{vmatrix} 1 & -3 \\ 2 & 4 \end{vmatrix}, \quad -\begin{vmatrix} 8 & -3 \\ -6 & -2 \end{vmatrix}$$

**1.61.** Given  $u = 3\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$ ,  $v = 2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ ,  $w = 4\mathbf{i} + 7\mathbf{j} + 2\mathbf{k}$ , find:

$$(a) u \times v, \quad (b) u \times w, \quad (c) v \times w.$$

**1.62.** Given  $u = [2, 1, 3]$ ,  $v = [4, -1, 2]$ ,  $w = [1, 1, 5]$ , find:

$$(a) u \times v, \quad (b) u \times w, \quad (c) v \times w.$$

**1.63.** Find the volume  $V$  of the parallelopiped formed by the vectors  $u, v, w$  appearing in:

(a) [Problem 1.61](#) (b) [Problem 1.62](#).

**1.64.** Find a unit vector  $u$  orthogonal to:

(a)  $v = [1, 2, 3]$  and  $w = [1, -1, 2]$ ;

(b)  $v = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$  and  $w = 4\mathbf{i} - 2\mathbf{j} - \mathbf{k}$ .

**1.65.** Prove the following properties of the cross product:

(a)  $u \times v = -(v \times u)$

(b)  $u \times u = 0$  for any vector  $u$

(c)  $(ku) \times v = k(u \times v) = u \times (kv)$

(d)  $u \times (v + w) = (u \times v) + (u \times w)$

(e)  $(v + w) \times u = (v \times u) + (w \times u)$

(f)  $(u \times v) \cdot w = (u \cdot w) v - (v \cdot w) u$

## Complex Numbers

---

**1.66.** Simplify:

(a)  $(4 - 7i)(9 + 2i)$ ;

(b)  $(3 - 5i)^2$ ;

(c)  $\frac{1}{4 - 7i}$ ;

(d)  $\frac{9 + 2i}{3 - 5i}$ ;

(e)  $(1 - i)^3$ .

**1.67.** Simplify:

(a)  $\frac{1}{2i}$ ; (b)  $\frac{2+3i}{7-3i}$ ; (c)  $i^{15}, i^{25}, i^{34}$ ; (d)  $\left(\frac{1}{3-i}\right)^2$ .

**1.68.** Let  $z = 2 - 5i$  and  $w = 7 + 3i$ . Find:

(a)  $v + w$ ; (b)  $zw$ ; (c)  $z/w$ ; (d)  $\bar{z}, \bar{w}$ ; (e)  $|z|, |w|$ .

**1.69.** Show that for complex numbers  $z$  and  $w$ :

(a)  $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$ , (b)  $\operatorname{Im} z = \frac{1}{2}(z - \bar{z})$ , (c)  $zw = 0$  implies  $z = 0$  or  $w = 0$ .

## Vectors in $\mathbf{C}^n$

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**1.70.** Let  $u = (1 + 7i, 2 - 6i)$  and  $v = (5 - 2i, 3 - 4i)$ . Find:

(a)  $u + v$  (b)  $(3 + i)u$  (c)  $2iu + (4 + 7i)v$  (d)  $u \cdot v$  (e)  $\|u\|$  and  $\|v\|$ .

**1.71.** Prove: For any vectors  $u, v, w$  in  $\mathbf{C}^n$ :

(a)  $(u + v) \cdot w = u \cdot w + v \cdot w$ , (b)  $w \cdot (u + v) = w \cdot u + w \cdot v$ .

**1.72.** Prove that the norm in  $\mathbf{C}^n$  satisfies the following laws:

[N<sub>1</sub>] For any vector  $u$ ,  $\|u\| \geq 0$ ; and  $\|u\| = 0$  if and only if  $u = 0$ .

[N<sub>2</sub>] For any vector  $u$  and complex number  $z$ ,  $\|zu\| = |z|\|u\|$ .

[N<sub>3</sub>] For any vectors  $u$  and  $v$ ,  $\|u + v\| \leq \|u\| + \|v\|$ .

## ANSWERS TO SUPPLEMENTARY PROBLEMS

- 1.41.** (a)  $(-3, -16, 10)$ ; (b)  $(6, 1, 35)$ ; (c)  $(-3, -12, 8)$ ; (d)  $\sqrt{21}, \sqrt{35}, \sqrt{14}$ ; (e)  $-3/\sqrt{21}\sqrt{35}$ ; (f)  $\sqrt{62}$ ; (g)  $-\frac{3}{35}(3, 5, 1) = (-\frac{9}{35}, -\frac{15}{35}, -\frac{3}{35})$
- 1.42.** (Column vectors) (a)  $(-1, 7, -22)$ ; (b)  $(-1, 26, -29)$ ; (c)  $(-15, -27, 34)$ ; (d)  $\sqrt{26}, \sqrt{30}$ ; (e)  $-15/(\sqrt{26}\sqrt{30})$ ; (f)  $\sqrt{86}$ ; (g)  $-\frac{15}{30}v = (-1, -\frac{1}{2}, -\frac{5}{2})$
- 1.43.** (a)  $(-7, -14, 13, 45, 0)$ ; (b)  $(20, -29, 22, 16, -23)$ ; (c)  $-6$ ; (d)  $\sqrt{90}, \sqrt{95}$ ; (e)  $-\frac{6}{95}v$ ; (f)  $\sqrt{197}$
- 1.44.** (a)  $(5/\sqrt{74}, -7/\sqrt{74})$ ; (b)  $(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}, \frac{4}{3})$ ; (c)  $(6/\sqrt{133}, -4/\sqrt{133}, 9/\sqrt{133})$
- 1.45.** (a)  $3, 13, \sqrt{120}, 9$
- 1.46.** (a)  $x = 3, y = 5$ ; (b)  $x = 0, y = 0$ , and  $x = 2, y = -4$
- 1.47.**  $x = -3, y = 3, z = \frac{3}{2}$
- 1.48.** (a)  $v = 5u_1 - u_2$ ; (b)  $v = 16u_1 - 23u_2$
- 1.49.**  $v = 3u_1 - u_2 + 2u_3$
- 1.50.** (a)  $6$ ; (b)  $3$ ; (c)  $\frac{3}{2}$
- 1.51.** (a)  $v = [-1, -9, 2]$ ; (b)  $[2, 3, 6, -10]$
- 1.52.** (a)  $2x_1 + 3x_2 - 5x_3 + 6x_4 = 35$ ; (b)  $2x_1 - 3x_2 + 5x_3 - 7x_4 = -16$
- 1.53.**  
(a)  $[2t + 1, -7t + 2, 6t + 1, -11t + 2]$ ; (b)  $[2t + 1, 4t + 1, 6t + 3, -8t + 3]$
- 1.54.**  
(a)  $-23\mathbf{j} + 13\mathbf{k}$ ; (b)  $9\mathbf{i} - 6\mathbf{j} - 10\mathbf{k}$ ; (c)  $-20, -12, 37$ ; (d)  $\sqrt{29}, \sqrt{38}, \sqrt{69}$
- 1.55.** (a)  $3x - 4y + 5z = 20$ ; (b)  $4x + 3y - 2z = 1$

- 1.56.** (a)  $[4t + 2, -5t + 5, 7t - 3]$ ; (b)  $[2t + 1, 3t - 5, 7t + 7]$
- 1.57.** (a)  $P = F(2) = 8\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ ; (b)  $Q = F(0) = -3\mathbf{k}$ ,  $Q' = F(5) = 125\mathbf{i} - 25\mathbf{j} + 7\mathbf{k}$ ; (c)  $\mathbf{T} = (6\mathbf{i} - 2\mathbf{j} + \mathbf{k})/\sqrt{41}$
- 1.58.** (a)  $\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ ; (b)  $2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ ; (c)  $\sqrt{17}$ ; (d)  $2\mathbf{i} + 6\mathbf{j}$
- 1.59.** (a)  $\mathbf{N} = 6\mathbf{i} + 7\mathbf{j} + 9\mathbf{k}$ ,  $6x + 7y + 9z = 45$ ; (b)  $\mathbf{N} = 6\mathbf{i} - 12\mathbf{j} - 10\mathbf{k}$ ,  $3x - 6y - 5z = 16$
- 1.60.** (a) 3, 6, 26; (b) -2, -10, 34
- 1.61.** (a)  $2\mathbf{i} + 13\mathbf{j} + 23\mathbf{k}$ ; (b)  $-22\mathbf{i} + 2\mathbf{j} + 37\mathbf{k}$ ; (c)  $31\mathbf{i} - 16\mathbf{j} - 6\mathbf{k}$
- 1.62.** (a)  $[5, 8, -6]$ ; (b)  $[2, -7, 1]$ ; (c)  $[-7, -18, 5]$
- 1.63.** (a) 145; (b) 17
- 1.64.** (a)  $(7, 1, -3)/\sqrt{59}$ ; (b)  $(5\mathbf{i} + 11\mathbf{j} - 2\mathbf{k})/\sqrt{150}$
- 1.66.** (a)  $50 - 55i$ ; (b)  $-16 - 30i$ ; (c)  $\frac{1}{65}(4 + 7i)$ ; (d)  $\frac{1}{2}(1 + 3i)$ ; (e)  $-2 - 2i$
- 1.67.** (a)  $-\frac{1}{2}i$ ; (b)  $\frac{1}{58}(5 + 27i)$ ; (c)  $-i, -1$ ; (d)  $\frac{1}{50}(4 + 3i)$
- 1.68.** (a)  $9 - 2i$ ; (b)  $29 - 29i$ ; (c)  $\frac{1}{58}(-1 - 41i)$ ; (d)  $2 + 5i, 7 - 3i$ ; (e)  $\sqrt{29}, \sqrt{58}$
- 1.69.** (c) *Hint:* If  $zw = 0$ , then  $|zw| = |z||w| = |0| = 0$
- 1.70.** (a)  $(6 + 5i, 5 - 10i)$ ; (b)  $(-4 + 22i, 12 - 16i)$ ; (c)  $(20 + 29i, 52 + 9i)$ ; (d)  $21 + 27i$ ; (e)  $\sqrt{90}, \sqrt{54}$

# CHAPTER 2

## *Algebra of Matrices*

### 2.1 Introduction

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This chapter investigates matrices and algebraic operations defined on them. These matrices may be viewed as rectangular arrays of elements where each entry depends on two subscripts (as compared with vectors, where each entry depended on only one subscript). Systems of linear equations and their solutions ([Chapter 3](#)) may be efficiently investigated using the language of matrices. Furthermore, certain abstract objects introduced in later chapters, such as “change of basis,” “linear transformations,” and “quadratic forms,” can be represented by these matrices (rectangular arrays). On the other hand, the abstract treatment of linear algebra presented later on will give us new insight into the structure of these matrices.

The entries in our matrices will come from some arbitrary, but fixed, field  $K$ . The elements of  $K$  are called *numbers* or *scalars*. Nothing essential is lost if the reader assumes that  $K$  is the real field  $\mathbf{R}$ .

## 2.2 Matrices

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A *matrix*  $A$  over a field  $K$  or, simply, a *matrix*  $A$  (when  $K$  is implicit) is a rectangular array of scalars usually presented in the following form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

The *rows* of such a matrix  $A$  are the  $m$  horizontal lists of scalars:

$$(a_{11}, a_{12}, \dots, a_{1n}), \quad (a_{21}, a_{22}, \dots, a_{2n}), \quad \dots, \quad (a_{m1}, a_{m2}, \dots, a_{mn})$$

and the *columns* of  $A$  are the  $n$  vertical lists of scalars:

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{m1} \end{bmatrix}, \quad \begin{bmatrix} a_{12} \\ a_{22} \\ \dots \\ a_{m2} \end{bmatrix}, \quad \dots, \quad \begin{bmatrix} a_{1n} \\ a_{2n} \\ \dots \\ a_{mn} \end{bmatrix}$$

Note that the element  $a_{ij}$ , called the *ij-entry* or *ij-element*, appears in row  $i$  and column  $j$ . We frequently denote such a matrix by simply writing  $A = [a_{ij}]$ .

A matrix with  $m$  rows and  $n$  columns is called an *m by n* matrix, written  $m \ n$ . The pair of numbers  $m$  and  $n$  is called the *size* of the matrix. Two matrices  $A$  and  $B$  are *equal*, written  $A = B$ , if they have the same size and if corresponding elements are equal. Thus, the equality of two  $m \ n$  matrices is equivalent to a system of  $mn$  equalities, one for each corresponding pair of elements.

A matrix with only one row is called a *row matrix* or *row vector*, and a matrix with only one column is called a *column*

*matrix* or *column vector*. A matrix whose entries are all zero is called a *zero matrix* and will usually be denoted by  $0$ .

Matrices whose entries are all real numbers are called *real matrices* and are said to be *matrices over  $\mathbf{R}$* . Analogously, matrices whose entries are all complex numbers are called *complex matrices* and are said to be *matrices over  $\mathbf{C}$* . This text will be mainly concerned with such real and complex matrices.

### EXAMPLE 2.1

(a) The rectangular array  $A = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 3 & -2 \end{bmatrix}$  is a  $2 \times 3$  matrix.

Its rows are  $(1, -4, 5)$  and  $(0, 3, -2)$ , and its columns are

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -4 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

(b) The  $2 \times 4$  zero matrix is the matrix  $0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ .

(c) Find  $x, y, z, t$  such that

$$\begin{bmatrix} x+y & 2z+t \\ x-y & z-t \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 1 & 5 \end{bmatrix}$$

By definition of equality of matrices, the four corresponding entries must be equal. Thus,

$$x+y=3, \quad x-y=1, \quad 2z+t=7, \quad z-t=5$$

Solving the above system of equations yields  $x=2, y=1, z=4, t=1$ .

## 2.3 Matrix Addition and Scalar Multiplication

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Let  $A = [a_{ij}]$  and  $B = [b_{ij}]$  be two matrices with the same size, say  $m \times n$  matrices. The *sum* of  $A$  and  $B$ , written  $A + B$ , is the matrix obtained by adding corresponding elements from  $A$  and  $B$ . That is,

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

The *product* of the matrix  $A$  by a scalar  $k$ , written  $k \cdot A$  or simply  $kA$ , is the matrix obtained by multiplying each element of  $A$  by  $k$ . That is,

$$kA = \begin{bmatrix} ka_{11} & ka_{12} & \dots & ka_{1n} \\ ka_{21} & ka_{22} & \dots & ka_{2n} \\ \dots & \dots & \dots & \dots \\ ka_{m1} & ka_{m2} & \dots & ka_{mn} \end{bmatrix}$$

Observe that  $A + B$  and  $kA$  are also  $m \times n$  matrices. We also define

$$-A = (-1)A \quad \text{and} \quad A - B = A + (-B)$$

The matrix  $-A$  is called the *negative* of the matrix  $A$ , and the matrix  $A - B$  is called the *difference* of  $A$  and  $B$ . The sum of matrices with different sizes is not defined.

**EXAMPLE 2.2** Let  $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$  and  $B = \begin{bmatrix} 4 & 6 & 8 \\ 1 & -3 & -7 \end{bmatrix}$ . Then

$$\begin{aligned}
 A + B &= \begin{bmatrix} 1+4 & -2+6 & 3+8 \\ 0+1 & 4+(-3) & 5+(-7) \end{bmatrix} = \begin{bmatrix} 5 & 4 & 11 \\ 1 & 1 & -2 \end{bmatrix} \\
 3A &= \begin{bmatrix} 3(1) & 3(-2) & 3(3) \\ 3(0) & 3(4) & 3(5) \end{bmatrix} = \begin{bmatrix} 3 & -6 & 9 \\ 0 & 12 & 15 \end{bmatrix} \\
 2A - 3B &= \begin{bmatrix} 2 & -4 & 6 \\ 0 & 8 & 10 \end{bmatrix} + \begin{bmatrix} -12 & -18 & -24 \\ -3 & 9 & 21 \end{bmatrix} = \begin{bmatrix} -10 & -22 & -18 \\ -3 & 17 & 31 \end{bmatrix}
 \end{aligned}$$

The matrix  $2A - 3B$  is called a *linear combination* of  $A$  and  $B$ .

Basic properties of matrices under the operations of matrix addition and scalar multiplication follow.

**THEOREM 2.1:** Consider any matrices  $A, B, C$  (with the same size) and any scalars  $k$  and  $k'$ . Then

- (i)  $(A + B) + C = A + (B + C)$ ,
- (ii)  $A + 0 = 0 + A = A$ ,
- (iii)  $A + (-A) = (-A) + A = 0$ ;
- (iv)  $A + B = B + A$ ,
- (v)  $k(A + B) = kA + kB$ ,
- (vi)  $(k + k')A = kA + k'A$ ,
- (vii)  $(kk')A = k(k'A)$ ,
- (viii)  $1 \cdot A = A$ .

Note first that the 0 in (ii) and (iii) refers to the zero matrix. Also, by (i) and (iv), any sum of matrices

$$A_1 + A_2 + \cdots + A_n$$

requires no parentheses, and the sum does not depend on the order of the matrices. Furthermore, using (vi) and (viii), we also have

$$A + A = 2A, \quad A + A + A = 3A, \quad \dots$$

and so on.

The proof of [Theorem 2.1](#) reduces to showing that the  $ij$ -entries on both sides of each matrix equation are equal. (See [Problem 2.3](#).)

Observe the similarity between [Theorem 2.1](#) for matrices and [Theorem 1.1](#) for vectors. In fact, the above operations for matrices may be viewed as generalizations of the corresponding operations for vectors.

## 2.4 Summation Symbol

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Before we define matrix multiplication, it will be instructive to first introduce the *summation symbol*  $\Sigma$  (the Greek capital letter sigma).

Suppose  $f(k)$  is an algebraic expression involving the letter  $k$ . Then the expression

$$\sum_{k=1}^n f(k) \quad \text{or equivalently} \quad \sum_{k=1}^n f(k)$$

has the following meaning. First we set  $k = 1$  in  $f(k)$ , obtaining

$$f(1)$$

Then we set  $k = 2$  in  $f(k)$ , obtaining  $f(2)$ , and add this to  $f(1)$ , obtaining

$$f(1) + f(2)$$

Then we set  $k = 3$  in  $f(k)$ , obtaining  $f(3)$ , and add this to the previous sum, obtaining

$$f(1) + f(2) + f(3)$$

We continue this process until we obtain the sum

$$f(1) + f(2) + \cdots + f(n)$$

Observe that at each step we increase the value of  $k$  by 1 until we reach  $n$ . The letter  $k$  is called the *index*, and 1 and  $n$  are called, respectively, the *lower* and *upper* limits. Other letters frequently used as indices are  $i$  and  $j$ .

We also generalize our definition by allowing the sum to range from any integer  $n_1$  to any integer  $n_2$ . That is, we define

$$\sum_{k=n_1}^{n_2} f(k) = f(n_1) + f(n_1 + 1) + f(n_1 + 2) + \cdots + f(n_2)$$

### EXAMPLE 2.3

- (a)  $\sum_{k=1}^5 x_k = x_1 + x_2 + x_3 + x_4 + x_5$  and  $\sum_{i=1}^n a_i b_i = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$
- (b)  $\sum_{j=2}^5 j^2 = 2^2 + 3^2 + 4^2 + 5^2 = 54$  and  $\sum_{i=0}^n a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$
- (c)  $\sum_{k=1}^p a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + a_{i3} b_{3j} + \cdots + a_{ip} b_{pj}$

## 2.5 Matrix Multiplication

---

The product of matrices  $A$  and  $B$ , written  $AB$ , is somewhat complicated. For this reason, we first begin with a special case.

The product  $AB$  of a row matrix  $A = [a_j]$  and a column matrix  $B = [b_i]$  with the same number of elements is defined to be the scalar (or  $1 \times 1$  matrix) obtained by multiplying corresponding entries and adding; that is,

$$AB = [a_1, a_2, \dots, a_n] \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{k=1}^n a_k b_k$$

We emphasize that  $AB$  is a scalar (or a  $1 \times 1$  matrix). The product  $AB$  is not defined when  $A$  and  $B$  have different numbers of elements.

#### EXAMPLE 2.4

$$(a) [7, -4, 5] \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} = 7(3) + (-4)(2) + 5(-1) = 21 - 8 - 5 = 8$$

$$(b) [6, -1, 8, 3] \begin{bmatrix} 4 \\ -9 \\ -2 \\ 5 \end{bmatrix} = 24 + 9 - 16 + 15 = 32$$

We are now ready to define matrix multiplication in general.

**DEFINITION:** Suppose  $A = [a_{ik}]$  and  $B = [b_{kj}]$  are matrices such that the number of columns of  $A$  is equal to the number of rows of  $B$ ; say,  $A$  is an  $m \times p$  matrix and  $B$  is a  $p \times n$  matrix. Then the product  $AB$  is the  $m \times n$  matrix whose  $ij$ -entry is obtained by multiplying the  $i$ th row of  $A$  by the  $j$ th column of  $B$ . That is,

$$\begin{bmatrix} a_{11} & \dots & a_{1p} \\ \vdots & \dots & \vdots \\ a_{i1} & \dots & a_{ip} \\ \vdots & \dots & \vdots \\ a_{m1} & \dots & a_{mp} \end{bmatrix} \begin{bmatrix} b_{11} & \dots & b_{1j} & \dots & b_{1n} \\ \vdots & \dots & \vdots & \dots & \vdots \\ \vdots & \dots & \vdots & \dots & \vdots \\ \vdots & \dots & \vdots & \dots & \vdots \\ b_{p1} & \dots & b_{pj} & \dots & b_{pn} \end{bmatrix} = \begin{bmatrix} c_{11} & \dots & c_{1n} \\ \vdots & \dots & \vdots \\ \vdots & c_{ij} & \vdots \\ \vdots & \dots & \vdots \\ c_{m1} & \dots & c_{mn} \end{bmatrix}$$

$$\text{where } c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}$$

The product  $AB$  is not defined if  $A$  is an  $m \times p$  matrix and  $B$  is a  $q \times n$  matrix, where  $p \neq q$ .

### EXAMPLE 2.5

(a) Find  $AB$  where  $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 0 & -4 \\ 5 & -2 & 6 \end{bmatrix}$ .

Because  $A$  is  $2 \times 2$  and  $B$  is  $2 \times 3$ , the product  $AB$  is defined and  $AB$  is a  $2 \times 3$  matrix. To obtain the first row of the product matrix  $AB$ , multiply the first row  $[1, 3]$  of  $A$  by each column of  $B$ ,

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \quad \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

respectively. That is,

$$AB = \begin{bmatrix} 2 + 15 & 0 - 6 & -4 + 18 \\ 4 - 5 & 0 + 2 & -8 - 6 \end{bmatrix} = \begin{bmatrix} 17 & -6 & 14 \\ -1 & 2 & -14 \end{bmatrix}$$

To obtain the second row of  $AB$ , multiply the second row  $[2, -1]$  of  $A$  by each column of  $B$ . Thus,

$$AB = \begin{bmatrix} 17 & -6 & 14 \\ 4 - 5 & 0 + 2 & -8 - 6 \end{bmatrix} = \begin{bmatrix} 17 & -6 & 14 \\ -1 & 2 & -14 \end{bmatrix}$$

(b) Suppose  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 5 & 6 \\ 0 & -2 \end{bmatrix}$ . Then

$$AB = \begin{bmatrix} 5 + 0 & 6 - 4 \\ 15 + 0 & 18 - 8 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 15 & 10 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 5 + 18 & 10 + 24 \\ 0 - 6 & 0 - 8 \end{bmatrix} = \begin{bmatrix} 23 & 34 \\ -6 & -8 \end{bmatrix}$$

The above example shows that matrix multiplication is not commutative—that is, in general,  $AB \neq BA$ . However, matrix multiplication does satisfy the following properties.

**THEOREM 2.2:** Let  $A, B, C$  be matrices. Then, whenever the products and sums are defined,

- (i)  $(AB)C = A(BC)$  (associative law),
- (ii)  $A(B + C) = AB + AC$  (left distributive law),
- (iii)  $(B + C)A = BA + CA$  (right distributive law),
- (iv)  $k(AB) = (kA)B = A(kB)$ , where  $k$  is a scalar.

We note that  $0A = 0$  and  $B0 = 0$ , where  $0$  is the zero matrix.

## 2.6 Transpose of a Matrix

---

The *transpose* of a matrix  $A$ , written  $A^T$ , is the matrix obtained by writing the columns of  $A$ , in order, as rows. For example,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad [1, -3, -5]^T = \begin{bmatrix} 1 \\ -3 \\ -5 \end{bmatrix}$$

In other words, if  $A = [a_{ij}]$  is an  $m \times n$  matrix, then  $A^T = [b_{ij}]$  is the  $n \times m$  matrix where  $b_{ij} = a_{ji}$ .

Observe that the transpose of a row vector is a column vector. Similarly, the transpose of a column vector is a row vector.

The next theorem lists basic properties of the transpose operation.

**THEOREM 2.3:** Let  $A$  and  $B$  be matrices and let  $k$  be a scalar. Then, whenever the sum and product are defined,

$$\begin{array}{ll}
 \text{(i)} & (A + B)^T = A^T + B^T, \\
 \text{(ii)} & (A^T)^T = A, \\
 \text{(iii)} & (kA)^T = kA^T, \\
 \text{(iv)} & (AB)^T = B^T A^T.
 \end{array}$$

We emphasize that, by (iv), the transpose of a product is the product of the transposes, but in the reverse order.

## 2.7 Square Matrices

---

A *square matrix* is a matrix with the same number of rows as columns. An  $n \times n$  square matrix is said to be of *order*  $n$  and is sometimes called an  *$n$ -square matrix*.

Recall that not every two matrices can be added or multiplied. However, if we only consider square matrices of some given order  $n$ , then this inconvenience disappears. Specifically, the operations of addition, multiplication, scalar multiplication, and transpose can be performed on any  $n \times n$  matrices, and the result is again an  $n \times n$  matrix.

**EXAMPLE 2.6** The following are square matrices of order 3:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & -4 & -4 \\ 5 & 6 & 7 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -5 & 1 \\ 0 & 3 & -2 \\ 1 & 2 & -4 \end{bmatrix}$$

The following are also matrices of order 3:

$$\begin{aligned}
 A + B &= \begin{bmatrix} 3 & -3 & 4 \\ -4 & -1 & -6 \\ 6 & 8 & 3 \end{bmatrix}, & 2A &= \begin{bmatrix} 2 & 4 & 6 \\ -8 & -8 & -8 \\ 10 & 12 & 14 \end{bmatrix}, & A^T &= \begin{bmatrix} 1 & -4 & 5 \\ 2 & -4 & 6 \\ 3 & -4 & 7 \end{bmatrix} \\
 AB &= \begin{bmatrix} 5 & 7 & -15 \\ -12 & 0 & 20 \\ 17 & 7 & -35 \end{bmatrix}, & BA &= \begin{bmatrix} 27 & 30 & 33 \\ -22 & -24 & -26 \\ -27 & -30 & -33 \end{bmatrix}
 \end{aligned}$$

## Diagonal and Trace

---

Let  $A = [a_{ij}]$  be an  $n$ -square matrix. The *diagonal* or *main diagonal* of  $A$  consists of the elements with the same subscripts—that is,

$$a_{11}, \quad a_{22}, \quad a_{33}, \quad \dots, \quad a_{nn}$$

The *trace* of  $A$ , written  $\text{tr}(A)$ , is the sum of the diagonal elements. Namely,

$$\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

The following theorem applies.

**THEOREM 2.4:** Suppose  $A = [a_{ij}]$  and  $B = [b_{ij}]$  are  $n$ -square matrices and  $k$  is a scalar. Then

$$\begin{array}{ll} \text{(i)} \quad \text{tr}(A + B) = \text{tr}(A) + \text{tr}(B), & \text{(iii)} \quad \text{tr}(A^T) = \text{tr}(A), \\ \text{(ii)} \quad \text{tr}(kA) = k \text{tr}(A), & \text{(iv)} \quad \text{tr}(AB) = \text{tr}(BA). \end{array}$$

### EXAMPLE 2.7

Let  $A$  and  $B$  be the matrices  $A$  and  $B$  in [Example 2.6](#). Then

$$\begin{array}{lll} \text{diagonal of } A = \{1, -4, 7\} & \text{and} & \text{tr}(A) = 1 - 4 + 7 = 4 \\ \text{diagonal of } B = \{2, 3, -4\} & \text{and} & \text{tr}(B) = 2 + 3 - 4 = 1 \end{array}$$

Moreover,

$$\begin{array}{ll} \text{tr}(A + B) = 3 - 1 + 3 = 5, & \text{tr}(2A) = 2 - 8 + 14 = 8, \quad \text{tr}(A^T) = 1 - 4 + 7 = 4 \\ \text{tr}(AB) = 5 + 0 - 35 = -30, & \text{tr}(BA) = 27 - 24 - 33 = -30 \end{array}$$

As expected from [Theorem 2.4](#),

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B), \quad \text{tr}(A^T) = \text{tr}(A), \quad \text{tr}(2A) = 2 \text{tr}(A)$$

Furthermore, although  $AB \neq BA$ , the traces are equal.

## Identity Matrix, Scalar Matrices

---

The  $n$ -square *identity* or *unit* matrix, denoted by  $I_n$ , or simply  $I$ , is the  $n$ -square matrix with 1's on the diagonal and 0's elsewhere. The identity matrix  $I$  is similar to the scalar 1 in that, for any  $n$ -square matrix  $A$ ,

$$AI = IA = A$$

More generally, if  $B$  is an  $m \times n$  matrix, then  $BI_n = I_m B = B$ .

For any scalar  $k$ , the matrix  $kI$  that contains  $k$ 's on the diagonal and 0's elsewhere is called the *scalar matrix* corresponding to the scalar  $k$ . Observe that

$$(kI)A = k(IA) = kA$$

That is, multiplying a matrix  $A$  by the scalar matrix  $kI$  is equivalent to multiplying  $A$  by the scalar  $k$ .

**EXAMPLE 2.8** The following are the identity matrices of orders 3 and 4 and the corresponding scalar matrices for  $k = 5$ :

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \quad \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \quad \begin{bmatrix} 5 & & & \\ & 5 & & \\ & & 5 & \\ & & & 5 \end{bmatrix}$$

**Remark 1:** It is common practice to omit blocks or patterns of 0's when there is no ambiguity, as in the above second and fourth matrices.

**Remark 2:** The *Kronecker delta function*  $\delta_{ij}$  is defined by

$$\delta_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Thus, the identity matrix may be defined by  $I = [\delta_{ij}]$ .

## 2.8 Powers of Matrices, Polynomials in Matrices

---

Let  $A$  be an  $n$ -square matrix over a field  $K$ . Powers of  $A$  are defined as follows:

$$A^2 = AA, \quad A^3 = A^2A, \quad \dots, \quad A^{n+1} = A^nA, \quad \dots, \quad \text{and} \quad A^0 = I$$

Polynomials in the matrix  $A$  are also defined. Specifically, for any polynomial

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

where the  $a_i$  are scalars in  $K$ ,  $f(A)$  is defined to be the following matrix:

$$f(A) = a_0I + a_1A + a_2A^2 + \dots + a_nA^n$$

[Note that  $f(A)$  is obtained from  $f(x)$  by substituting the matrix  $A$  for the variable  $x$  and substituting the scalar matrix  $a_0I$  for the scalar  $a_0$ .] If  $f(A)$  is the zero matrix, then  $A$  is called a *zero* or *root* of  $f(x)$ .

**EXAMPLE 2.9** Suppose  $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ . Then

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix} \quad \text{and} \quad A^3 = A^2A = \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} -11 & 38 \\ 57 & -106 \end{bmatrix}$$

Suppose  $f(x) = 2x^2 - 3x + 5$  and  $g(x) = x^2 + 3x - 10$ . Then

$$f(A) = 2 \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix} - 3 \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 16 & -18 \\ -27 & 61 \end{bmatrix}$$

$$g(A) = \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix} + 3 \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} - 10 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Thus,  $A$  is a zero of the polynomial  $g(x)$ .

## 2.9 Invertible (Nonsingular) Matrices

A square matrix  $A$  is said to be *invertible* or *nonsingular* if there exists a matrix  $B$  such that

$$AB = BA = I$$

where  $I$  is the identity matrix. Such a matrix  $B$  is unique. That is, if  $AB_1 = B_1A = I$  and  $AB_2 = B_2A = I$ , then

$$B_1 = B_1I = B_1(AB_2) = (B_1A)B_2 = IB_2 = B_2$$

We call such a matrix  $B$  the *inverse* of  $A$  and denote it by  $A^{-1}$ . Observe that the above relation is symmetric; that is, if  $B$  is the inverse of  $A$ , then  $A$  is the inverse of  $B$ .

**EXAMPLE 2.10** Suppose that  $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$ .

Then

$$AB = \begin{bmatrix} 6-5 & -10+10 \\ 3-3 & -5+6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 6-5 & 15-15 \\ -2+2 & -5+6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus,  $A$  and  $B$  are inverses.

It is known ([Theorem 3.18](#)) that  $AB = I$  if and only if  $BA = I$ . Thus, it is necessary to test only one product to determine whether or not two given matrices are inverses. (See [Problem 2.17](#).)

Now suppose  $A$  and  $B$  are invertible. Then  $AB$  is invertible and  $(AB)^{-1} = B^{-1}A^{-1}$ . More generally, if  $A_1, A_2, \dots, A_k$  are invertible, then their product is invertible and

$$(A_1 A_2 \dots A_k)^{-1} = A_k^{-1} \dots A_2^{-1} A_1^{-1}$$

the product of the inverses in the reverse order.

## Inverse of a $2 \times 2$ Matrix

---

Let  $A$  be an arbitrary  $2 \times 2$  matrix, say  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ . We want to derive a formula for  $A^{-1}$ , the inverse of  $A$ . Specifically, we seek  $2^2 = 4$  scalars, say  $x_1, y_1, x_2, y_2$ , such that

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} ax_1 + by_1 & ax_2 + by_2 \\ cx_1 + dy_1 & cx_2 + dy_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Setting the four entries equal to the corresponding entries in the identity matrix yields four equations, which can be partitioned into two  $2 \times 2$  systems as follows:

$$\begin{aligned} ax_1 + by_1 &= 1, & ax_2 + by_2 &= 0 \\ cx_1 + dy_1 &= 0, & cx_2 + dy_2 &= 1 \end{aligned}$$

Suppose we let  $|A| = ad - bc$  (called the *determinant* of  $A$ ). Assuming  $|A| \neq 0$ , we can solve uniquely for the above unknowns  $x_1, y_1, x_2, y_2$ , obtaining

$$x_1 = \frac{d}{|A|}, \quad y_1 = \frac{-c}{|A|}, \quad x_2 = \frac{-b}{|A|}, \quad y_2 = \frac{a}{|A|}$$

Accordingly,

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \begin{bmatrix} d/|A| & -b/|A| \\ -c/|A| & a/|A| \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

In other words, when  $|A| \neq 0$ , the inverse of a  $2 \times 2$  matrix  $A$  may be obtained from  $A$  as follows:

- (1) Interchange the two elements on the diagonal.
- (2) Take the negatives of the other two elements.
- (3) Multiply the resulting matrix by  $1/|A|$  or, equivalently, divide each element by  $|A|$ .

In case  $|A| = 0$ , the matrix  $A$  is not invertible.

**EXAMPLE 2.11** Find the inverse of  $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ .

First evaluate  $|A| = 2(5) - 3(4) = 10 - 12 = -2$ . Because  $|A| \neq 0$ , the matrix  $A$  is invertible and

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} -\frac{5}{2} & \frac{3}{2} \\ 2 & -1 \end{bmatrix}$$

Now evaluate  $|B| = 1(6) - 3(2) = 6 - 6 = 0$ . Because  $|B| = 0$ , the matrix  $B$  has no inverse.

**Remark:** The above property that a matrix is invertible if and only if  $A$  has a nonzero determinant is true for square matrices of any order. (See [Chapter 8](#).)

### **Inverse of an $n \times n$ Matrix**

---

Suppose  $A$  is an arbitrary  $n$ -square matrix. Finding its inverse  $A^{-1}$  reduces, as above, to finding the solution of a collection of  $n \times n$  systems of linear equations. The solution of such systems and an efficient way of solving such a collection of systems is treated in [Chapter 3](#).

## **2.10 Special Types of Square Matrices**

---

This section describes a number of special kinds of square matrices.

## **Diagonal and Triangular Matrices**

---

A square matrix  $D = [d_{ij}]$  is *diagonal* if its nondiagonal entries are all zero. Such a matrix is sometimes denoted by

$$D = \text{diag}(d_{11}, d_{22}, \dots, d_{nn})$$

where some or all the  $d_{ij}$  may be zero. For example,

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad \begin{bmatrix} 4 & 0 \\ 0 & -5 \end{bmatrix}, \quad \begin{bmatrix} 6 & & & \\ & 0 & & \\ & & -9 & \\ & & & 8 \end{bmatrix}$$

are diagonal matrices, which may be represented, respectively, by

$$\text{diag}(3, -7, 2), \quad \text{diag}(4, -5), \quad \text{diag}(6, 0, -9, 8)$$

(Observe that patterns of 0's in the third matrix have been omitted.)

A square matrix  $A = [a_{ij}]$  is *upper triangular* or simply *triangular* if all entries below the (main) diagonal are equal to 0—that is, if  $a_{ij} = 0$  for  $i > j$ . Generic upper triangular matrices of orders 2, 3, 4 are as follows:

$$\begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{bmatrix}, \quad \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ & b_{22} & b_{23} \\ & & b_{33} \end{bmatrix}, \quad \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ & c_{22} & c_{23} & c_{24} \\ & & c_{33} & c_{34} \\ & & & c_{44} \end{bmatrix}$$

(As with diagonal matrices, it is common practice to omit patterns of 0's.)

The following theorem applies.

**THEOREM 2.5:** Suppose  $A = [a_{ij}]$  and  $B = [b_{ij}]$  are  $n \times n$  (upper) triangular matrices. Then

(i)  $A + B$ ,  $kA$ ,  $AB$  are triangular with respective diagonals:

$$(a_{11} + b_{11}, \dots, a_{nn} + b_{nn}), \quad (ka_{11}, \dots, ka_{nn}), \quad (a_{11}b_{11}, \dots, a_{nn}b_{nn})$$

(ii) For any polynomial  $f(x)$ , the matrix  $f(A)$  is triangular with diagonal

$$(f(a_{11}), f(a_{22}), \dots, f(a_{nn}))$$

(iii)  $A$  is invertible if and only if each diagonal element  $a_{ii} \neq 0$ , and when  $A^{-1}$  exists it is also triangular.

A *lower triangular matrix* is a square matrix whose entries above the diagonal are all zero. We note that [Theorem 2.5](#) is true if we replace “triangular” by either “lower triangular” or “diagonal.”

**Remark:** A nonempty collection  $A$  of matrices is called an *algebra* (of matrices) if  $A$  is closed under the operations of matrix addition, scalar multiplication, and matrix multiplication. Clearly, the square matrices with a given order form an algebra of matrices, but so do the scalar, diagonal, triangular, and lower triangular matrices.

### **Special Real Square Matrices: Symmetric, Orthogonal, Normal [Optional until [Chapter 12](#)]**

Suppose now  $A$  is a square matrix with real entries—that is, a real square matrix. The relationship between  $A$  and its transpose  $A^T$  yields important kinds of matrices.

## (a) Symmetric Matrices

---

A matrix  $A$  is *symmetric* if  $A^T = A$ . Equivalently,  $A = [a_{ij}]$  is symmetric if *symmetric elements* (mirror elements with respect to the diagonal) are equal—that is, if each  $a_{ij} = a_{ji}$ .

A matrix  $A$  is *skew-symmetric* if  $A^T = -A$  or, equivalently, if each  $a_{ij} = -a_{ji}$ . Clearly, the diagonal elements of such a matrix must be zero, because  $a_{ii} = -a_{ii}$  implies  $a_{ii} = 0$ .

(Note that a matrix  $A$  must be square if  $A^T = A$  or  $A^T = -A$ .)

**EXAMPLE 2.12** Let

$$A = \begin{bmatrix} 2 & -3 & 5 \\ -3 & 6 & 7 \\ 5 & 7 & -8 \end{bmatrix}, B = \begin{bmatrix} 0 & 3 & -4 \\ -3 & 0 & 5 \\ 4 & -5 & 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (a) By inspection, the symmetric elements in  $A$  are equal, or  $A^T = A$ . Thus,  $A$  is symmetric.
- (b) The diagonal elements of  $B$  are 0 and symmetric elements are negatives of each other, or  $B^T = -B$ . Thus,  $B$  is skew-symmetric.
- (c) Because  $C$  is not square,  $C$  is neither symmetric nor skew-symmetric.

## (b) Orthogonal Matrices

---

A real matrix  $A$  is *orthogonal* if  $A^T = A^{-1}$ —that is, if  $AA^T = A^TA = I$ . Thus,  $A$  must necessarily be square and invertible.

**EXAMPLE 2.13** Let  $A = \begin{bmatrix} \frac{1}{9} & \frac{8}{9} & -\frac{4}{9} \\ \frac{4}{9} & -\frac{4}{9} & -\frac{7}{9} \\ \frac{8}{9} & \frac{1}{9} & \frac{4}{9} \end{bmatrix}$ . Multiplying  $A$  by  $A^T$

yields  $I$ ; that is,  $AA^T = I$ . This means  $A^TA = I$ , as well. Thus,  $A^T = A^{-1}$ ; that is,  $A$  is orthogonal.

Now suppose  $A$  is a real orthogonal  $3 \times 3$  matrix with rows

$$u_1 = (a_1, a_2, a_3), \quad u_2 = (b_1, b_2, b_3), \quad u_3 = (c_1, c_2, c_3)$$

Because  $A$  is orthogonal, we must have  $AA^T = I$ . Namely,

$$AA^T = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Multiplying  $A$  by  $A^T$  and setting each entry equal to the corresponding entry in  $I$  yields the following nine equations:

$$\begin{array}{lll} a_1^2 + a_2^2 + a_3^2 = 1, & a_1b_1 + a_2b_2 + a_3b_3 = 0, & a_1c_1 + a_2c_2 + a_3c_3 = 0 \\ b_1a_1 + b_2a_2 + b_3a_3 = 0, & b_1^2 + b_2^2 + b_3^2 = 1, & b_1c_1 + b_2c_2 + b_3c_3 = 0 \\ c_1a_1 + c_2a_2 + c_3a_3 = 0, & c_1b_1 + c_2b_2 + c_3b_3 = 0, & c_1^2 + c_2^2 + c_3^2 = 1 \end{array}$$

Accordingly,  $u_1 \cdot u_1 = 1$ ,  $u_2 \cdot u_2 = 1$ ,  $u_3 \cdot u_3 = 1$ , and  $u_i \cdot u_j = 0$  for  $i \neq j$ . Thus, the rows  $u_1$ ,  $u_2$ ,  $u_3$  are unit vectors and are orthogonal to each other.

Generally speaking, vectors  $u_1, u_2, \dots; u_m$  in  $\mathbf{R}^n$  are said to form an *orthonormal set* of vectors if the vectors are unit vectors and are orthogonal to each other; that is,

$$u_i \cdot u_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

In other words,  $u_i \cdot u_j = \delta_{ij}$  where  $\delta_{ij}$  is the Kronecker delta function:

We have shown that the condition  $AA^T = I$  implies that the rows of  $A$  form an orthonormal set of vectors. The condition  $A^TA = I$  similarly implies that the columns of  $A$  also form an orthonormal set of vectors. Furthermore, because each step is reversible, the converse is true.

The above results for  $3 \times 3$  matrices are true in general. That is, the following theorem holds.

**THEOREM 2.6:** Let  $A$  be a real matrix. Then the following are equivalent:

- (a)  $A$  is orthogonal.
- (b) The rows of  $A$  form an orthonormal set.
- (c) The columns of  $A$  form an orthonormal set.

For  $n = 2$ , we have the following result (proved in [Problem 2.28](#)).

**THEOREM 2.7:** Let  $A$  be a real  $2 \times 2$  orthogonal matrix. Then, for some real number  $\theta$ ,

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$$

### (c) Normal Matrices

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A real matrix  $A$  is *normal* if it *commutes* with its transpose  $A^T$ —that is, if  $AA^T = A^T A$ . If  $A$  is symmetric, orthogonal, or skew-symmetric, then  $A$  is normal. There are also other normal matrices.

**EXAMPLE 2.14** Let  $A = \begin{bmatrix} 6 & -3 \\ 3 & 6 \end{bmatrix}$ . Then

$$AA^T = \begin{bmatrix} 6 & -3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 6 & 3 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 45 & 0 \\ 0 & 45 \end{bmatrix} \quad \text{and} \quad A^T A = \begin{bmatrix} 6 & 3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} 6 & -3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 45 & 0 \\ 0 & 45 \end{bmatrix}$$

Because  $AA^T = A^T A$ , the matrix  $A$  is normal.

## 2.11 Complex Matrices

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Let  $A$  be a complex matrix—that is, a matrix with complex entries. Recall (Section 1.7) that if  $z = a + bi$  is a complex number, then  $\bar{z} = a - bi$  is its conjugate. The *conjugate* of a

complex matrix  $A$ , written  $\bar{A}$ , is the matrix obtained from  $A$  by taking the conjugate of each entry in  $A$ . That is, if  $A = [a_{ij}]$ , then  $\bar{A} = [\bar{a}_{ij}]$ , where  $\bar{a}_{ij} = \overline{a_{ij}}$ . (We denote this fact by writing  $\bar{A} = [\bar{a}_{ij}]$ .)

The two operations of transpose and conjugation commute for any complex matrix  $A$ , and the special notation  $A^H$  is used for the conjugate transpose of  $A$ . That is,

$$A^H = (\bar{A})^T = \overline{(A^T)}$$

Note that if  $A$  is real, then  $A^H = A^T$ . [Some texts use  $A^*$  instead of  $A^H$ .]

**EXAMPLE 2.15** Let  $A = \begin{bmatrix} 2+8i & 5-3i & 4-7i \\ 6i & 1-4i & 3+2i \end{bmatrix}$ . Then

$$A^H = \begin{bmatrix} 2-8i & -6i \\ 5+3i & 1+4i \\ 4+7i & 3-2i \end{bmatrix}.$$

### Special Complex Matrices: Hermitian, Unitary, Normal [Optional until [Chapter 12](#)]

---

Consider a complex matrix  $A$ . The relationship between  $A$  and its conjugate transpose  $A^H$  yields important kinds of complex matrices (which are analogous to the kinds of real matrices described above).

A complex matrix  $A$  is said to be *Hermitian* or *skew-Hermitian* according as to whether

$$A^H = A \quad \text{or} \quad A^H = -A.$$

Clearly,  $A = [a_{ij}]$  is Hermitian if and only if symmetric elements are conjugate—that is, if each  $a_{ij} = \bar{a}_{ji}$ —in which case each diagonal element  $a_{ii}$  must be real. Similarly, if  $A$  is

skew-symmetric, then each diagonal element  $a_{ii} = 0$ . (Note that  $A$  must be square if  $A^H = A$  or  $A^H = -A$ .)

A complex matrix  $A$  is *unitary* if  $A^H A^{-1} = A^{-1} A^H = I$ —that is, if

$$A^H = A^{-1}.$$

Thus,  $A$  must necessarily be square and invertible. We note that a complex matrix  $A$  is unitary if and only if its rows (columns) form an orthonormal set relative to the dot product of complex vectors.

A complex matrix  $A$  is said to be *normal* if it commutes with  $A^H$ —that is, if

$$AA^H = A^H A$$

(Thus,  $A$  must be a square matrix.) This definition reduces to that for real matrices when  $A$  is real.

**EXAMPLE 2.16** Consider the following complex matrices:

$$A = \begin{bmatrix} 3 & 1-2i & 4+7i \\ 1+2i & -4 & -2i \\ 4-7i & 2i & 5 \end{bmatrix} \quad B = \frac{1}{2} \begin{bmatrix} 1 & -i & -1+i \\ i & 1 & 1+i \\ 1+i & -1+i & 0 \end{bmatrix} \quad C = \begin{bmatrix} 2+3i & 1 \\ i & 1+2i \end{bmatrix}$$

(a) By inspection, the diagonal elements of  $A$  are real, and the symmetric elements  $1 - 2i$  and  $1 + 2i$  are conjugate,  $4 + 7i$  and  $4 - 7i$  are conjugate, and  $-2i$  and  $2i$  are conjugate.

Thus,  $A$  is Hermitian.

(b) Multiplying  $B$  by  $B^H$  yields  $I$ ; that is,  $BB^H = I$ . This implies  $B^H B = I$ , as well. Thus,  $B^H = B^{-1}$ , which means  $B$  is unitary.

(c) To show  $C$  is normal, we evaluate  $CC^H$  and  $C^H C$ :

$$CC^H = \begin{bmatrix} 2+3i & 1 \\ i & 1+2i \end{bmatrix} \begin{bmatrix} 2-3i & -i \\ 1 & 1-2i \end{bmatrix} = \begin{bmatrix} 14 & 4-4i \\ 4+4i & 6 \end{bmatrix}$$

and similarly  $C^H C = \begin{bmatrix} 14 & 4 - 4i \\ 4 + 4i & 6 \end{bmatrix}$ . Because  $CC^H = C^H C$ , the complex matrix  $C$  is normal.

We note that when a matrix  $A$  is real, Hermitian is the same as symmetric, and unitary is the same as orthogonal.

## 2.12 Block Matrices

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Using a system of horizontal and vertical (dashed) lines, we can partition a matrix  $A$  into submatrices called *blocks* (or *cells*) of  $A$ . Clearly a given matrix may be divided into blocks in different ways. For example,

$$\begin{bmatrix} 1 & -2 & | & 0 & 1 & | & 3 \\ 2 & -3 & | & 5 & 7 & | & -2 \\ 3 & 1 & | & 4 & 5 & | & 9 \\ 4 & 6 & | & -3 & 1 & | & 8 \end{bmatrix}, \quad \begin{bmatrix} 1 & -2 & | & 0 & 1 & | & 3 \\ 2 & 3 & | & 5 & 7 & | & -2 \\ 3 & 1 & | & 4 & 5 & | & 9 \\ 4 & 6 & | & -3 & 1 & | & 8 \end{bmatrix}, \quad \begin{bmatrix} 1 & -2 & 0 & | & 1 & 3 \\ 2 & -3 & 5 & | & 7 & -2 \\ 3 & 1 & 4 & | & 5 & 9 \\ 4 & 6 & -3 & | & 1 & 8 \end{bmatrix}$$

The convenience of the partition of matrices, say  $A$  and  $B$ , into blocks is that the result of operations on  $A$  and  $B$  can be obtained by carrying out the computation with the blocks, just as if they were the actual elements of the matrices. This is illustrated below, where the notation  $A = [A_{ij}]$  will be used for a block matrix  $A$  with blocks  $A_{ij}$ .

Suppose that  $A = [A_{ij}]$  and  $B = [B_{ij}]$  are block matrices with the same numbers of row and column blocks, and suppose that corresponding blocks have the same size. Then adding the corresponding blocks of  $A$  and  $B$  also adds the corresponding elements of  $A$  and  $B$ , and multiplying each block of  $A$  by a scalar  $k$  multiplies each element of  $A$  by  $k$ . Thus,

$$A + B = \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} & \cdots & A_{1n} + B_{1n} \\ A_{21} + B_{21} & A_{22} + B_{22} & \cdots & A_{2n} + B_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ A_{m1} + B_{m1} & A_{m2} + B_{m2} & \cdots & A_{mn} + B_{mn} \end{bmatrix}$$

and

$$kA = \begin{bmatrix} kA_{11} & kA_{12} & \cdots & kA_{1n} \\ kA_{21} & kA_{22} & \cdots & kA_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ kA_{m1} & kA_{m2} & \cdots & kA_{mn} \end{bmatrix}$$

The case of matrix multiplication is less obvious, but still true. That is, suppose that  $U = [U_{ik}]$  and  $V = [V_{kj}]$  are block matrices such that the number of columns of each block  $U_{ik}$  is equal to the number of rows of each block  $V_{kj}$ . (Thus, each product  $U_{ik}V_{kj}$  is defined.) Then

$$UV = \begin{bmatrix} W_{11} & W_{12} & \cdots & W_{1n} \\ W_{21} & W_{22} & \cdots & W_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ W_{m1} & W_{m2} & \cdots & W_{mn} \end{bmatrix}, \quad \text{where } W_{ij} = U_{i1}V_{1j} + U_{i2}V_{2j} + \cdots + U_{ip}V_{pj}$$

The proof of the above formula for  $UV$  is straightforward but detailed and lengthy. It is left as an exercise ([Problem 2.85](#)).

## Square Block Matrices

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Let  $M$  be a block matrix. Then  $M$  is called a *square block matrix* if

- (i)  $M$  is a square matrix.
- (ii) The blocks form a square matrix.
- (iii) The diagonal blocks are also square matrices.

The latter two conditions will occur if and only if there are the same number of horizontal and vertical lines and they are placed symmetrically.

Consider the following two block matrices:

$$A = \left[ \begin{array}{cc|cc|c} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \\ \hline 9 & 8 & 7 & 6 & 5 \\ 4 & 4 & 4 & 4 & 4 \\ \hline 3 & 5 & 3 & 5 & 3 \end{array} \right] \quad \text{and} \quad B = \left[ \begin{array}{cc|cc|c} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \\ \hline 9 & 8 & 7 & 6 & 5 \\ 4 & 4 & 4 & 4 & 4 \\ \hline 3 & 5 & 3 & 5 & 3 \end{array} \right]$$

The block matrix  $A$  is not a square block matrix, because the second and third diagonal blocks are not square. On the other hand, the block matrix  $B$  is a square block matrix.

## Block Diagonal Matrices

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Let  $M = [A_{ij}]$  be a square block matrix such that the nondiagonal blocks are all zero matrices; that is,  $A_{ij} = 0$  when  $i \neq j$ . Then  $M$  is called a *block diagonal matrix*. We sometimes denote such a block diagonal matrix by writing

$$M = \text{diag}(A_{11}, A_{22}, \dots, A_{rr}) \quad \text{or} \quad M = A_{11} \oplus A_{22} \oplus \dots \oplus A_{rr}$$

The importance of block diagonal matrices is that the algebra of the block matrix is frequently reduced to the algebra of the individual blocks. Specifically, suppose  $f(x)$  is a polynomial and  $M$  is the above block diagonal matrix. Then  $f(M)$  is a block diagonal matrix, and

$$f(M) = \text{diag}(f(A_{11}), f(A_{22}), \dots, f(A_{rr}))$$

Also,  $M$  is invertible if and only if each  $A_{ii}$  is invertible, and, in such a case,  $M^{-1}$  is a block diagonal matrix, and

$$M^{-1} = \text{diag}(A_{11}^{-1}, A_{22}^{-1}, \dots, A_{rr}^{-1})$$

Analogously, a square block matrix is called a *block upper triangular matrix* if the blocks below the diagonal are zero matrices and a *block lower triangular matrix* if the blocks above the diagonal are zero matrices.

**EXAMPLE 2.17** Determine which of the following square block matrices are upper diagonal, lower diagonal, or diagonal:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 4 \\ 5 & 0 & 6 \\ 0 & 7 & 8 & 9 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 0 & 6 & 7 \end{bmatrix}$$

- (a)  $A$  is upper triangular because the block below the diagonal is a zero block.
- (b)  $B$  is lower triangular because all blocks above the diagonal are zero blocks.
- (c)  $C$  is diagonal because the blocks above and below the diagonal are zero blocks.
- (d)  $D$  is neither upper triangular nor lower triangular. Also, no other partitioning of  $D$  will make it into either a block upper triangular matrix or a block lower triangular matrix.

## SOLVED PROBLEMS

### Matrix Addition and Scalar Multiplication

**2.1.** Given  $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 5 & -6 \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & 0 & 2 \\ -7 & 1 & 8 \end{bmatrix}$ , find:

(a)  $A + B$ ,

(b)  $2A - 3B$ .

(a) Add the corresponding elements:

$$A + B = \begin{bmatrix} 1+3 & -2+0 & 3+2 \\ 4-7 & 5+1 & -6+8 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 5 \\ -3 & 6 & 2 \end{bmatrix}$$

(b) First perform the scalar multiplication and then a matrix addition:

$$2A - 3B = \begin{bmatrix} 2 & -4 & 6 \\ 8 & 10 & -12 \end{bmatrix} + \begin{bmatrix} -9 & 0 & -6 \\ 21 & -3 & -24 \end{bmatrix} = \begin{bmatrix} -7 & -4 & 0 \\ 29 & 7 & -36 \end{bmatrix}$$

(Note that we multiply  $B$  by  $-3$  and then add, rather than multiplying  $B$  by  $3$  and subtracting. This usually prevents errors.)

**2.2.** Find  $x, y, z, t$  where  $3 \begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2t \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+t & 3 \end{bmatrix}$ .

Write each side as a single equation:

$$\begin{bmatrix} 3x & 3y \\ 3z & 3t \end{bmatrix} = \begin{bmatrix} x+4 & x+y+6 \\ z+t-1 & 2t+3 \end{bmatrix}$$

Set corresponding entries equal to each other to obtain the following system of four equations:

$$3x = x + 4, \quad 3y = x + y + 6, \quad 3z = z + t - 1, \quad 3t = 2t + 3$$

$$\text{or } 2x = 4, \quad 2y = 6 + x, \quad 2z = t - 1, \quad t = 3$$

The solution is  $x = 2, y = 4, z = 1, t = 3$ .

**2.3.** Prove [Theorem 2.1](#) (i) and (v): (i)  $(A + B) + C = A + (B + C)$ , (v)  $k(A + B) = kA + kB$ . Suppose  $A = [a_{ij}]$ ,  $B = [b_{ij}]$ ,  $C = [c_{ij}]$ . The proof reduces to showing that corresponding  $ij$ -entries in each side of each matrix equation are equal. [We prove only (i) and (v), because the other parts of [Theorem 2.1](#) are proved similarly.]

- (i) The  $ij$ -entry of  $A + B$  is  $a_{ij} + b_{ij}$ ; hence, the  $ij$ -entry of  $(A + B) + C$  is  $(a_{ij} + b_{ij}) + c_{ij}$ . On the other hand, the  $ij$ -entry of  $B + C$  is  $b_{ij} + c_{ij}$ ; hence, the  $ij$ -entry of  $A + (B + C)$  is  $a_{ij} + (b_{ij} + c_{ij})$ . However, for scalars in  $K$ ,

$$(a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij})$$

Thus,  $(A + B) + C$  and  $A + (B + C)$  have identical  $ij$ -entries. Therefore,  $(A + B) + C = A + (B + C)$ .

- (v) The  $ij$ -entry of  $A + B$  is  $a_{ij} + b_{ij}$ ; hence,  $k(a_{ij} + b_{ij})$  is the  $ij$ -entry of  $k(A + B)$ . On the other hand, the  $ij$ -entries of  $kA$  and  $kB$  are  $ka_{ij}$  and  $kb_{ij}$ , respectively. Thus,  $ka_{ij} + kb_{ij}$  is the  $ij$ -entry of  $kA + kB$ . However, for scalars in  $K$ ,

$$k(a_{ij} + b_{ij}) = ka_{ij} + kb_{ij}$$

Thus,  $k(A + B)$  and  $kA + kB$  have identical  $ij$ -entries. Therefore,  $k(A + B) = kA + kB$ .

## Matrix Multiplication

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**2.4.** Calculate:

$$(a) \quad [8, -4, 5] \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}, \quad (b) \quad [6, -1, 7, 5] \begin{bmatrix} 4 \\ -9 \\ -3 \\ 2 \end{bmatrix}, \quad (c) \quad [3, 8, -2, 4] \begin{bmatrix} 5 \\ -1 \\ 6 \end{bmatrix}$$

(a) Multiply the corresponding entries and add:

$$[8, -4, 5] \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} = 8(3) + (-4)(2) + 5(-1) = 24 - 8 - 5 = 11$$

(b) Multiply the corresponding entries and add:

$$[6, -1, 7, 5] \begin{bmatrix} 4 \\ -9 \\ -3 \\ 2 \end{bmatrix} = 24 + 9 - 21 + 10 = 22$$

(c) The product is not defined when the row matrix and the column matrix have different numbers of elements.

**2.5.** Let  $(r \times s)$  denote an  $r \times s$  matrix. Find the sizes of those matrix products that are defined:

- (a)  $(2 \times 3)(3 \times 4)$ ,
- (b)  $(4 \times 1)(1 \times 2)$ ,
- (c)  $(1 \times 2)(3 \times 1)$ ,
- (d)  $(5 \times 2)(2 \times 3)$ ,
- (e)  $(4 \times 4)(3 \times 3)$
- (f)  $(2 \times 2)(2 \times 4)$

In each case, the product is defined if the inner numbers are equal, and then the product will have the size of the outer numbers in the given order.

- (a)  $2 \times 4$ ,
- (b)  $4 \times 2$ ,
- (c) not defined,
- (d)  $5 \times 3$ ,
- (e) not defined
- (f)  $2 \times 4$

**2.6.** Let  $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 0 & -4 \\ 3 & -2 & 6 \end{bmatrix}$ . Find: (a)  $AB$ , (b)  $BA$ .

- (a) Because  $A$  is a  $2 \times 2$  matrix and  $B$  a  $2 \times 3$  matrix, the product  $AB$  is defined and is a  $2 \times 3$  matrix. To obtain the entries in the first row of  $AB$ , multiply the first row  $[1, 3]$  of  $A$  by the columns  $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ ,  $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$ ,  $\begin{bmatrix} -4 \\ 6 \end{bmatrix}$  of  $B$ , respectively, as follows:

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -4 \\ 3 & -2 & 6 \end{bmatrix} = \begin{bmatrix} 2+9 & 0-6 & -4+18 \\ 4-3 & 0+2 & -8+6 \end{bmatrix} = \begin{bmatrix} 11 & -6 & 14 \\ 1 & 2 & -14 \end{bmatrix}$$

To obtain the entries in the second row of  $AB$ , multiply the second row  $[2, -1]$  of  $A$  by the columns of  $B$ :

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -4 \\ 3 & -2 & 6 \end{bmatrix} = \begin{bmatrix} 11 & -6 & 14 \\ 4-3 & 0+2 & -8+6 \end{bmatrix}$$

Thus,

$$AB = \begin{bmatrix} 11 & -6 & 14 \\ 1 & 2 & -14 \end{bmatrix}.$$

- (b) The size of  $B$  is  $2 \times 3$  and that of  $A$  is  $2 \times 2$ . The inner numbers 3 and 2 are not equal; hence, the product  $BA$  is not defined.

**2.7.** Find  $AB$ , where  $A = \begin{bmatrix} 2 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix}$  and

$$B = \begin{bmatrix} 2 & -1 & 0 & 6 \\ 1 & 3 & -5 & 1 \\ 4 & 1 & -2 & 2 \end{bmatrix}.$$

Because  $A$  is a  $2 \times 3$  matrix and  $B$  a  $3 \times 4$  matrix, the product  $AB$  is defined and is a  $2 \times 4$  matrix. Multiply the rows of  $A$  by the columns of  $B$  to obtain

$$AB = \begin{bmatrix} 4+3-4 & -2+9-1 & 0-15+2 & 12+3-2 \\ 8-2+20 & -4-6+5 & 0+10-10 & 24-2+10 \end{bmatrix} = \begin{bmatrix} 3 & 6 & -13 & 13 \\ 26 & -5 & 0 & 32 \end{bmatrix}.$$

**2.8.** Find: (a)  $\begin{bmatrix} 1 & 6 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ -7 \end{bmatrix}$ , (b)  $\begin{bmatrix} 2 \\ -7 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ -3 & 5 \end{bmatrix}$ , (c)  $[2, -7] \begin{bmatrix} 1 & 6 \\ -3 & 5 \end{bmatrix}$ .

(a) The first factor is  $2 \times 2$  and the second is  $2 \times 1$ , so the product is defined as a  $2 \times 1$  matrix:

$$\begin{bmatrix} 1 & 6 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ -7 \end{bmatrix} = \begin{bmatrix} 2 - 42 \\ -6 - 35 \end{bmatrix} = \begin{bmatrix} -40 \\ -41 \end{bmatrix}$$

(b) The product is not defined, because the first factor is  $2 \times 1$  and the second factor is  $2 \times 2$ .

(c) The first factor is  $1 \times 2$  and the second factor is  $2 \times 2$ , so the product is defined as a  $1 \times 2$  (row) matrix:

$$[2, -7] \begin{bmatrix} 1 & 6 \\ -3 & 5 \end{bmatrix} = [2 + 21, 12 - 35] = [23, -23]$$

**2.9.** Clearly,  $0A = 0$  and  $A0 = 0$ , where the 0's are *zero matrices* (with possibly different sizes). Find matrices  $A$  and  $B$  with no zero entries such that  $AB = 0$ .

Let  $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 6 & 2 \\ -3 & -1 \end{bmatrix}$ . Then  $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ .

**2.10.** Prove [Theorem 2.2\(i\)](#):  $(AB)C = A(BC)$ .

Let  $A = [a_{ij}]$ ,  $B = [b_{jk}]$ ,  $C = [c_{kl}]$ , and let  $AB = S = [s_{ik}]$ ,  $BC = T = [t_{jl}]$ . Then

$$s_{ik} = \sum_{j=1}^m a_{ij}b_{jk} \quad \text{and} \quad t_{jl} = \sum_{k=1}^n b_{jk}c_{kl}$$

Multiplying  $S = AB$  by  $C$ , the  $il$ -entry of  $(AB)C$  is

$$s_{i1}c_{1l} + s_{i2}c_{2l} + \cdots + s_{in}c_{nl} = \sum_{k=1}^n s_{ik}c_{kl} = \sum_{k=1}^n \sum_{j=1}^m (a_{ij}b_{jk})c_{kl}$$

On the other hand, multiplying  $A$  by  $T = BC$ , the  $il$ -entry of  $A(BC)$  is

$$a_{i1}t_{1l} + a_{i2}t_{2l} + \cdots + a_{im}t_{ml} = \sum_{j=1}^m a_{ij}t_{jl} = \sum_{j=1}^m \sum_{k=1}^n a_{ij}(b_{jk}c_{kl})$$

The above sums are equal; that is, corresponding elements in  $(AB)C$  and  $A(BC)$  are equal. Thus,  $(AB)C = A(BC)$ .

**2.11.** Prove [Theorem 2.2\(ii\)](#):  $A(B + C) = AB + AC$ .

Let  $A = [a_{ij}]$ ,  $B = [b_{jk}]$ ,  $C = [c_{jk}]$ , and let  $D = B + C = [d_{jk}]$ ,  $E = AB = [e_{ik}]$ ,  $F = AC = [f_{ik}]$ . Then

$$d_{jk} = b_{jk} + c_{jk}, \quad e_{ik} = \sum_{j=1}^m a_{ij}b_{jk}, \quad f_{ik} = \sum_{j=1}^m a_{ij}c_{jk}$$

Thus, the  $ik$ -entry of the matrix  $AB + AC$  is

$$e_{ik} + f_{ik} = \sum_{j=1}^m a_{ij}b_{jk} + \sum_{j=1}^m a_{ij}c_{jk} = \sum_{j=1}^m a_{ij}(b_{jk} + c_{jk})$$

On the other hand, the  $ik$ -entry of the matrix  $AD = A(B + C)$  is

$$a_{i1}d_{1k} + a_{i2}d_{2k} + \cdots + a_{im}d_{mk} = \sum_{j=1}^m a_{ij}d_{jk} = \sum_{j=1}^m a_{ij}(b_{jk} + c_{jk})$$

Thus,  $A(B + C) = AB + AC$ , because the corresponding elements are equal.

## **Transpose**

---

**2.12.** Find the transpose of each matrix:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 7 & 8 & -9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}, \quad C = [1, -3, 5, -7], \quad D = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$$

Rewrite the rows of each matrix as columns to obtain the transpose of the matrix:

$$A^T = \begin{bmatrix} 1 & 7 \\ -2 & 8 \\ 3 & -9 \end{bmatrix}, \quad B^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 \\ -3 \\ 5 \\ -7 \end{bmatrix}, \quad D^T = [2, -4, 6]$$

(Note that  $B^T = B$ ; such a matrix is said to be *symmetric*. Note also that the transpose of the row vector  $C$  is a column vector, and the transpose of the column vector  $D$  is a row vector.)

**2.13.** Prove [Theorem 2.3\(iv\)](#):  $(AB)^T = B^T A^T$ .

Let  $A = [a_{ik}]$  and  $B = [b_{kj}]$ . Then the  $ij$ -entry of  $AB$  is

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}$$

This is the  $ji$ -entry (reverse order) of  $(AB)^T$ . Now column  $j$  of  $B$  becomes row  $j$  of  $B^T$ , and row  $i$  of  $A$  becomes column  $i$  of  $A^T$ . Thus, the  $ij$ -entry of  $B^T A^T$  is

$$[b_{1j}, b_{2j}, \dots, b_{mj}][a_{i1}, a_{i2}, \dots, a_{im}]^T = b_{1j}a_{i1} + b_{2j}a_{i2} + \cdots + b_{mj}a_{im}$$

Thus,  $(AB)^T = B^T A^T$  on because the corresponding entries are equal.

## Square Matrices

---

**2.14.** Find the diagonal and trace of each matrix:

$$(a) \quad A = \begin{bmatrix} 1 & 3 & 6 \\ 2 & -5 & 8 \\ 4 & -2 & 9 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 2 & 4 & 8 \\ 3 & -7 & 9 \\ -5 & 0 & 2 \end{bmatrix}, \quad (c) \quad C = \begin{bmatrix} 1 & 2 & -3 \\ 4 & -5 & 6 \end{bmatrix}.$$

(a) The diagonal of  $A$  consists of the elements from the upper left corner of  $A$  to the lower right corner of  $A$  or, in other words, the elements  $a_{11}$ ,  $a_{22}$ ,  $a_{33}$ . Thus, the diagonal of  $A$  consists of the numbers 1,  $-5$ , and 9.

The trace of  $A$  is the sum of the diagonal elements.  
Thus,

$$\text{tr}(A) = 1 - 5 + 9 = 5$$

(b) The diagonal of  $B$  consists of the numbers 2,  $-7$ , and 2. Hence,

$$\text{tr}(B) = 2 - 7 + 2 = -3$$

(c) The diagonal and trace are only defined for square matrices.

**2.15.** Let  $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$ , and let  $f(x) = 2x^3 - 4x + 5$  and  $g(x) = x^2 + 2x + 11$ . Find

(a)  $A^2$ , (b)  $A^3$ , (c)  $f(A)$ , (d)  $g(A)$ .

$$(a) \quad A^2 = AA = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} = \begin{bmatrix} 1+8 & 2-6 \\ 4-12 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix}$$

(b)

$$A^3 = AA^2 = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} = \begin{bmatrix} 9-16 & -4+34 \\ 36+24 & -16-51 \end{bmatrix} = \begin{bmatrix} -7 & 30 \\ 60 & -67 \end{bmatrix}$$

(c) First substitute  $A$  for  $x$  and 5/ for the constant in  $f(x)$ , obtaining

$$f(A) = 2A^3 - 4A + 5I = 2 \begin{bmatrix} -7 & 30 \\ 60 & -67 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now perform the scalar multiplication and then the matrix addition:

$$f(A) = \begin{bmatrix} -14 & 60 \\ 120 & -134 \end{bmatrix} + \begin{bmatrix} -4 & -8 \\ -16 & 12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} -13 & 52 \\ 104 & -117 \end{bmatrix}$$

(d) Substitute  $A$  for  $x$  and 11/ for the constant in  $g(x)$ , and then calculate as follows:

$$\begin{aligned}
 g(A) &= A^2 + 2A - 11I = \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} + 2\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} - 11\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} + \begin{bmatrix} 2 & 4 \\ 8 & -6 \end{bmatrix} + \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

Because  $g(A)$  is the zero matrix,  $A$  is a root of the polynomial  $g(x)$ .

**2.16.** Let  $A = \begin{bmatrix} 1 & 3 \\ 4 & -3 \end{bmatrix}$ . (a) Find a nonzero column vector  $u = \begin{bmatrix} x \\ y \end{bmatrix}$  such that  $Au = 3u$ .

(b) Describe all such vectors.

(a) First set up the matrix equation  $Au = 3u$ , and then write each side as a single matrix (column vector) as follows:

$$\begin{bmatrix} 1 & 3 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 3 \begin{bmatrix} x \\ y \end{bmatrix}, \quad \text{and then} \quad \begin{bmatrix} x + 3y \\ 4x - 3y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

Set the corresponding elements equal to each other to obtain a system of equations:

$$\begin{array}{ccccc}
 x + 3y = 3x & & 2x - 3y = 0 & & \\
 4x - 3y = 3y & \text{or} & 4x - 6y = 0 & \text{or} & 2x - 3y = 0
 \end{array}$$

The system reduces to one nondegenerate linear equation in two unknowns, and so has an infinite number of solutions. To obtain a nonzero solution, let, say,  $y = 2$ ; then  $x = 3$ . Thus,  $u = (3, 2)^T$  is a desired nonzero vector.

(b) To find the general solution, set  $y = a$ , where  $a$  is a parameter. Substitute  $y = a$  into  $2x - 3y = 0$  to obtain  $x = \frac{3}{2}a$ . Thus,  $u = (\frac{3}{2}a, a)^T$  represents all such solutions.

## Invertible Matrices, Inverses

---

**2.17.** Show that  $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$  and  $B = \begin{bmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{bmatrix}$  are inverses.

Compute the product  $AB$ , obtaining

$$AB = \begin{bmatrix} -11 + 0 + 12 & 2 + 0 - 2 & 2 + 0 - 2 \\ -22 + 4 + 18 & 4 + 0 - 3 & 4 - 1 - 3 \\ -44 - 4 + 48 & 8 + 0 - 8 & 8 + 1 - 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Because  $AB = I$ , we can conclude ([Theorem 3.18](#)) that  $BA = I$ . Accordingly,  $A$  and  $B$  are inverses.

**2.18.** Find the inverse, if possible, of each matrix:

(a)  $A = \begin{bmatrix} 5 & 3 \\ 4 & 2 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 2 & -3 \\ 1 & 3 \end{bmatrix}$ , (c)  $\begin{bmatrix} -2 & 6 \\ 3 & -9 \end{bmatrix}$ .

Use the formula for the inverse of a  $2 \times 2$  matrix appearing in Section 2.9.

(a) First find  $|A| = 5(2) - 3(4) = 10 - 12 = -2$ . Next interchange the diagonal elements, take the negatives of the nondiagonal elements, and multiply by  $1/|A|$ :

$$A^{-1} = -\frac{1}{2} \begin{bmatrix} 2 & -3 \\ -4 & 5 \end{bmatrix} = \begin{bmatrix} -1 & \frac{3}{2} \\ 2 & -\frac{5}{2} \end{bmatrix}$$

(b) First find  $|B| = 2(3) - (-3)(1) = 6 + 3 = 9$ . Next interchange the diagonal elements, take the negatives of the nondiagonal elements, and multiply by  $1/|B|$ :

$$B^{-1} = \frac{1}{9} \begin{bmatrix} 3 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{9} & \frac{2}{9} \end{bmatrix}$$

(c) First find  $[C] = -2(-9) - 6(3) = 18 - 18 = 0$ .  
Because  $[C] = 0$ ,  $C$  has no inverse.

**2.19.** Let  $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$ . Find  $A^{-1} = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$ .

Multiplying  $A$  by  $A^{-1}$  and setting the nine entries equal to the nine entries of the identity matrix  $I$  yields the following three systems of three equations in three of the unknowns:

$$\begin{array}{lll} x_1 + y_1 + z_1 = 1 & x_2 + y_2 + z_2 = 0 & x_3 + y_3 + z_3 = 0 \\ y_1 + 2z_1 = 0 & y_2 + 2z_2 = 1 & y_3 + 2z_3 = 0 \\ x_1 + 2y_1 + 4z_1 = 0 & x_2 + 2y_2 + 4z_2 = 0 & x_3 + 2y_3 + 4z_3 = 1 \end{array}$$

[Note that  $A$  is the coefficient matrix for all three systems.] Solving the three systems for the nine unknowns yields

$$x_1 = 0, \quad y_1 = 2, \quad z_1 = -1; \quad x_2 = -2, \quad y_2 = 3, \quad z_2 = -1; \quad x_3 = 1, \quad y_3 = -2, \quad z_3 = 1$$

Thus,

$$A^{-1} = \begin{bmatrix} 0 & -2 & 1 \\ 2 & 3 & -2 \\ -1 & -1 & 1 \end{bmatrix}$$

(**Remark:** [Chapter 3](#) gives an efficient way to solve the three systems.)

- 2.20.** Let  $A$  and  $B$  be invertible matrices (with the same size). Show that  $AB$  is also invertible and  $(AB)^{-1} = B^{-1}A^{-1}$ . [Thus, by induction,  $(A_1A_2 \dots A_m)^{-1} = A_m^{-1} \dots A_2^{-1}A_1^{-1}$ .]

Using the associativity of matrix multiplication, we get

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Thus,  $(AB)^{-1} = B^{-1}A^{-1}$ .

## **Diagonal and Triangular Matrices**

- 2.21.** Write out the diagonal matrices  $A = \text{diag}(4, -3, 7)$ ,  $B = \text{diag}(2, -6)$ ,  $C = \text{diag}(3, -8, 0, 5)$ .

Put the given scalars on the diagonal and 0's elsewhere:

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & -6 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & & & \\ & -8 & & \\ & & 0 & \\ & & & 5 \end{bmatrix}$$

- 2.22.** Let  $A = \text{diag}(2, 3, 5)$  and  $B = \text{diag}(7, 0, -4)$ . Find

(a)  $AB$ ,  $A^2$ ,  $B^2$ ;

(b)  $f(A)$ , where  $f(x) = x^2 + 3x - 2$ ;

(c)  $A^{-1}$  and  $B^{-1}$ .

(a) The product matrix  $AB$  is a diagonal matrix obtained by multiplying corresponding diagonal entries; hence,

$$AB = \text{diag}(2(7), 3(0), 5(-4)) = \text{diag}(14, 0, -20)$$

Thus, the squares  $A^2$  and  $B^2$  are obtained by squaring each diagonal entry; hence,

$$A^2 = \text{diag}(2^2, 3^2, 5^2) = \text{diag}(4, 9, 25) \quad \text{and} \quad B^2 = \text{diag}(49, 0, 16)$$

(b)  $f(A)$  is a diagonal matrix obtained by evaluating  $f(x)$  at each diagonal entry. We have

$$f(2) = 4 + 6 \times 2 = 8, \quad f(3) = 9 + 9 \times 2 = 16; \quad f(5) = 25 + 15 \times 2 = 38$$

Thus,  $f(A) = \text{diag}(8, 16, 38)$ .

(c) The inverse of a diagonal matrix is a diagonal matrix obtained by taking the inverse (reciprocal) of each diagonal entry. Thus,  $A^{-1} = \text{diag}(\frac{1}{2}, \frac{1}{3}, \frac{1}{5})$  but  $B$  has no inverse because there is a 0 on the diagonal.

**2.23.** Find a  $2 \times 2$  matrix  $A$  such that  $A^2$  is diagonal but not  $A$ .

Let  $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ . Then  $A^2 = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$ , which is diagonal.

**2.24.** Find an upper triangular matrix  $A$  such that

$$A^3 = \begin{bmatrix} 8 & -57 \\ 0 & 27 \end{bmatrix}.$$

Set  $A = \begin{bmatrix} x & y \\ 0 & z \end{bmatrix}$ . Then  $x^3 = 8$ , so  $x = 2$ ; and  $z^3 = 27$ , so  $z = 3$ . Next calculate  $A^3$  using  $x = 2$  and  $y = 3$ :

$$A^2 = \begin{bmatrix} 2 & y \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & y \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 5y \\ 0 & 9 \end{bmatrix} \quad \text{and} \quad A^3 = \begin{bmatrix} 2 & y \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 & 5y \\ 0 & 9 \end{bmatrix} = \begin{bmatrix} 8 & 19y \\ 0 & 27 \end{bmatrix}$$

Thus,  $19y = 57$ , or  $y = 3$ . Accordingly,  $A = \begin{bmatrix} 2 & -3 \\ 0 & 3 \end{bmatrix}$ .

**2.25.** Let  $A = [a_{ij}]$  and  $B = [b_{ij}]$  be upper triangular matrices. Prove that  $AB$  is upper triangular with diagonal  $a_{11}b_{11}$ ,  $a_{22}b_{22}$ ,  $\dots$ ;  $a_{nn}b_{nn}$ .

Let  $AB = [c_{ij}]$ . Then  $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$  and  $c_{ii} = \sum_{k=1}^n a_{ik}b_{ki}$ . Suppose  $i > j$ . Then, for any  $k$ , either  $i > k$  or  $k > j$ , so that either  $a_{ik} = 0$  or  $b_{kj} = 0$ . Thus,  $c_{ij} = 0$ , and  $AB$  is upper triangular. Suppose  $i = j$ . Then, for  $k < i$ , we have  $a_{ik} = 0$ ; and, for  $k > i$ , we have  $b_{ki} = 0$ . Hence,  $c_{ii} = a_{ii}b_{ii}$ , as claimed. [This proves one part of [Theorem 2.5\(i\)](#); the statements for  $A + B$  and  $kA$  are left as exercises.]

## Special Real Matrices: Symmetric and Orthogonal

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**2.26.** Determine whether or not each of the following matrices is *symmetric*—that is,  $A^T = A$ —or *skew-symmetric*—that is,  $A^T = -A$ :

$$(a) \quad A = \begin{bmatrix} 5 & -7 & 1 \\ -7 & 8 & 2 \\ 1 & 2 & -4 \end{bmatrix}, \quad (b) \quad B = \begin{bmatrix} 0 & 4 & -3 \\ -4 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}, \quad (c) \quad C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- (a) By inspection, the symmetric elements (mirror images in the diagonal) are  $-7$  and  $-7$ ,  $1$  and  $1$ ,  $2$  and  $2$ . Thus,  $A$  is symmetric, because symmetric elements are equal.
- (b) By inspection, the diagonal elements are all  $0$ , and the symmetric elements,  $4$  and  $-4$ ,  $-3$  and  $3$ , and  $5$  and  $-5$ , are negatives of each other. Hence,  $B$  is skew-symmetric.
- (c) Because  $C$  is not square,  $C$  is neither symmetric nor skew-symmetric.

**2.27.** Suppose  $B = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$  is symmetric. Find  $x$  and  $B$ .

Set the symmetric elements  $x+2$  and  $2x-3$  equal to each other, obtaining  $2x-3 = x+2$  or  $x = 5$ . Hence,

$$B = \begin{bmatrix} 4 & 7 \\ 7 & 6 \end{bmatrix}.$$

**2.28.** Let  $A$  be an arbitrary  $2 \times 2$  (real) orthogonal matrix.

(a) Prove: If  $(a, b)$  is the first row of  $A$ , then  $a^2 + b^2 = 1$  and

$$A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}.$$

(b) Prove [Theorem 2.7](#): For some real number  $\theta$ ,

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$$

(a) Suppose  $(x, y)$  is the second row of  $A$ . Because the rows of  $A$  form an orthonormal set, we get

$$a^2 + b^2 = 1, \quad x^2 + y^2 = 1, \quad ax + by = 0$$

Similarly, the columns form an orthogonal set, so

$$a^2 + x^2 = 1, \quad b^2 + y^2 = 1, \quad ab + xy = 0$$

Therefore,  $x^2 = 1 - a^2 = b^2$ , whence  $x = \pm b$ :

Case (i):  $x = b$ . Then  $b(a + y) = 0$ , so  $y = -a$ .

Case (ii):  $x = -b$ . Then  $b(y - a) = 0$ , so  $y = a$ .

This means, as claimed,

$$A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$$

(b) Because  $a^2 + b^2 = 1$ , we have  $-1 \leq a \leq 1$ . Let  $a = \cos \theta$ . Then  $b^2 = 1 - \cos^2 \theta$ , so  $b = \sin \theta$ . This proves the theorem.

**2.29.** Find a  $2 \times 2$  orthogonal matrix  $A$  whose first row is a (positive) multiple of  $(3, 4)$ . Normalize  $(3, 4)$  to get  $(\frac{3}{5}, \frac{4}{5})$ . Then, by [Problem 2.28](#),

$$A = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix}.$$

**2.30.** Find a  $3 \times 3$  orthogonal matrix  $P$  whose first two rows are multiples of  $u_1 = (1, 1, 1)$  and  $u_2 = (0, -1, 1)$ , respectively. (Note that, as required,  $u_1$  and  $u_2$  are orthogonal.)

First find a nonzero vector  $u_3$  orthogonal to  $u_1$  and  $u_2$ ; say (cross product)  $u_3 = u_1 \times u_2 = (2, -1, -1)$ . Let  $A$  be the matrix whose rows are  $u_1, u_2, u_3$ ; and let  $P$  be the matrix obtained from  $A$  by normalizing the rows of  $A$ . Thus,

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & -1 & -1 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 2/\sqrt{6} & -1/\sqrt{6} & -1/\sqrt{6} \end{bmatrix}$$

## Complex Matrices: Hermitian and Unitary Matrices

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**2.31.** Find  $A^H$  where (a)  $A = \begin{bmatrix} 3-5i & 2+4i \\ 6+7i & 1+8i \end{bmatrix}$ , (b)

$$A = \begin{bmatrix} 2-3i & 5+8i \\ -4 & 3-7i \\ -6-i & 5i \end{bmatrix}$$

Recall that  $A^H = \bar{A}^T$ , the conjugate transpose of  $A$ . Thus,

$$(a) \quad A^H = \begin{bmatrix} 3+5i & 6-7i \\ 2-4i & 1-8i \end{bmatrix}, \quad (b) \quad A^H = \begin{bmatrix} 2+3i & -4 & -6+i \\ 5-8i & 3+7i & -5i \end{bmatrix}$$

**2.32.** Show that  $A = \begin{bmatrix} \frac{1}{3} - \frac{2}{3}i & \frac{2}{3}i \\ -\frac{2}{3}i & -\frac{1}{3} - \frac{2}{3}i \end{bmatrix}$  is unitary.

The rows of  $A$  form an orthonormal set:

$$\begin{aligned}\left(\frac{1}{3} - \frac{2}{3}i, \frac{2}{3}i\right) \cdot \left(\frac{1}{3} - \frac{2}{3}i, \frac{2}{3}i\right) &= \left(\frac{1}{9} + \frac{4}{9}\right) + \frac{4}{9} = 1 \\ \left(\frac{1}{3} - \frac{2}{3}i, \frac{2}{3}i\right) \cdot \left(-\frac{2}{3}i, -\frac{1}{3} - \frac{2}{3}i\right) &= \left(\frac{2}{9}i + \frac{4}{9}\right) + \left(-\frac{2}{9}i - \frac{4}{9}\right) = 0 \\ \left(-\frac{2}{3}i, -\frac{1}{3} - \frac{2}{3}i\right) \cdot \left(-\frac{2}{3}i, -\frac{1}{3} - \frac{2}{3}i\right) &= \frac{4}{9} + \left(\frac{1}{9} + \frac{4}{9}\right) = 1\end{aligned}$$

Thus,  $A$  is unitary.

**2.33.** Prove the complex analogue of [Theorem 2.6](#): Let  $A$  be a complex matrix. Then the following are equivalent: (i)  $A$  is unitary. (ii) The rows of  $A$  form an orthonormal set. (iii) The columns of  $A$  form an orthonormal set.

(The proof is almost identical to the proof on page 37 for the case when  $A$  is a  $3 \times 3$  real matrix.) First recall that the vectors  $u_1, u_2, \dots, u_n$  in  $\mathbf{C}^n$  form an orthonormal set if they are unit vectors and are orthogonal to each other, where the dot product in  $\mathbf{C}^n$  is defined by

$$(a_1, a_2, \dots, a_n) \cdot (b_1, b_2, \dots, b_n) = a_1 \bar{b}_1 + a_2 \bar{b}_2 + \dots + a_n \bar{b}_n$$

Suppose  $A$  is unitary, and  $R_1, R_2, \dots, R_n$  are its rows.

Then  $\bar{R}_1^T, \bar{R}_2^T, \dots, \bar{R}_n^T$  are the columns of  $A^H$ . Let  $AA^H = [c_{ij}]$ .

By matrix multiplication,  $c_{ij} = R_i \bar{R}_j^T = R_i \cdot R_j$ . Because  $A$  is unitary, we have  $AA^H = I$ . Multiplying  $A$  by  $A^H$  and setting each entry  $c_{ij}$  equal to the corresponding entry in  $I$  yields the following  $n^2$  equations:

$$R_1 \cdot R_1 = 1, \quad R_2 \cdot R_2 = 1, \quad \dots, \quad R_n \cdot R_n = 1, \quad \text{and} \quad R_i \cdot R_j = 0, \quad \text{for } i \neq j$$

Thus, the rows of  $A$  are unit vectors and are orthogonal to each other; hence, they form an orthonormal set of vectors. The condition  $A^T A = I$  similarly shows that the

columns of  $A$  also form an orthonormal set of vectors. Furthermore, because each step is reversible, the converse is true. This proves the theorem.

## Block Matrices

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**2.34.** Consider the following block matrices (which are partitions of the same matrix):

$$(a) \left[ \begin{array}{cc|cc|c} 1 & -2 & 0 & 1 & 3 \\ 2 & -3 & 5 & -7 & -2 \\ 3 & 1 & 4 & 5 & 9 \end{array} \right], \quad (b) \left[ \begin{array}{cc|cc|c} 1 & -2 & 0 & 1 & 3 \\ 2 & -3 & 5 & -7 & -2 \\ 3 & 1 & 4 & 5 & 9 \end{array} \right]$$

Find the size of each block matrix and also the size of each block.

- (a) The block matrix has two rows of matrices and three columns of matrices; hence, its size is  $2 \times 3$ . The block sizes are  $2 \times 2$ ,  $2 \times 2$ , and  $2 \times 1$  for the first row; and  $1 \times 2$ ,  $1 \times 2$ , and  $1 \times 1$  for the second row.
- (b) The size of the block matrix is  $3 \times 2$ ; and the block sizes are  $1 \times 3$  and  $1 \times 2$  for each of the three rows.

**2.35.** Compute  $AB$  using block multiplication, where

$$A = \left[ \begin{array}{cc|c} 1 & 2 & 1 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{array} \right] \quad \text{and} \quad B = \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

Here  $A = \begin{bmatrix} E & F \\ 0_{1 \times 2} & G \end{bmatrix}$  and  $B = \begin{bmatrix} R & S \\ 0_{1 \times 3} & T \end{bmatrix}$ , where  $E, F, G, R, S, T$  are the given blocks, and  $0_{1 \times 2}$  and  $0_{1 \times 3}$  are zero matrices of the indicated sizes. Hence,

$$AB = \begin{bmatrix} ER & ES + FT \\ 0_{1 \times 3} & GT \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 9 & 12 & 15 \\ 19 & 26 & 33 \\ 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 3 \\ 7 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} & 2 \end{bmatrix} = \begin{bmatrix} 9 & 12 & 15 & 4 \\ 19 & 26 & 33 & 7 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

**2.36.** Let  $M = \text{diag}(A, B, C)$ , where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = [5], C = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}. \text{ Find } M^2.$$

Because  $M$  is block diagonal, square each block:

$$A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}, \quad B^2 = [25], \quad C^2 = \begin{bmatrix} 16 & 24 \\ 40 & 64 \end{bmatrix},$$

so

$$M^2 = \begin{bmatrix} 7 & 10 & & & \\ & 22 & & & \\ & 15 & 25 & & \\ & & & 16 & 24 \\ & & & 40 & 64 \end{bmatrix}$$

## Miscellaneous Problem

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**2.37.** Let  $f(x)$  and  $g(x)$  be polynomials and let  $A$  be a square matrix. Prove

(a)  $(f + g)(A) = f(A) + g(A),$

(b)  $(f \cdot g)(A) = f(A)g(A),$

(c)  $f(A)g(A) = g(A)f(A).$

Suppose  $f(x) = \sum_{i=1}^r a_i x^i$  and  $g(x) = \sum_{j=1}^s b_j x^j$ .

(a) We can assume  $r = s = n$  by adding powers of  $x$  with 0 as their coefficients. Then

$$f(x) + g(x) = \sum_{i=1}^n (a_i + b_i) x^i$$

Hence,

$$(f + g)(A) = \sum_{i=1}^n (a_i + b_i) A^i = \sum_{i=1}^n a_i A^i + \sum_{i=1}^n b_i A^i = f(A) + g(A)$$

(b) We have  $f(x)g(x) = \sum_{ij} a_i b_j x^{i+j}$ . Then

$$f(A)g(A) = \left( \sum_i a_i A^i \right) \left( \sum_j b_j A^j \right) = \sum_{ij} a_i b_j A^{i+j} = (fg)(A)$$

(c) Using  $f(x)g(x) = g(x)f(x)$ , we have

$$f(A)g(A) = (fg)(A) = (gf)(A) = g(A)f(A)$$

## SUPPLEMENTARY PROBLEMS

### Algebra of Matrices

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Problems 2.38–2.41 refer to the following matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 0 \\ -6 & 7 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -3 & 4 \\ 2 & 6 & -5 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 7 & -1 \\ 4 & -8 & 9 \end{bmatrix}$$

**2.38.** Find (a)  $5A - 2B$ , (b)  $2A + 3B$ , (c)  $2C - 3D$ .

**2.39.** Find (a)  $AB$  and  $(AB)C$ , (b)  $BC$  and  $A(BC)$ . [Note that  $(AB)C = A(BC)$ .]

**2.40.** Find (a)  $A^2$  and  $A^3$ , (b)  $AD$  and  $BD$ , (c)  $CD$ .

**2.41.** Find (a)  $A^T$ , (b)  $B^T$ , (d)  $A^T B^T$ . [Note that  $A^T B^T \neq (AB)^T$ .]

Problems 2.42 and 2.43 refer to the following matrices:

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 0 & -3 \\ -1 & -2 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -3 & 0 & 1 \\ 5 & -1 & -4 & 2 \\ -1 & 0 & 0 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

**2.42.** Find (a)  $3A - 4B$ , (b)  $AC$ , (c)  $BC$ , (d)  $AD$ , (e)  $BD$ , (f)  $CD$ .

**2.43.** Find (a)  $A^T$ , (b)  $A^T B$ , (c)  $A^T C$ .

- 2.44.** Let  $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ . Find a  $2 \times 3$  matrix  $B$  with distinct nonzero entries such that  $AB = 0$ .
- 2.45.** Let  $e_1 = [1, 0, 0]$ ,  $e_2 = [0, 1, 0]$ ,  $e_3 = [0, 0, 1]$ , and  $A = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \end{bmatrix}$ . Find  $e_1A$ ,  $e_2A$ ,  $e_3A$ .
- 2.46.** Let  $e_i = [0, \dots, 0, 1, 0, \dots, 0]$ , where 1 is the  $i$ th entry. Show
- (a)  $e_iA = A_i$ ,  $i$ th row of  $A$ .
  - (c) If  $e_iA = e_iB$ , for each  $i$ , then  $A = B$ .
  - (b)  $Be_j^T = B^j$ ,  $j$ th column of  $B$ .
  - (d) If  $Ae_j^T = Be_j^T$ , for each  $j$ , then  $A = B$ .
- 2.47.** Prove [Theorem 2.2](#)(iii) and (iv): (iii)  $(B + C)A = BA + CA$ , (iv)  $k(AB) = (kA)B = A(kB)$ .
- 2.48.** Prove [Theorem 2.3](#):(i)  $(A + B)^T = A^T + B^T$ , (ii)  $(A^T)^T = A$ , (iii)  $(kA)^T = kA^T$ .
- 2.49.** Show (a) If  $A$  has a zero row, then  $AB$  has a zero row. (b) If  $B$  has a zero column, then  $AB$  has a zero column.

## **Square Matrices, Inverses**

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- 2.50.** Find the diagonal and trace of each of the following matrices:

$$(a) \ A = \begin{bmatrix} 2 & -5 & 8 \\ 3 & -6 & -7 \\ 4 & 0 & -1 \end{bmatrix}, \quad (b) \ B = \begin{bmatrix} 1 & 3 & -4 \\ 6 & 1 & 7 \\ 2 & -5 & -1 \end{bmatrix}, \quad (c) \ C = \begin{bmatrix} 4 & 3 & -6 \\ 2 & -5 & 0 \end{bmatrix}$$

Problems 2.51–2.53 refer to

$$A = \begin{bmatrix} 2 & -5 \\ 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 4 & -2 \\ 1 & -6 \end{bmatrix}, C = \begin{bmatrix} 6 & -4 \\ 3 & -2 \end{bmatrix}.$$

**2.51.** Find (a)  $A^2$  and  $A^3$ , (b)  $f(A)$  and  $g(A)$ , where

$$f(x) = x^3 - 2x^2 - 5, \quad g(x) = x^2 - 3x + 17.$$

**2.52.** Find (a)  $B^2$  and  $B^3$ , (b)  $f(B)$  and  $g(B)$ , where

$$f(x) = x^2 + 2x - 22, \quad g(x) = x^2 - 3x - 6.$$

**2.53.** Find a nonzero column vector  $u$  such that  $Cu = 4u$ .

**2.54.** Find the inverse of each of the following matrices (if it exists):

$$A = \begin{bmatrix} 7 & 4 \\ 5 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & -6 \\ -2 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & -2 \\ 6 & -3 \end{bmatrix}$$

**2.55.** Find the inverses of  $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \\ 1 & 3 & 7 \end{bmatrix}$  and

$$B = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 1 & 3 & -2 \end{bmatrix}. \text{ [Hint: See Problem 2.19.]}$$

**2.56.** Suppose  $A$  is invertible. Show that if  $AB = AC$ , then  $B = C$ . Give an example of a nonzero matrix  $A$  such that  $AB = AC$  but  $B \neq C$ .

**2.57.** Find  $2 \times 2$  invertible matrices  $A$  and  $B$  such that  $A + B \neq 0$  and  $A + B$  is not invertible.

**2.58.** Show (a)  $A$  is invertible if and only if  $A^T$  is invertible. (b) The operations of inversion and transpose commute;

that is,  $(A^T)^{-1} = (A^{-1})^T$ . (c) If  $A$  has a zero row or zero column, then  $A$  is not invertible.

## Diagonal and triangular matrices

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**2.59.** Let  $A = \text{diag}(1, 2, -3)$  and  $B = \text{diag}(2, -5, 0)$ . Find (a)  $AB$ ,  $A^2$ ,  $B^2$ ;

(b)  $f(A)$ , where  $f(x) = x^2 + 4x - 3$ ; (c)  $A^{-1}$  and  $B^{-1}$ .

**2.60.** Let  $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ . (a) Find  $A^n$ . (b) Find  $B^n$ .

**2.61.** Find all real triangular matrices  $A$  such that  $A^2 = B$ , where (a)  $B = \begin{bmatrix} 4 & 21 \\ 0 & 25 \end{bmatrix}$ , (b)  $B = \begin{bmatrix} 1 & 4 \\ 0 & -9 \end{bmatrix}$ .

**2.62.** Let  $A = \begin{bmatrix} 5 & 2 \\ 0 & k \end{bmatrix}$ . Find all numbers  $k$  for which  $A$  is a root of the polynomial:

(a)  $f(x) = x^2 - 7x + 10$ , (b)  $g(x) = x^2 - 25$ , (c)  $h(x) = x^2 - 4$ .

**2.63.** Let  $B = \begin{bmatrix} 1 & 0 \\ 26 & 27 \end{bmatrix}$ . Find a matrix  $A$  such that  $A^3 = B$ .

**2.64.** Let  $B = \begin{bmatrix} 1 & 8 & 5 \\ 0 & 9 & 5 \\ 0 & 0 & 4 \end{bmatrix}$ . Find a triangular matrix  $A$  with positive diagonal entries such that  $A^2 = B$ .

**2.65.** Using only the elements 0 and 1, find the number of  $3 \times 3$  matrices that are (a) diagonal, (b) upper triangular,

(c) nonsingular and upper triangular. Generalize to  $n \times n$  matrices.

**2.66.** Let  $D_k = kI$ , the scalar matrix belonging to the scalar  $k$ . Show

(a)  $D_k A = kA$ , (b)  $BD_k = kB$ , (c)  $D_k + D_{k'} = D_{k+k'}$ , (d)  $D_k D_{k'} = D_{kk'}$

**2.67.** Suppose  $AB = C$ , where  $A$  and  $C$  are upper triangular.

(a) Find  $2 \times 2$  nonzero matrices  $A, B, C$ , where  $B$  is not upper triangular.

(b) Suppose  $A$  is also invertible. Show that  $B$  must also be upper triangular.

### **Special Types of Real Matrices**

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**2.68.** Find  $x, y, z$  such that  $A$  is symmetric, where

(a)  $A = \begin{bmatrix} 2 & x & 3 \\ 4 & 5 & y \\ z & 1 & 7 \end{bmatrix}$ , (b)  $A = \begin{bmatrix} 7 & -6 & 2x \\ y & z & -2 \\ x & -2 & 5 \end{bmatrix}$ .

**2.69.** Suppose  $A$  is a square matrix. Show (a)  $A + A^T$  is symmetric, (b)  $A - A^T$  is skew-symmetric, (c)  $A = B + C$ , where  $B$  is symmetric and  $C$  is skew-symmetric.

**2.70.** Write  $A = \begin{bmatrix} 4 & 5 \\ 1 & 3 \end{bmatrix}$  as the sum of a symmetric matrix  $B$  and a skew-symmetric matrix  $C$ .

**2.71.** Suppose  $A$  and  $B$  are symmetric. Show that the following are also symmetric:

(a)  $A + B$ ;

(b)  $kA$ , for any scalar  $k$ ;

- (c)  $A^2$ ;
- (d)  $A^n$ , for  $n > 0$ ;
- (e)  $f(A)$ , for any polynomial  $f(x)$ .

**2.72.** Find a  $2 \times 2$  orthogonal matrix  $P$  whose first row is a multiple of (a)  $(3, -4)$ , (b)  $(1, 2)$ .

**2.73.** Find a  $3 \times 3$  orthogonal matrix  $P$  whose first two rows are multiples of

- (a)  $(1, 2, 3)$  and  $(0, -2, 3)$ , (b)  $(1, 3, 1)$  and  $(1, 0, -1)$ .

**2.74.** Suppose  $A$  and  $B$  are orthogonal matrices. Show that  $A^T$ ,  $A^{-1}$ ,  $AB$  are also orthogonal.

**2.75.** Which of the following matrices are normal?

$$A = \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

## Complex Matrices

---

**2.76.** Find real numbers  $x, y, z$  such that  $A$  is Hermitian,

$$\text{where } A = \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+zi \\ yi & 1-xi & -1 \end{bmatrix}.$$

**2.77.** Suppose  $A$  is a complex matrix. Show that  $AA^H$  and  $A^HA$  are Hermitian.

**2.78.** Let  $A$  be a square matrix. Show that (a)  $A + A^H$  is Hermitian, (b)  $A - A^H$  is skew-Hermitian, (c)  $A = B + C$ , where  $B$  is Hermitian and  $C$  is skew-Hermitian.

**2.79.** Determine which of the following matrices are unitary:

$$A = \begin{bmatrix} i/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -i/2 \end{bmatrix}, \quad B = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}, \quad C = \frac{1}{2} \begin{bmatrix} 1 & -i & -1+i \\ i & 1 & 1+i \\ 1+i & -1+i & 0 \end{bmatrix}$$

**2.80.** Suppose  $A$  and  $B$  are unitary. Show that  $A^H$ ,  $A^{-1}$ ,  $AB$  are unitary.

**2.81.** Determine which of the following matrices are normal:

$$A = \begin{bmatrix} 3+4i & 1 \\ i & 2+3i \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 \\ 1-i & i \end{bmatrix}.$$

## Block Matrices

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$$\mathbf{2.82.} \text{ Let } U = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 \\ 0 & 0 & 5 & 1 & 2 \\ 0 & 0 & 3 & 4 & 1 \end{bmatrix} \text{ and } V = \begin{bmatrix} 3 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & -4 & 1 \end{bmatrix}.$$

(a) Find  $UV$  using block multiplication.

(b) Are  $U$  and  $V$  block diagonal matrices?

(c) Is  $UV$  block diagonal?

**2.83.** Partition each of the following matrices so that it becomes a square block matrix with as many diagonal blocks as possible:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

**2.84.** Find  $M^2$  and  $M^3$  for (a)  $M = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ , (b)

$$M = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 4 & 5 \end{bmatrix}.$$

**2.85.** For each matrix  $M$  in Problem 2.84, find  $f(M)$  where  $f(x) = x^2 + 4x - 5$ .

**2.86.** Suppose  $U = [U_{ik}]$  and  $V = [V_{kj}]$  are block matrices for which  $UV$  is defined and the number of columns of each block  $U_{ik}$  is equal to the number of rows of each block  $V_{kj}$ . Show that  $UV = [W_{ij}]$ , where  $W_{ij} = \sum_k U_{ik} V_{kj}$ .

**2.87.** Suppose  $M$  and  $N$  are block diagonal matrices where corresponding blocks have the same size, say  $M = \text{diag}(A_i)$  and  $N = \text{diag}(B_i)$ . Show

(i)  $M + N = \text{diag}(A_i + B_i)$ ,

(ii)  $kM = \text{diag}(kA_i)$ ,

(iii)  $MN = \text{diag}(A_i B_i)$ ,

(iv)  $f(M) = \text{diag}(f(A_i))$  for any polynomial  $f(x)$ .

## ANSWERS TO SUPPLEMENTARY PROBLEMS

**Notation:**  $A = [R_1; R_2; \dots]$  denotes a matrix  $A$  with rows  $R_1, R_2, \dots$

**2.38.** (a)  $[5, 10; 27, -34]$ , (b)  $[17, 4; -12, 13]$ , (c)  $[7, -27, 11; -8, 36, -37]$

- 2.39.** (a)  $[7, 14; 39, -28], [21, 105, -98; -17, -285, 296]$   
 (b)  $[5, -15, 20; 8, 60, -59], [21, 105, -98; -17, -285, 296]$
- 2.40.** (a)  $[7, -6; -9, 22], [11, 38; 57, -106 \times;$   
 (b)  $[11, -9, 17; -7, 53, -39], [15, 35, -5; 10, -98, 69];$   
 (c) not defined
- 2.41.** (a)  $[1, 3; 2, -4],$  (b)  $[5, -6; 0, 7],$  (c)  $[7, 39; 14, -28],$   
 (d)  $[5, 15; 10, -40]$
- 2.42.** (a)  $[13, -3, 18; 4, 17, 0],$  (b)  $[5, -2, 4, 5; 11, -3, -12, 18],$  (c)  $[11, -12, 0, -5; -15, 5, 8, 4],$  (d)  $[9; 9],$  (e)  $[1; 9],$  (f) not defined
- 2.43.** (a)  $[1, 0; -1, 3; 2, 4],$  (b)  $[4, 0, -3; -7, -6, 12; 4, -8, 6],$  (c) not defined
- 2.44.**  $[2, 4, 6; -1, -2, -3-]$
- 2.45.**  $[a_1, a_2, a_3, a_4], [b_1, b_2, b_3, b_4], [c_1, c_2, c_3, c_4]$
- 2.50.** (a)  $2, -6, -1, \text{tr}(A) = 5,$  (b)  $1, 1, -1, \text{tr}(B) = 1,$  (c) not defined
- 2.51.** (a)  $[11, -15; 9, -14], [67, 40; -24, -59],$  (b)  $[50, 70; -42, -36], g(A) = 0$
- 2.52.** (a)  $[14, 4; -2, 34], [60, -52; 26, -200],$  (b)  $f(B) = 0, [4, 10; -5, 46]$
- 2.53.**  $u = [2a, a]^T$
- 2.54.**  $[3, -4; -5, 7], [-\frac{5}{2}, \frac{3}{2}; 2, -1],$  not defined,  $[1, -\frac{2}{3}; 2, -\frac{5}{3}]$
- 2.55.**  $[1, 1, -1; 2, -5, 3; -1, 2, -1], [1, 1, 0; -1, -3, 1; 1, -4, 1]$
- 2.56.**  $A = [1, 2; 1, 2], B = [0, 0; 1, 1], C = [2, 2; 0, 0]$

**2.57.**  $A = [1, 2; 0, 3]; B = [4, 3; 3, 0]$

**2.58.** (c) *Hint:* Use [Problem 2.48](#)

**2.59.** (a)  $AB = \text{diag}(2, -10, 0)$ ,  $A^2 = \text{diag}(1, 4, 9)$ ,  $B^2 = \text{diag}(4, 25, 0)$ ;

(b)  $f(A) = \text{diag}(2, 9, -6)$ ;

(c)  $A^{-1} = \text{diag}(1, \frac{1}{2}, -\frac{1}{3})$ ,  $B^{-1}$  does not exist

**2.60.** (a)  $[1, 2n; 0, 1]$ , (b)  $[1, n, \frac{1}{2}n(n-1); 0, 1, n; 0, 0, 1]$

**2.61.** (a)  $[2, 3; 0, 5]$ ,  $[2, -3; 0, -5]$ ,  $[2, -7; 0, -5]$ ,  $[2, 7; 0, 5]$ , (b) none

**2.62.** (a)  $k = 2$ , (b)  $k = -5$ , (c) none

**2.63.**  $[1, 0; 2, 3]$

**2.64.**  $[1, 2, 1; 0, 3, 1; 0, 0, 2]$

**2.65.** All entries below the diagonal must be 0 to be upper triangular, and all diagonal entries must be 1 to be nonsingular. (a)  $8(2^n)$ , (b)  $2^6(2^{n(n+1)/2})$ , (c)  $2^3(2^{n(n-1)/2})$

**2.67.** (a)  $A = [1, 1; 0, 0]$ ,  $B = [1, 2; 3, 4]$ ,  $C = [4, 6; 0, 0]$

**2.68.** (a)  $x = 4$ ,  $y = 1$ ,  $z = 3$ ; (b)  $x = 0$ ,  $y = 6$ ,  $z$  any real number

**2.69.** (c) *Hint:* Let  $B = \frac{1}{2}(A + A^T)$  and  $C = \frac{1}{2}(A - A^T)$ .

**2.70.**  $B = [4, 3; 3, 3]$ ,  $C = [0, 2; 2, 0]$

**2.72.** (a)  $[\frac{3}{5}, -\frac{4}{5}; \frac{4}{5}, \frac{3}{5}]$ , (b)  $[1/\sqrt{5}, 2/\sqrt{5}; 2/\sqrt{5}, -1/\sqrt{5}]$

**2.73.**

(a)  $[1/\sqrt{14}, 2/\sqrt{14}, 3/\sqrt{14}; 0, -2/\sqrt{13}, 3/\sqrt{13}; 12/\sqrt{157}, -3/\sqrt{157}, -2/\sqrt{157}]$

(b)  $[1/\sqrt{11}, 3/\sqrt{11}, 1/\sqrt{11}; 1/\sqrt{2}, 0, -1/\sqrt{2}; 3/\sqrt{22}, -2/\sqrt{22}, 3/\sqrt{22}]$

**2.75.**  $A, C$

**2.76.**  $x = 3, y = 0, z = 3$

**2.78.** (c) *Hint:* Let  $B = \frac{1}{2}(A + A^H)$  and  $C = \frac{1}{2}(A - A^H)$ .

**2.79.**  $A, B, C$

**2.81.**  $A$

**2.82.** (a)  $UV = \text{diag}([7, 6; 17, 10]; [1, 9; 7, -5]);$  (b) no; (c) yes

**2.83.**  $A$ : line between first and second rows (columns);  
 $B$ : line between second and third rows (columns) and between fourth and fifth rows (columns);  
 $C$ :  $C$  itself—no further partitioning of  $C$  is possible.

**2.84.** (a)  $M^2 = \text{diag}([4, [9, 8; 4, 9], [9]),$

$$M^3 = \text{diag}([8], [25, 44; 22, 25], [27])$$

(b)  $M^2 = \text{diag}([3, 4; 8, 11], [9, 12; 24, 33])$

$$M^3 = \text{diag}([11, 15; 30, 41], [57, 78; 156, 213])$$

**2.85.** (a)  $\text{diag}([7], [8, 24; 12, 8, [16]),$  (b)  $\text{diag}([2, 8; 16, 181], [8, 20; 40, 48])$